# ENTIRE NONLOCAL MINIMAL GRAPHS OVER $\mathbb{R}^8$: AN ANALYTIC CONSTRUCTION FOR EVERY FRACTIONAL EXPONENT

Abstract. For every $s \in (0,1)$, we construct a smooth nonaffine function $u_s : \mathbb{R}^8 \to \mathbb{R}$ whose subgraph $E_{u_s}$ locally minimizes the fractional $s$-perimeter against arbitrary measurable competitors in bounded open sets. The function is invariant under $O(4) \times O(4)$ and odd under interchange of the two factors. It is positive on $D = \{(x,y) : |y| > |x|\}$ and has zero set $\partial D$. As $R \to \infty$, the indicators of $R^{-1}E_{u_s}$ converge locally in $L^1$ in $\mathbb{R}^9$ to the indicator of $D \times \mathbb{R}$.

A quaternionic change of variables reduces the curvature estimate for an explicit cap to a weighted fractional-Laplacian bound and a scalar inequality. Comparing the resulting negative cap curvature bound and the positive curvature estimate for a hyperboloid with their Jacobi expansions places the smaller critical radial exponent of the Jacobi operator on $\partial D$ in $(1, 5/2)$. We construct two profiles of opposite curvature signs with this common leading decay and faster terms of opposite signs. Dilating the profiles at scales that vary with height gives ordered graphs on $D$. The height dependence of the critical term and the radial Jacobi gap at the faster exponent provide margins that control the nonlinear curvature and give the barrier signs. An angular kernel inequality gives comparison on $D$ and allows Dirichlet exhaustion between the barriers.

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## 1. Introduction

The classical Bernstein problem asks whether an entire minimal graph must be affine. The answer is affirmative over $\mathbb{R}^n$ for $n \le 7$. Bombieri, De Giorgi, and Giusti constructed nonaffine entire minimal graphs over $\mathbb{R}^8$ [1, Theorem B and Section IV]. Their construction is closely related to the balanced cone

$$\Sigma = \{(x,y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x| = |y|\}.$$

We study the corresponding existence problem for fractional perimeter with the same symmetry under rotations of the two factors.

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1.1. **Research background.** Caffarelli, Roquejoffre, and Savin developed the variational theory of fractional perimeter. They proved existence of minimizers, derived their Euler–Lagrange equation in the viscosity sense, and established interior regularity results [3]. Figalli and Valdinoci showed that flatness of nontrivial minimizing $s$-cones in $\mathbb{R}^n$ implies that every entire minimizing $s$-graph over $\mathbb{R}^n$ is affine [11, Theorem 1.2]. Savin and Valdinoci's classification of planar minimizing cones supplies the cone input in dimension two [12, Theorem 1]. Bernstein rigidity over $\mathbb{R}$ and $\mathbb{R}^2$ for every $s \in (0,1)$ is recorded in [11, Corollary 1.3]. Combining the same reduction with Caffarelli and Valdinoci's cone classification [4, Theorem 2] gives rigidity in base dimensions $n \le 7$ for $s$ sufficiently close to one. For $s$ sufficiently close to zero, Tsiamis proves flatness of minimizing cones in $\mathbb{R}^3$ and obtains Bernstein rigidity for entire minimizing graphs over $\mathbb{R}^3$ [13, Theorem 1.1 and Corollary 1.2(iv)].

One can also obtain rigidity from a bound on the graph itself. In any base dimension, Cozzi, Farina, and Lombardini proved that an entire nonlocal minimal graph is affine if $u(X) \ge -C(1+|X|)$ for all $X$ and some $C > 0$, or if $u(X) \le C(1+|X|)$ for all $X$ [6, Theorem 1.3].

Dávila, del Pino, and Wei constructed stationary nonlocal Lawson cones for every $s \in (0,1)$ and proved their stability in ambient dimension seven for sufficiently small $s$ [8, Theorems 3 and 4]. Stability concerns the nonnegativity of the second variation, whereas perimeter minimality compares against arbitrary competitors. Caselli, Florit-Simon, and Serra established a rigidity theorem for $C^2$ boundaries stationary for fractional perimeter with finite Morse index in $\mathbb{R}^d$. Such boundaries are flat for $3 \le d < d_s^*$, where $d_s^*$ is the smallest ambient dimension admitting a nonflat stable $s$-cone smooth away from the vertex [5, Theorem 1.31 and Definition 1.11, arXiv v3].

1.2. **Definitions and the main result.** Fix $0 < s < 1$. For a measurable set $E \subset \mathbb{R}^d$ and a bounded open set $\mathcal{O} \subset \mathbb{R}^d$, define

$$\mathrm{Per}_s(E;\mathcal{O}) = \frac{1}{2}\iint_{Q_{\mathcal{O}}} \frac{|\chi_E(X)-\chi_E(Y)|}{|X-Y|^{d+s}}\,dX\,dY, \quad Q_{\mathcal{O}} = (\mathbb{R}^d\times\mathbb{R}^d)\setminus(\mathcal{O}^c\times\mathcal{O}^c). \tag{1.1}$$

A set $E$ is locally $s$-minimal in an open set $\mathcal{U} \subset \mathbb{R}^d$ if for every open $\mathcal{O} \Subset \mathcal{U}$, $\mathrm{Per}_s(E;\mathcal{O}) < \infty$ and $\mathrm{Per}_s(E;\mathcal{O}) \le \mathrm{Per}_s(F;\mathcal{O})$ for every measurable set $F$ that agrees with $E$ almost everywhere on $\mathcal{O}^c$. Equalities and inclusions of measurable sets are understood up to null sets.

At a smooth boundary point, our curvature convention is

$$\mathcal{H}_s[E](P) = \mathrm{PV}\int_{\mathbb{R}^d} \frac{\chi_{E^c}(Y)-\chi_E(Y)}{|P-Y|^{d+s}}\,dY. \tag{1.2}$$

The principal value deletes $B_\varepsilon(P)$ and then lets $\varepsilon \downarrow 0$. With this convention a ball has positive curvature. We use $d = 8$ for the sets in the base space and $d = 9$ for the subgraphs.

For $X = (x,y) \in \mathbb{R}^4 \times \mathbb{R}^4$, set

$$D = \{(x,y) : |y| > |x|\}, \qquad T(x,y) = (y,x).$$

Thus $\partial D = \Sigma$, and $TD$ is the opposite chamber. For $u : \mathbb{R}^8 \to \mathbb{R}$, write $E_u = \{(X,z) \in \mathbb{R}^8 \times \mathbb{R} : z < u(X)\}$.

**Theorem 1.1** (Entire graph for each fractional exponent)**.** *For every $s \in (0,1)$, there exists a nonaffine $u_s \in C^\infty(\mathbb{R}^8)$ with the following properties.*

(1) *The subgraph $E_{u_s}$ is locally $s$-minimal in $\mathbb{R}^9$.*

(2) *For every $x, y \in \mathbb{R}^4$ and $Q_1, Q_2 \in O(4)$,*

$$u_s(Q_1x, Q_2y) = u_s(x,y), \qquad u_s(TX) = -u_s(X), \quad X = (x,y).$$

*The function $u_s$ is positive on $D$ and has zero set $\Sigma$.*

(3) *With $R^{-1}E_{u_s} = \{(X/R, z/R) : (X,z) \in E_{u_s}\}$,*

$$\chi_{R^{-1}E_{u_s}} \longrightarrow \chi_{D\times\mathbb{R}} \quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^9) \quad \text{as } R \to \infty.$$

*The boundary of the limiting set is $\Sigma \times \mathbb{R}$.*

1.3. **The cap estimate and the proof structure.** The starting cap is the explicit subset of $D$ given by

$$h(r) = \left(1 + \frac{2r^2}{5}\right)^{-3/8}, \qquad E_* = \{(x,y) : |y|^2 - |x|^2 > h(2|x||y|)\}. \tag{1.3}$$

For $P = (x,y) \in \partial E_*$, write $R = 2|x||y|$, $a = |x|^2 + |y|^2 = \sqrt{R^2 + h(R)^2}$, and $\alpha = s/2$. Theorem 4.9 gives

$$-\mathcal{H}_s[E_*](P) > \frac{\mathcal{A}_\alpha}{2000}\frac{h(R)}{a^{1+\alpha}}, \qquad \mathcal{A}_\alpha = \frac{\pi^{7/2}}{2}\frac{\Gamma(1+\alpha)\Gamma(1/2-\alpha)}{\Gamma(4+\alpha)\Gamma(2-\alpha)}. \tag{1.4}$$

In Section 3, the quaternionic Hopf quotient turns the quadratic comparison set into a half-space. Inversion pairs the negative part of the curvature difference with part of the positive contribution, giving a lower bound in terms of a weighted fractional Laplacian. Section 4 proves the scalar inequality that gives the strict margin in (1.4).

Section 2 derives the radial Jacobi coefficient $A_s - H_s(\lambda)$ for the power $\rho^{-\lambda}$ and obtains $H_s(1) < A_s$ from a hyperboloid of positive curvature. In the normal coordinates of the cone, the cap has displacement $f_0(\rho) \sim -B\rho^{-\kappa}$, with $\kappa = 5/2$ and $B > 0$. Section 5 expands its curvature to order $\rho^{-\kappa-1-s}$, with coefficient $-2B(H_s(\kappa) - A_s)$. On the cap, $h(R) = -2\rho f_0(\rho)$ and $a \sim \rho^2$, so the lower bound in (1.4) has the same decay rate. This forces $H_s(\kappa) > A_s$. Since $H_s$ is strictly increasing on $(0,(6-s)/2)$, the two inequalities place the smaller root $\gamma$ of $H_s(\gamma) = A_s$ in $(1,\kappa)$. Section 5 then constructs two caps with negative and positive curvature, respectively, and exact normal tails

$$f_\pm(\rho) = -A_\pm\rho^{-\gamma} \pm B_\pm\rho^{-\kappa}, \qquad A_\pm, B_\pm > 0,$$

for all sufficiently large $\rho$. The first power is a Jacobi field, and the nonlinear remainder decays faster than the curvature from the second power. Thus the sign of the second term determines the curvature sign in the tail. The dilates of each cap boundary fill $D$ without overlap. A point $X$ on the dilate by $\lambda$ has label $V_\pm(X) = \lambda^p$, with $p = 1 + \gamma$.

Section 6 builds the barriers by choosing increasing functions $g_\pm$ of the height $z > 0$ and taking the horizontal boundary sections $V_\pm(X) = \varepsilon g_\pm(z)$. This makes the coefficients of the two normal powers depend on height. The height dependence of the critical power $\rho^{-\gamma}$ supplies one curvature margin. The faster power $\rho^{-\kappa}$ supplies a second margin through $H_s(\kappa) - A_s > 0$. The proof uses these margins to control the full nonlinear curvature in the exterior regions and uses the strict cap signs on rescaled cores. This yields continuous odd barriers $0 \le L \le U$ on $D$. The lower curvature is nonpositive and the upper curvature nonnegative in the weak sense, including across the edge of the lower barrier's zero plateau.

Section 7 takes Dirichlet graph minimizers on expanding balls with exterior datum $L$, using the existence theorem of Cozzi and Lombardini [7, Theorem 1.3]. Since oddness reverses the barrier order on $TD$, comparison is carried out on $D$. Taking the absolute value on $D$ and extending it oddly to $TD$ preserves minimality. Uniqueness then gives nonnegative heights on $D$. For nonnegative heights, the magnitude of their difference is at most their sum. For fixed base radii and vertical separation, the averaged kernel is larger within $D$ than across the chambers, and it decreases as the vertical separation grows. The resulting comparison traps the Dirichlet minimizers between $L$ and $U$. The interior gradient estimate of Cabré and Cozzi gives local compactness [2, Theorem 1.1]. The bounded graph flux allows passage to the weak equation by dominated convergence. The limit is smooth and its subgraph is locally minimizing [7, Theorem 1.10]. Along each fixed ray in $D$, $L(RX)/R \to +\infty$. The chamber bounds and oddness then give $u_s(RX)/R \to +\infty$ on $D$ and $u_s(RX)/R \to -\infty$ on $TD$. The dilated subgraph has height $u_s(RX)/R$, which gives the full limit in Theorem 1.1.

The constants $c_s, C_s$ are positive, may depend on the fixed exponent $s$, and may change between estimates.

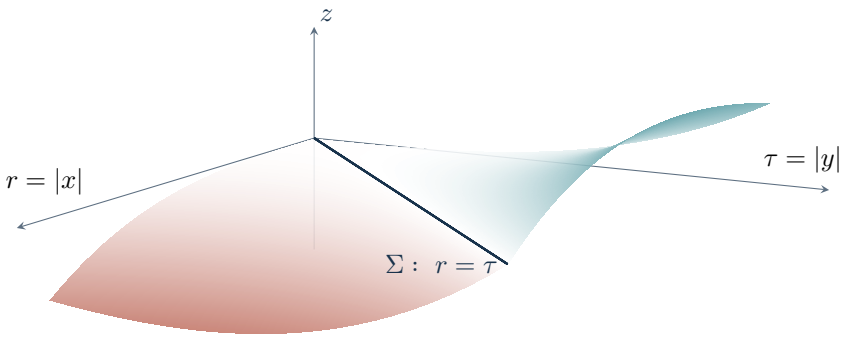



FIGURE 1. A three-dimensional picture in the reduced variables $(r, \tau, z)$. The plotted function is the schematic model $z = 0.27(\tau - r)(r + \tau)^2$. It illustrates oddness across $r = \tau$, rotational reduction, and superlinear growth.

## 2. The balanced cone and its radial operator

Fix $0 < s < 1$. In this section the ambient space is $\mathbb{R}^8$ and the curvature is given by (1.2). We use the chamber $D = \{(x, y) : |y| > |x|\}$, its boundary $\Sigma$, and the link $\Lambda = \Sigma \cap S^7$.

At a $C^2$ boundary point $P$, subtract the sign function of the tangent halfspace from the curvature integrand. Its integral against the radially truncated kernel is zero by reflection across the tangent plane. At tangential distance $r$ from $P$, the set and its tangent halfspace differ in a slab of thickness $O(r^2)$. The local contribution after subtraction is therefore absolutely integrable, with radial bound $Cr^6 r^2 r^{-8-s}\, dr = Cr^{-s}\, dr$. The tail of the kernel is integrable by $s > 0$.

2.1. **Normal coordinates and stationarity.** For $Z = \rho(\omega_1, \omega_2)/\sqrt{2}$ with $\rho > 0$ and $\omega_1, \omega_2 \in S^3$, the exterior unit normal to $D$ is

$$\nu(Z) = \frac{(\omega_1, -\omega_2)}{\sqrt{2}}.$$

It is the normalized gradient of $|x|^2 - |y|^2$ at $Z$. The normal coordinate map is

$$(Z, \tau) \longmapsto Z + \tau\nu(Z) = \left(\frac{\rho + \tau}{\sqrt{2}}\omega_1, \frac{\rho - \tau}{\sqrt{2}}\omega_2\right), \qquad -\rho < \tau < \rho. \tag{2.1}$$

It is bijective off the two factor axes, since the factor directions are $\omega_1 = x/|x|$, $\omega_2 = y/|y|$ and the radii determine

$$\rho = \frac{|x| + |y|}{\sqrt{2}}, \qquad \tau = \frac{|x| - |y|}{\sqrt{2}}, \qquad D = \{\tau < 0\} \tag{2.2}$$

in these coordinates. The omitted axes are Lebesgue null sets.

The $\rho$ direction on the cone has length one and is orthogonal to the angular directions. Each sphere contributes a surface factor $(\rho/\sqrt{2})^3$. Thus $d\sigma_\Lambda = 2^{-3} d\sigma_{S^3}(\omega_1) d\sigma_{S^3}(\omega_2)$. In the ambient product polar coordinates the angular factor is

$$|x|^3|y|^3 = \frac{(\rho^2 - \tau^2)^3}{8}, \qquad \left|\det \frac{\partial(|x|, |y|)}{\partial(\rho, \tau)}\right| = \left|\det \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\right| = 1.$$

Consequently, writing $\omega = Z/|Z|$, we have

$$d\sigma_\Sigma(Z) = \rho^6\, d\rho\, d\sigma_\Lambda(\omega), \qquad dY = \left(1 - \frac{\tau^2}{\rho^2}\right)^3 d\sigma_\Sigma(Z)\, d\tau. \tag{2.3}$$

For $P = (x, y) \in \Sigma \setminus \{0\}$, choose $U \in O(4)$ with $Uy = x$. The map $(X, Y) \mapsto (UY, U^{-1}X)$ fixes $P$ and interchanges the two chambers. It leaves the truncated kernel centered at $P$ invariant and reverses the sign $\chi_{D^c} - \chi_D$ almost everywhere. A change of variables in each truncated integral therefore gives

$$\mathcal{H}_s[D](P) = 0 \qquad (P \in \Sigma \setminus \{0\}). \tag{2.4}$$

2.2. **The radial spectrum.** Fix $P_1 \in \Lambda$, and write $d\sigma = d\sigma_\Sigma$. The normal $\nu$ is constant along each ray of $\Sigma$. Set

$$H_s(\lambda) = \mathrm{PV} \int_\Sigma \frac{1 - |Z|^{-\lambda}}{|P_1 - Z|^{8+s}}\, d\sigma(Z), \qquad -1 - s < \lambda < 7, \tag{2.5}$$

$$A_s = \int_\Sigma \frac{1 - \nu(P_1)\cdot\nu(Z)}{|P_1 - Z|^{8+s}}\, d\sigma(Z). \tag{2.6}$$

The group $O(4) \times O(4)$ acts transitively on $\Lambda$ and preserves the two integrals. The numerator of $A_s$ is $\frac{1}{2}|\nu(P_1) - \nu(Z)|^2 = O(|P_1 - Z|^2)$ near $P_1$, so the local radial bound on the seven-dimensional surface is $Cr^{-s}\, dr$. Near the vertex the distance is bounded below and the area factor is $\rho^6\, d\rho$. At infinity the radial bound is $C\rho^{-2-s}\, d\rho$, by (2.3). The normal varies with the angles, giving a positive numerator on a set of positive surface measure. Hence $0 < A_s < \infty$.

Put

$$\ell_s = 6 - s, \qquad m_s = \frac{\ell_s}{2}, \qquad \mathcal{A}_s(r) = \int_\Lambda |P_1 - r\omega|^{-8-s}\, d\sigma_\Lambda(\omega).$$

**Lemma 2.1** (Folded radial formula)**.** *The principal value in* (2.5) *exists throughout its stated domain, and*

$$H_s(\lambda) = \int_0^1 r^s(1 - r^\lambda)(1 - r^{\ell_s - \lambda})\mathcal{A}_s(r)\, dr. \tag{2.7}$$

*The integral on the right is absolutely convergent. The function $H_s$ is strictly concave on $(-1-s, 7)$, is symmetric about $m_s$, and satisfies $H_s(0) = H_s(\ell_s) = 0$. Its derivative is positive on $(0, m_s)$ and negative on $(m_s, \ell_s)$.*

*Proof.* We use radial inversion as in [8, Eq. (11.7), accepted manuscript] and track its effect on the deleted ball. Split the truncated integral at $|Z| = 1$. Near $|Z| = 0$ the radial bound is $C(r^6 + r^{6-\lambda})$, and at infinity it is $C(r^{-2-s} + r^{-2-s-\lambda})$. Both bounds are integrable in the stated range. In the outer part put $Z = \omega/r$, with $0 < r < 1$. Write $d(r,\omega) = |P_1 - r\omega|$ and $R = |Z| = 1/r$ in the outer integral. Then

$$|P_1 - \omega/r| = d(r,\omega)/r, \qquad R^6\, |dR| = r^{-8}\, dr.$$

The outer integrand becomes $r^s(1 - r^\lambda)d(r,\omega)^{-8-s}$, and its cutoff is $d(r,\omega) > r\varepsilon$. The inner integrand is $r^6(1-r^{-\lambda})d(r,\omega)^{-8-s}$, with cutoff $d(r,\omega) > \varepsilon$. Over the common region $d(r,\omega) > \varepsilon$, the radial numerator is

$$r^6(1 - r^{-\lambda}) + r^s(1 - r^\lambda) = r^6 + r^s - r^{6-\lambda} - r^{s+\lambda} = r^s(1 - r^\lambda)(1 - r^{\ell_s - \lambda}).$$

The only contribution outside this common region comes from the outer integrand on $r\varepsilon < d(r,\omega) \le \varepsilon$. There $1 - r \le d(r,\omega) \le \varepsilon$, so $\varepsilon(1-\varepsilon) < d(r,\omega) \le \varepsilon$. In a tangent chart $Z = P_1 + u + g(u)\nu(P_1)$, we have $g(0) = Dg(0) = 0$, so

$$|Z - P_1| = \sqrt{|u|^2 + g(u)^2} = |u| + O(|u|^3), \qquad d\sigma(Z) = \sqrt{1 + |Dg(u)|^2}\, du.$$

The shell thus has surface measure $O(\varepsilon^6\varepsilon^2)$. The angular-radial measure $dr\, d\sigma_\Lambda$ is comparable to surface measure there, since $r \to 1$ on the shell. The numerator $r^s(1 - r^\lambda)$ is $O(\varepsilon)$, so the contribution is $O(\varepsilon^{1-s})$.

It remains to integrate the paired numerator. For $r \le 1/2$, $\mathcal{A}_s(r)$ is bounded and the four powers in the expanded numerator all have exponents greater than $-1$. For $1/2 < r < 1$, use

$$d(r,\omega)^2 = (1 - r)^2 + r|P_1 - \omega|^2.$$

On a six-dimensional chart of $\Lambda$ at $P_1$, $|P_1 - \omega|$ is comparable to the chart radius. The contribution there to $\mathcal{A}_s(r)$ is bounded by

$$C \int_{\mathbb{R}^6} \frac{du}{((1-r)^2 + |u|^2)^{(8+s)/2}} = C(1-r)^{-2-s} \int_{\mathbb{R}^6} \frac{dv}{(1 + |v|^2)^{(8+s)/2}}.$$

The rest of the link contributes a bounded integral. Both factors $1 - r^\lambda$ and $1 - r^{\ell_s - \lambda}$ are $O(1 - r)$ near one. The paired integrand is thus $O((1-r)^{-s})$. Dominated convergence on the common region and the vanishing shell error give (2.7).

On compact subintervals of $(-1-s, 7)$, differentiating the numerator up to twice at $r = 0$ adds only integrable logarithmic factors to the power bounds. Near $r = 1$, the first derivative contains $\log r\,(r^{\ell_s-\lambda} - r^{\lambda})$ and the second contains $(\log r)^2(r^{\lambda} + r^{\ell_s-\lambda})$. Both are $O((1-r)^2)$ uniformly on those compact subintervals. We may therefore differentiate under the integral to get

$$H_s'(\lambda) = \int_0^1 r^s \log r\,(r^{\ell_s-\lambda} - r^{\lambda})\mathcal{A}_s(r)\,dr,$$
$$H_s''(\lambda) = -\int_0^1 r^s(\log r)^2(r^{\lambda} + r^{\ell_s-\lambda})\mathcal{A}_s(r)\,dr < 0.$$

If $\lambda < m_s$, then $\ell_s - \lambda > \lambda$ and $r^{\ell_s-\lambda} < r^{\lambda}$ for $0 < r < 1$. Thus the first integrand is positive. The folded expression is invariant under $\lambda \mapsto \ell_s - \lambda$, which gives the derivative sign on the right half and the two endpoint values. □

The Jacobi operator on the punctured cone is

$$J_s w(P) = \mathrm{PV}\int_{\Sigma} \frac{w(Z) - w(P)}{|P - Z|^{8+s}}\,d\sigma(Z) + A_s|P|^{-1-s}w(P). \tag{2.8}$$

For a smooth $w$ near $P$ with integrable tails, the linear term of $w(Z) - w(P)$ cancels in a centered tangent chart. In the same graph coordinates as above, centered now at $P$, Taylor's formula gives $w(Z) - w(P) = L(u) + O(|u|^2)$ with $L$ linear. The distance and area estimates show that the kernel times surface density is $|u|^{-8-s}(1 + O(|u|^2))$. The integral of $L(u)|u|^{-8-s}$ over a centered tangent annulus is zero. After subtracting it, the remainder is bounded by $C|u|^{-6-s}$ against $du$. The ball cutoff differs from the tangent cutoff only on a shell of thickness $O(\varepsilon^3)$, where the integral of the linear term is $O(\varepsilon^{2-s})$. This gives the principal value in (2.8).

For $P = RP_1$, $R = |P| > 0$, put $Z = RZ'$. The surface measure contributes $R^7$, the kernel $R^{-8-s}$, and the power $R^{-\lambda}$. The deleted ball becomes $|Z' - P_1| > \varepsilon/R$. The first integral in (2.8) is therefore $-R^{-\lambda-1-s}H_s(\lambda)$ when $w(Z) = |Z|^{-\lambda}$. The same scaling without the factor $R^{-\lambda}$ gives $A_sR^{-1-s}$ for the integral of $1 - \nu(P)\cdot\nu(Z)$ against the kernel. Hence

$$J_s(\rho^{-\lambda})(P) = (A_s - H_s(\lambda))|P|^{-\lambda-1-s} \qquad (-1-s < \lambda < 7). \tag{2.9}$$

We use the first-variation formula for a smooth normal displacement $w$ that vanishes near the vertex and decays with its first two derivatives as a positive inverse power at infinity. In the coordinates (2.1) let $E_{\varepsilon w} = \{\tau < \varepsilon w(Z)\}$, and set $P_\varepsilon = P + \varepsilon w(P)\nu(P)$. Since $\sup_\Sigma |w(Z)|/|Z| < \infty$, we may take $|\varepsilon w(Z)| < |Z|/2$. Then

$$\frac{d}{d\varepsilon}\mathcal{H}_s[E_{\varepsilon w}](P_\varepsilon)\Big|_{\varepsilon=0} = -2J_s w(P). \tag{2.10}$$

For smooth boundaries and smooth bounded displacements we use [8, Proposition B.1, accepted manuscript]. Their Jacobi operator has the sign in (2.8). Their curvature numerator is $\chi_E - \chi_{E^c}$, giving the minus sign in (2.10) with our convention. To apply the formula here, smooth the cone near the vertex using

$$D_\delta = \{(x, y) : |y|^2 > |x|^2 + \delta^2\beta(|x|^2/\delta^2)\},$$

where $0 \le \beta \le 1$ is smooth, equals one near zero, and is zero for arguments at least one. On the boundary the $y$ gradient of the defining function is nonzero, because $|y|^2 = |x|^2 + \delta^2\beta(|x|^2/\delta^2) > 0$. The change of set is confined to $|x| \le \delta$, $|y| \le \sqrt{2}\,\delta$, so $D_\delta = D$ outside $B_{\sqrt{3}\,\delta}$. Since $D_\delta = \delta D_1$, the surface area of the rounded portion is $O(\delta^7)$. The corresponding cone portion has the same bound by (2.3).

Take $\delta$ small enough that $P$ lies outside that ball and the ball lies within the region where the displacement is zero. On $\partial D_\delta$ use $w$ on the common boundary and zero on the rounded portion. This gives a smooth bounded displacement on a smooth boundary, with the same

point $P_\varepsilon$. Denote the perturbed set by $D_{\delta,\varepsilon}$. The two perturbations differ by the fixed set $D \setminus D_\delta$ in the ball. Thus, for small $\varepsilon$,

$$\mathcal{H}_s[D_{\delta,\varepsilon}](P_\varepsilon) - \mathcal{H}_s[E_{\varepsilon w}](P_\varepsilon) = 2\int_{D\setminus D_\delta} |P_\varepsilon - Y|^{-8-s}\,dY.$$

The right side has derivative bounded by $C_{s,P}|w(P)|\delta^8$, since $P_\varepsilon$ stays a positive distance from the ball. In the Jacobi formula on $\partial D_\delta$, the integral away from the ball is identical to that on the cone. In the ball the source displacement is zero, so the combined numerator on either surface is $-w(P)\nu(P)\cdot\nu_{\text{surf}}(Z)$, where $\nu_{\text{surf}}$ is the exterior unit normal of that surface. The change in the Jacobi integral is therefore bounded by $C_{s,P}|w(P)|\delta^7$. Applying the smooth formula and sending $\delta \downarrow 0$ proves (2.10).

### 2.3. **A positive hyperboloid for every exponent.**

**Proposition 2.2** (Hyperboloid sign)**.** *For every $c > 0$, the set*

$$E_c = \{(x,y) : |y|^2 - |x|^2 > c\}$$

*satisfies $\mathcal{H}_s[E_c](P) > 0$ at every boundary point. Moreover,*

$$H_s(1) < A_s. \tag{2.11}$$

*Proof.* Let $Q(X) = |X_y|^2 - |X_x|^2$ and $B(P,\omega) = P_y\cdot\omega_y - P_x\cdot\omega_x$. For $Q(P) = c$ and $h > 0$,

$$Q(P \pm h\omega) - c = h(hQ(\omega) \pm 2B(P,\omega)).$$

The two signs on the rays $\omega$ and $-\omega$ cancel for $h < 2|B(P,\omega)|/|Q(\omega)|$. Above that threshold, both signs equal $\operatorname{sign} Q(\omega)$. These formulas hold away from a null set of directions, and directions with $Q(\omega) = 0$ contribute zero. The exterior-minus-interior curvature uses the negative of these signs. Averaging the two rays and integrating $h^{-1-s}$ above the threshold gives

$$\mathcal{H}_s[E_c](P) = -\frac{2^{-s}}{s}\int_{S^7} \operatorname{sign} Q(\omega)\frac{|Q(\omega)|^s}{|B(P,\omega)|^s}\,d\omega. \tag{2.12}$$

With the cutoff $h > \varepsilon$, the radial integral starts at the maximum of the threshold and $\varepsilon$. Its absolute value is bounded by $C_s|B(P,\omega)|^{-s}$. The linear functional $B(P,\cdot)$ has nonzero coefficient vector, so its absolute negative power is integrable on $S^7$ by the one-dimensional bound $\int_0^1 u^{-s}\,du < \infty$ near its equator. This justifies removal of the cutoff.

Rotate the factors so that $P = (re_1, te_1)$ with $t > r \ge 0$. Use $\omega = (a\xi, b\eta)$, where $a = \cos\theta$, $b = \sin\theta$, $0 < \theta < \pi/2$, and $\xi, \eta \in S^3$. The $\theta$ direction is unit and orthogonal to the two angular factors, so the area element is $a^3b^3\,d\theta\,d\sigma(\xi)\,d\sigma(\eta)$. Write Av for normalized spherical averages, and let $c_{4,s} = \operatorname{Av}_{e\in S^3}|e_1\cdot e|^{-s}$. For any fixed $e \in S^3$, simultaneous rotational invariance gives

$$\operatorname{Av}_{\xi,\eta}|(A\eta - B\xi)\cdot e_1|^{-s} = \operatorname{Av}_{\xi,\eta}|(A\eta - B\xi)\cdot e|^{-s}.$$

Averaging in $e$ and using Tonelli's theorem gives

$$\operatorname{Av}_{\xi,\eta}|A\eta_1 - B\xi_1|^{-s} = c_{4,s}\operatorname{Av}_{\xi,\eta}|A\eta - B\xi|^{-s}. \tag{2.13}$$

Pair each $\theta > \pi/4$ with $\pi/2 - \theta$ in (2.12). On the upper half put $q = b^2 - a^2 > 0$ and $u = \eta\cdot\xi$. The two squared vector lengths from (2.13) are

$$D_+ = t^2b^2 + r^2a^2 - 2rtabu, \qquad D_- = t^2a^2 + r^2b^2 - 2rtabu.$$

Here $D_+$ corresponds to the positive value $Q(\omega) = q$. Their difference is $D_+ - D_- = (t^2 - r^2)q = cq$. For almost every angle the fundamental theorem of calculus gives

$$D_-^{-s/2} - D_+^{-s/2} = \frac{s}{2}cq\int_0^1 \left((1-v)D_- + vD_+\right)^{-1-s/2}\,dv.$$

Thus the paired form of (2.12) is

$$\frac{\mathcal{H}_s[E_c](P)}{c} = \frac{c_{4,s}|S^3|^2}{2^{1+s}}\int_{\pi/4}^{\pi/2} a^3b^3q^{s+1}\operatorname{Av}_{\xi,\eta}\int_0^1 \left((1-v)D_- + vD_+\right)^{-1-s/2}\,dv\,d\theta. \tag{2.14}$$

The original angular integral is absolutely convergent, and the paired difference is nonnegative. The right side is strictly positive, including when $r = 0$, because both squared lengths are positive for almost every angle and $q > 0$ on the interval. This proves the curvature sign.

We now let $c \downarrow 0$ at normal radius one. By (2.1), $Q(Z + \tau\nu(Z)) = -2\rho\tau$. The boundary point corresponding to $P_1$ is therefore $P_c = P_1 - (c/2)\nu(P_1)$, with factor radii

$$r_c = (1 - c/2)/\sqrt{2}, \qquad t_c = (1 + c/2)/\sqrt{2}.$$

For these radii $D_\pm$ tend to $\frac{1}{2} - abu$. For any $A, B \geq 0$ and $-1 \leq u \leq 1$,

$$A^2 + B^2 - 2ABu \;\geq\; \tfrac{1}{2}(A^2 + B^2)(1 - u).$$

For $u \geq 0$ this follows from $2AB \leq A^2 + B^2$, and for $u < 0$ the left side is at least $A^2 + B^2$. For small $c$, both $r_c$ and $t_c$ are bounded below. Applied to the coefficient pairs $(t_c b, r_c a)$ and $(t_c a, r_c b)$, the inequality gives $D_\pm \geq C(1 - u) = C|\eta - \xi|^2/2$ uniformly in $\theta$. Their convex combinations have the same bound. The majorant $C|\eta - \xi|^{-2-s}$ is integrable on $S^3 \times S^3$, since the radial bound in a chart centered at $\xi$ is $Cr^{-s}\,dr$. Dominated convergence in (2.14), and $\mathcal{H}_s[D](P_1) = 0$, give

$$\frac{d}{dc}\mathcal{H}_s[E_c](P_c)\Big|_{c=0+} = \frac{c_{4,s}|S^3|^2}{2^{1+s}} \int_{\pi/4}^{\pi/2} a^3 b^3 q^{s+1}\, \mathrm{Av}_{\xi,\eta}\big(\tfrac{1}{2} - ab\,\eta\cdot\xi\big)^{-1-s/2}\, d\theta > 0. \tag{2.15}$$

To identify this derivative with the Jacobi coefficient, let $\chi_\delta$ be a smooth radial cutoff, zero for $\rho \leq \delta$ and one for $\rho \geq 2\delta$, with values in $[0, 1]$. Apply (2.10) to the displacement $w_\delta(Z) = -\chi_\delta(\rho)/(2\rho)$. For fixed $0 < \delta < 1/8$ and $0 < c < \delta^2$, the perturbed cone and the hyperboloid both pass through $P_c$ and agree for $\rho \geq 2\delta$. Inside that radius the hyperboloid has threshold $-\min\{\rho, c/(2\rho)\}$ in the $\tau$ coordinate, including the case of an empty slice. The cutoff threshold is $cw_\delta(Z)$. Both thresholds lie in $[-c/(2\rho), 0]$, so the symmetric difference has volume at most

$$C\int_0^{2\delta} \rho^6 \frac{c}{2\rho}\, d\rho \leq Cc\delta^6$$

by (2.3). Since $|Y|^2 = \rho^2 + \tau^2$ and $|\tau| \leq \rho$, the difference lies in $|Y| \leq 2\sqrt{2}\,\delta$. Its contribution to curvature at $P_c$, divided by $c$, is $O_s(\delta^6)$. For the cutoff cone the quotient tends to $-2J_s w_\delta(P_1)$ by (2.10) and stationarity. The cutoff changes only the source term of $J_s(\rho^{-1})(P_1)$, with error bounded by $C_s\int_0^{2\delta}\rho^5\,d\rho$. Letting $c \downarrow 0$ first and then $\delta \downarrow 0$, we conclude that (2.15) equals $J_s(\rho^{-1})(P_1) = A_s - H_s(1)$. This proves (2.11). □

We will use the fixed intermediate exponent

$$\kappa = \frac{5}{2} < m_s = \frac{6 - s}{2}. \tag{2.16}$$

**Lemma 2.3** (Spectrum under an intermediate gap)**.** *If $H_s(\kappa) > A_s$, there are exactly two roots of $H_s(\lambda) = A_s$ in $(0, 6 - s)$. They are simple and obey*

$$1 < \gamma_-(s) < \kappa < m_s < \gamma_+(s) < 6 - s, \qquad \gamma_+(s) = 6 - s - \gamma_-(s).$$

*In particular $\rho^{-\gamma_-(s)}$ is a positive radial Jacobi field on the punctured cone.*

*Proof.* Equation (2.11) and the hypothesis bracket a root in $(1, \kappa)$. Lemma 2.1 gives $H_s' > 0$ on $(0, m_s)$, so this root is unique there and simple. It also gives $H_s(m_s) \geq H_s(\kappa) > A_s$. Reflection about $m_s$ gives the unique root $\gamma_+ = 6 - s - \gamma_-$ in $(m_s, 6 - s)$, with $H_s'(\gamma_+) = -H_s'(\gamma_-) < 0$. Equation (2.9) then gives $J_s(\rho^{-\gamma_-}) = 0$. □

The strict intermediate gap will follow by comparing the explicit cap's curvature sign with its normal asymptotic expansion.

## 3. An explicit cap with a strict curvature sign

This section reduces the curvature of one explicit cap to a scalar inequality. We use (1.2) in dimension eight, so

$$\mathcal{H}_s[E](P) = \mathrm{PV}\int_{\mathbb{R}^8} \frac{\mathbf{1}_{E^c}(X) - \mathbf{1}_E(X)}{|X-P|^{8+s}}\,dX. \tag{3.1}$$

The interior-minus-exterior signed integral used below is $-\mathcal{H}_s[E](P)$.

Fix the parameters

$$\alpha = \frac{s}{2}, \qquad \theta = \frac{1}{2}+\alpha, \qquad \sigma = 2\theta = 1+s, \qquad \beta = \frac{1-\alpha}{2}, \qquad q = \frac{3}{4}, \qquad c = \frac{5}{2}. \tag{3.2}$$

Define

$$h(r) = \left(1+\frac{r^2}{c}\right)^{-q/2} = \left(1+\frac{2r^2}{5}\right)^{-3/8}, \tag{3.3}$$
$$E = \left\{(x,y)\in\mathbb{R}^4\times\mathbb{R}^4 : |y|^2 - |x|^2 > h(2|x||y|)\right\}.$$

The variables used after the Hopf change of variables are different from the original variables $(x,y)$. We write them as $(z,Y)\in\mathbb{R}\times\mathbb{R}^4$.

Here are the constants in the reduction. Their normalizations will be derived below. For $0<v<4-\sigma$, set

$$\mathfrak{c}_\sigma(v) = 2^\sigma \frac{\Gamma((v+\sigma)/2)\Gamma((4-v)/2)}{\Gamma(v/2)\Gamma((4-v-\sigma)/2)}. \tag{3.4}$$

Set

$$\mathcal{A}_\alpha = \frac{\pi^{7/2}}{2}\frac{\Gamma(1+\alpha)\Gamma(\frac{1}{2}-\alpha)}{\Gamma(4+\alpha)\Gamma(2-\alpha)}, \tag{3.5}$$

$$Q_\alpha = \frac{4\Gamma(2-\alpha)}{(1+2\alpha)\Gamma(1+\alpha)}\left[\mathfrak{c}_\sigma(q+\beta) - \mathfrak{c}_\sigma(\beta)\right], \tag{3.6}$$

$$\ell^2 = \frac{q^2}{c}\frac{(q+1)^{q+1}}{(q+2)^{q+2}}, \qquad c_L(\alpha) = (1+\ell^2)^{-(5+2\alpha)/2}. \tag{3.7}$$

Finally, put $c_0 = q/2 = 3/8$ and define

$$C_\alpha = \frac{c_0\,2^{\alpha-1}}{\frac{3}{2}-\alpha} + \frac{c_0^{(1+\alpha)/2}-c_0}{(1-\alpha)/2} + \frac{c_0^{(1+\alpha)/2}}{(1+\alpha)/2}, \tag{3.8}$$

$$R_\alpha = \frac{\Gamma(\frac{1}{2}+\alpha)\Gamma(1-\alpha)}{\Gamma(\frac{1}{2}-\alpha)\Gamma(1+\alpha)}, \tag{3.9}$$

$$E_\alpha = 2^{\alpha-1}\alpha(1-\alpha^2)\left(\frac{1}{2}+\alpha\right)R_\alpha C_\alpha. \tag{3.10}$$

The letter $E$ denotes the set in (3.3); the subscripted quantity $E_\alpha$ is a scalar error bound.

**Proposition 3.1** (The curvature reduction)**.** *Let $P\in\partial E$. Write*

$$R = 2|x||y|, \qquad h_0 = h(R), \qquad a = |x|^2+|y|^2 = \sqrt{R^2+h_0^2}, \qquad P = (x,y).$$

*Then*

$$-\mathcal{H}_s[E](P) \geq \mathcal{A}_\alpha \frac{h_0}{a^{1+\alpha}}\left[c_L(\alpha)Q_\alpha - E_\alpha - 1\right]. \tag{3.11}$$

In the quotient the quadratic comparison set is a horizontal halfspace. The curvature difference is a signed integral over two strips. Inversion pairs the negative strip with part of the positive one and gives a lower bound in terms of a weighted fractional Laplacian. Theorem 4.1 proves positivity of the resulting bracket.

3.1. **Elementary properties of the cap.**

**Lemma 3.2.** *The function $h$ is positive and strictly decreasing on $(0,\infty)$. The function $r \mapsto \sqrt{r^2+h(r)^2}$ is strictly increasing there. Moreover,*

$$\mathrm{Lip}(h)^2 = \ell^2 < \frac{3}{80}, \qquad h(Rt) - h(R) \le t^{-q}\big[h(R) - h(R/t)\big] \quad (0<t<1, R>0). \tag{3.12}$$

*Proof.* Differentiation gives

$$h'(r) = -\frac{qrh(r)}{c+r^2}, \qquad h''(r) = \frac{qh(r)((q+1)r^2-c)}{(c+r^2)^2}. \tag{3.13}$$

Since $h \le 1$ and $c > q$,

$$\frac{d}{dr}(r^2+h(r)^2) = 2r\left(1 - \frac{qh(r)^2}{c+r^2}\right) > 0.$$

Writing $u = r^2/c$, we obtain

$$h'(r)^2 = \frac{q^2}{c}\, u(1+u)^{-q-2}.$$

Since

$$\frac{d}{du}\big[u(1+u)^{-q-2}\big] = (1+u)^{-q-3}\big[1-(q+1)u\big],$$

the maximum occurs at $u = 1/(q+1)$. Substitution gives (3.7). For the stated parameters,

$$\ell^2 = \frac{9}{10}\left(\frac{7^7}{11^{11}}\right)^{1/4} < \frac{3}{80},$$

because

$$11^{11} - 24^4\, 7^7 = 12\,079\,868\,243 > 0.$$

For the second inequality, put $B(r) = -rh'(r) = qc^{q/2}r^2(c+r^2)^{-1-q/2}$. If $0<t<1$, then $c + t^2r^2 \ge t^2(c+r^2)$ and hence

$$\frac{B(tr)}{B(r)} = t^2\left(\frac{c+r^2}{c+t^2r^2}\right)^{1+q/2} \le t^2 t^{-2-q} = t^{-q}.$$

Consequently,

$$\begin{aligned} h(Rt) - h(R) &= \int_R^{R/t} B(tu)\,\frac{du}{u} \\ &\le t^{-q}\int_R^{R/t} B(u)\,\frac{du}{u} = t^{-q}\big[h(R) - h(R/t)\big]. \end{aligned}$$

□

The defining function $|y|^2 - |x|^2 - h(2|x||y|)$ is smooth at the axes, since $h(2|x||y|) = (1 + 4|x|^2|y|^2/c)^{-q/2}$. Its zero set has $|y| > |x|$. With $\tau = |y| > 0$ and $|x|$ fixed, the derivative is $2\tau - 2|x|h'(2|x|\tau) \ge 2\tau > 0$. The implicit function theorem gives a smooth boundary, and the tangent-half-space argument in Section 2 gives existence of the curvature principal value there.

3.2. **The quaternionic quotient and its exact kernel.** Identify $\mathbb{R}^4$ with the quaternions $\mathbb{H}$. Quaternionic conjugation is denoted by a bar, and the Euclidean inner product is $\langle u, v\rangle = \mathrm{Re}(\overline{u}v)$. Consider

$$\Pi(x,y) = (|y|^2 - |x|^2,\ 2y\overline{x}) \in \mathbb{R}\times\mathbb{H} = \mathbb{R}^5. \tag{3.14}$$

Multiplicativity of the quaternion norm gives

$$|\Pi(x,y)|^2 = (|y|^2-|x|^2)^2 + 4|y|^2|x|^2 = (|x|^2+|y|^2)^2.$$

Thus

$$|\Pi(x,y)| = |x|^2 + |y|^2. \tag{3.15}$$

The set $E$ is the inverse image under $\Pi$ of the epigraph

$$\mathcal{E} = \{(z,Y) \in \mathbb{R}\times\mathbb{R}^4 : z > h(|Y|)\}.$$

**Lemma 3.3** (Coarea and the fiber average)**.** *Let $\eta \neq 0$ and put $b = |\eta|$. In quotient and fiber coordinates, Lebesgue measure decomposes as*

$$dX = \frac{\pi^2}{16b}\, d\eta\, d\omega, \tag{3.16}$$

*where $d\omega$ is the normalized surface measure on the fiber.*

*Let $\xi = \Pi(P)$, $a = |\xi|$, and $d = |\xi - \eta|$. For $d > 0$ the fiber average of $|X - P|^{-8-s}$ is*

$$(a+b)^{1+\alpha} d^{-5-2\alpha} F_\alpha\left(1 - \frac{d^2}{(a+b)^2}\right), \tag{3.17}$$

*where*

$$F_\alpha(v) = {}_2F_1\left(-\frac{\alpha}{2}, -\frac{1+\alpha}{2}; 2; v\right). \tag{3.18}$$

*Proof.* For this proof write $(u, v) = (y, x)$. At $(u, v)$, a variation $(\dot u, \dot v)$ gives

$$D\Pi(u,v)(\dot u, \dot v) = 2(\langle u, \dot u\rangle - \langle v, \dot v\rangle,\ \dot u\, \overline{v} + u\, \overline{\dot v}).$$

Taking the adjoint uses the identity $\mathrm{Re}(AB) = \mathrm{Re}(BA)$, which also permits cyclic permutations under the real part. The adjoint is

$$D\Pi(u,v)^*(\lambda, W) = 2(\lambda u + Wv, -\lambda v + \overline{W} u).$$

Here the cross terms in the squared norm cancel because $\langle u, Wv\rangle = \langle v, \overline{W} u\rangle$. The diagonal terms give

$$|D\Pi(u,v)^*(\lambda, W)|^2 = 4(|u|^2 + |v|^2)(\lambda^2 + |W|^2).$$

Thus $D\Pi D\Pi^* = 4b\, I_5$ when $\Pi(u, v) = \eta$, and the coarea Jacobian is $(2\sqrt{b})^5 = 32 b^{5/2}$.

For $\eta = (z, Y)$ the component norms satisfy $|v|^2 = (b - z)/2$ and $|u|^2 = (b + z)/2$. If $b > z$, choose $v = \sqrt{(b-z)/2}$ real and $u = Y/(2v)$ to obtain a representative. If $b = z > 0$, then $Y = 0$ and one can take $(u, v) = (\sqrt{b}, 0)$. Right multiplication of both factors by a unit quaternion preserves $\Pi$. Conversely, for another representative $(u', v')$ with $v \neq 0$, put $\omega = v^{-1} v'$. Then $|\omega| = 1$ and $u'\overline{\omega}\,\overline{v} = u\overline{v}$ gives $u' = u\omega$. If $v = 0$, both $v$ and $v'$ vanish and the same conclusion follows by taking $\omega = u^{-1} u'$. The real-linear map $\omega \mapsto (u\omega, v\omega)$ multiplies lengths by $\sqrt{b}$, so the fiber has surface measure $2\pi^2 b^{3/2}$. Dividing this measure by the coarea Jacobian proves (3.16).

Choose a representative $X = (u, v)$ of the fiber over $\eta$, and write $P = (u_P, v_P)$ in the same order. Set

$$Z = \overline{u_P} u + \overline{v_P} v.$$

Writing $\xi = (z_P, Y_P)$ and $\eta = (z, Y)$, the scalar components give

$$ab + z_P z = 2|u_P|^2 |u|^2 + 2|v_P|^2 |v|^2.$$

The remaining part is $\langle Y_P, Y\rangle = 4\,\mathrm{Re}(v_P \overline{u_P} u \overline{v})$, which is twice the cross term in $|Z|^2$. Hence $2|Z|^2 = ab + \langle \xi, \eta\rangle$. Thus

$$|P - (u\omega, v\omega)|^2 = a + b - 2\,\mathrm{Re}(Z\omega).$$

For a uniform unit quaternion, the coordinate $t = \cos\vartheta$ has density $(2/\pi)\sqrt{1-t^2}$ on $[-1, 1]$, since the area factor $\sin^2\vartheta\, d\vartheta$ becomes $\sqrt{1-t^2}\,|dt|$ and $|S^2|/|S^3| = 2/\pi$. Put

$$A = a + b, \qquad B = 2|Z|, \qquad A^2 - B^2 = d^2.$$

The required average is

$$\frac{2}{\pi} \int_{-1}^{1} (A - Bt)^{-4-\alpha} \sqrt{1-t^2}\, dt. \tag{3.19}$$

We use the notation

$${}_2F_1(a_1, a_2; b; z) = \sum_{n=0}^{\infty} \frac{(a_1)_n (a_2)_n}{(b)_n n!}\, z^n, \qquad (a)_n = a(a+1)\cdots(a+n-1).$$

The substitution $u = t^2$ in the moment integral gives

$$\frac{2}{\pi}\int_{-1}^{1} t^{2n}\sqrt{1-t^2}\,dt = \frac{2}{\pi}\,\mathrm{B}(n+\tfrac{1}{2},\tfrac{3}{2}) = \frac{(1/2)_n}{(2)_n},$$

where B is the beta function. Since $d > 0$, we have $B/A < 1$ and may integrate the binomial expansion of (3.19) term by term. The odd moments vanish. Using $(2n)! = 4^n n!(1/2)_n$, the coefficient of $(B^2/A^2)^n$ after factoring out $A^{-4-\alpha}$ is

$$\frac{(4+\alpha)_{2n}}{(2n)!}\frac{(1/2)_n}{(2)_n} = \frac{(2+\alpha/2)_n(5/2+\alpha/2)_n}{(2)_n n!}.$$

For $\zeta = B^2/A^2 \in [0,1)$, we use Euler's transformation

$${}_2F_1(a_1,a_2;2;\zeta) = (1-\zeta)^{2-a_1-a_2}\,{}_2F_1(2-a_1,2-a_2;2;\zeta)$$

in the normalization above (DLMF, Eq. 15.8.1).

Here $2-a_1-a_2 = -5/2-\alpha$ and $1-\zeta = d^2/A^2$. Thus the prefactor is $A^{-4-\alpha}(d^2/A^2)^{-5/2-\alpha} = A^{1+\alpha}d^{-5-2\alpha}$, as claimed. □

**Lemma 3.4** (The kernel factor)**.** *For $0 \le v \le 1$,*

$$0 \le F_\alpha(1) - F_\alpha(v) \le F_\alpha'(1)(1-v). \tag{3.20}$$

*Moreover,*

$$F_\alpha(1) = \frac{\Gamma(\frac{5}{2}+\alpha)}{\Gamma(2+\alpha/2)\Gamma(\frac{5}{2}+\alpha/2)}, \tag{3.21}$$

$$\frac{F_\alpha'(1)}{F_\alpha(1)} = \frac{\alpha(1+\alpha)}{6+4\alpha}. \tag{3.22}$$

*Proof.* Write $p_1 = -\alpha/2$ and $p_2 = -(1+\alpha)/2$. For $n \ge 1$ each of $(p_1)_n$ and $(p_2)_n$ has exactly one negative factor. All nonconstant coefficients are therefore positive, and $F_\alpha$ and $F_\alpha'$ increase on $[0,1)$. Differentiating the series gives

$$F_\alpha'(v) = \frac{p_1p_2}{2}\,{}_2F_1(p_1+1,p_2+1;3;v).$$

At unity we use Gauss's summation formula

$${}_2F_1(a_1,a_2;b;1) = \frac{\Gamma(b)\Gamma(b-a_1-a_2)}{\Gamma(b-a_1)\Gamma(b-a_2)} \qquad (\mathrm{Re}(b-a_1-a_2)>0)$$

(DLMF, Eq. 15.4.20).

The parameter differences $2-p_1-p_2 = 5/2+\alpha$ and $3-(p_1+1)-(p_2+1) = 3/2+\alpha$ are both positive. All gamma factors are finite. Applying the formula to the two series gives (3.21) and

$$\frac{F_\alpha'(1)}{F_\alpha(1)} = p_1p_2\frac{\Gamma(1-p_1-p_2)}{\Gamma(2-p_1-p_2)} = \frac{\alpha(1+\alpha)}{6+4\alpha}.$$

Integrating $0 \le F_\alpha'(w) \le F_\alpha'(1)$ over $v < w < 1$ proves (3.20). □

The kernel depends on the quotient points through $a$, $b$, and $d$. Rotating the last four quotient coordinates also preserves the radial epigraph, so we may assume

$$\xi = (h_0, X_0), \qquad X_0 = Re_1.$$

For $\eta = (z, Y)$ put

$$r = |Y|, \quad b = \sqrt{r^2+z^2}, \quad d = \sqrt{|Y-X_0|^2+(z-h_0)^2}, \quad M_\alpha(a,b) = \frac{(a+b)^{1+\alpha}}{b}. \tag{3.23}$$

The exact curvature formula is

$$-\mathcal{H}_s[E](P) = \frac{\pi^2}{16}\,\mathrm{PV}\int_{\mathbb{R}^5} \mathrm{sign}(z-h(|Y|))M_\alpha(a,b)d^{-5-2\alpha}F_\alpha\left(1-\frac{d^2}{(a+b)^2}\right)dz\,dY. \tag{3.24}$$

We first record the local cancellation used for the quotient and for the strip integrals below. In dimension $n$, write $w$ for coordinates centered at the singular point and $r = |w|$. Suppose $g(w) = L(w) + O(r^2)$, where $L$ is a nonzero linear form, and $k(w) = k_0 + O(r)$. In the integrand $\operatorname{sign} g(w)\, k(w) r^{-n-s}$ subtract the model

$$k_0 \chi(r) \operatorname{sign} L(w)\, r^{-n-s}, \tag{3.25}$$

where $\chi$ is radial, compactly supported in the coordinate neighborhood, and equal to one near zero. The coefficient error is bounded by $Cr^{-n-s+1}$. The sign discrepancy is confined to $|L(w)| \le Cr^2$, which occupies a fraction $O(r)$ of each small sphere centered at zero. As in the tangent-half-space argument in Section 2, the remainder is absolutely integrable with radial bound $Cr^{-s}\, dr$. The model is odd under $w \mapsto -w$, so its integral under a centered spherical or horizontal deletion is zero. This also applies to differences of two sign functions by linearity.

To compare ambient and quotient deletions, put $K = \ker D\Pi(P)$. The adjoint calculation in Lemma 3.3 gives $D\Pi(P) D\Pi(P)^* = 4aI_5$. Thus the map

$$X \longmapsto \left( \frac{\Pi(X) - \xi}{2\sqrt{a}}, \operatorname{pr}_K(X - P) \right)$$

has an orthogonal derivative at $P$ after identifying $K$ with $\mathbb{R}^3$. Its inverse is a local chart $X = \Phi(u, v)$ for $u \in \mathbb{R}^5$ and $v \in \mathbb{R}^3$. With $r = (|u|^2 + |v|^2)^{1/2}$ it satisfies

$$\Pi(\Phi(u,v)) = \xi + 2\sqrt{a}\, u, \qquad |\Phi(u,v) - P| = r + O(r^2), \qquad |\det D\Phi(u,v)| = 1 + O(r).$$

The sign of $E$ is $\operatorname{sign} g(u)$, where $g(u) = z(u) - h(|Y(u)|)$ and $(z(u), Y(u)) = \xi + 2\sqrt{a}\, u$. Its linear part is nonzero because the derivative in the $z$ direction is one before the rescaling. Thus (3.25) applies with $w = (u, v)$ and $k_0 = 1$, and the model cancels for both $r > \varepsilon$ and $|u| > \varepsilon/(2\sqrt{a})$. The original ambient deletion $|\Phi(u,v) - P| > \varepsilon$ differs from $r > \varepsilon$ only on a shell of thickness $O(\varepsilon^2)$ at radius $\varepsilon$. The model's absolute integral there is $O(\varepsilon^7 \varepsilon^2 \varepsilon^{-8-s}) = O(\varepsilon^{1-s})$. The absolute remainder therefore has the same limit for the ambient and quotient deletions. Outside this chart we are away from $P$, so the kernel is integrable there and dominated convergence applies. The quotient deletion tends to one almost everywhere outside the chart, since the omitted fiber is a three-dimensional sphere and has zero ambient measure.

For each positive quotient cutoff $d > \varepsilon$, apply (3.16) and (3.17) to the ambient integral with this cutoff. This gives the truncated right side of (3.24). The preceding comparison proves the equality and existence of its ordinary five-dimensional principal value. It applies also to the horizontal epigraph $z > h_0$, with $g(u) = z(u) - h_0$. The factor $b^{-1}$ at the quotient origin is locally integrable in five dimensions. At infinity $d \asymp b$ and the quotient kernel is $O(b^{-5-\alpha})$.

3.3. **A lower bound for the quadratic boundary term.** The horizontal quotient half-space $\{z > h_0\}$ lifts to

$$E_{h_0}^{\mathrm{quad}} = \{(x, y) : |y|^2 - |x|^2 > h_0\}.$$

The following estimate is the negative term that the cap gain must overcome.

**Lemma 3.5.** *At a point $P \in \partial E_{h_0}^{\mathrm{quad}}$ with $a = |P|^2$ and $h_0 > 0$,*

$$-\mathcal{H}_s[E_{h_0}^{\mathrm{quad}}](P) \ge -\mathcal{A}_\alpha h_0 a^{-1-\alpha}. \tag{3.26}$$

*Proof.* Put $\delta = h_0/a \in (0, 1]$. We will use only ordinary Gaussian integrals. For a function $f$ on $\mathbb{R}^4 \times \mathbb{R}^4$, write

$$\langle f \rangle_G = (2\pi)^{-4} \int_{\mathbb{R}^4 \times \mathbb{R}^4} f(U, V) e^{-(|U|^2 + |V|^2)/2}\, dU\, dV.$$

For $S > 0$ set

$$\lambda = \frac{S}{2}(1 + \delta), \qquad \mu = \frac{S}{2}(1 - \delta),$$

$$Q_{S,\delta}(U, V) = |U + \sqrt{\lambda} e_1|^2 - |V + \sqrt{\mu} e_1|^2 - S\delta, \qquad B(S, \delta) = \langle \operatorname{sign} Q_{S,\delta} \rangle_G.$$

In particular, $|B(S,\delta)| \le 1$. The Gaussian representation of the Riesz kernel gives

$$-\mathcal{H}^{}_s[E^{\mathrm{quad}}_{h_0}](P) = \frac{\pi^4 a^{-\alpha}}{2^\alpha \Gamma(4+\alpha)} \int_0^\infty S^{\alpha-1} B(S,\delta)\, dS. \tag{3.27}$$

Here are the details, including the principal-value issue. The identity

$$|X-P|^{-8-s} = \frac{1}{\Gamma(4+\alpha)} \int_0^\infty u^{3+\alpha} e^{-u|X-P|^2}\, du$$

follows by the change of variable $v = u|X-P|^2$ in the gamma integral. Independent rotations in the two four-dimensional factors place $P^{}_y$ and $P^{}_x$ on their respective first coordinate axes. Substitute $X = P + (2u)^{-1/2}(V,U)$ and put $S = 2ua$. Since $2u|P^{}_y|^2 = S(1+\delta)/2$ and $2u|P^{}_x|^2 = S(1-\delta)/2$, the sign of $|y|^2 - |x|^2 - h_0$ becomes the sign of $Q^{}_{S,\delta}$. The Gaussian mass is $(\pi/u)^4$, and changing from $u$ to $S$ gives the constant in (3.27). For the interchange, subtract the sign of the tangent half-space through $P$. The difference is absolutely integrable against $|X-P|^{-8-s}$ by the tangent-half-space estimate of Section 2. Applying Tonelli to its absolute value in the Gaussian representation justifies Fubini for the signed difference. The half-space has zero integral against each centered Gaussian by reflection, so the resulting averages are exactly $B(S,\delta)$.

To calculate the Gaussian characteristic function, use

$$(2\pi)^{-1/2} \int_{\mathbb{R}} e^{-Aw^2/2 + Bw}\, dw = A^{-1/2} e^{B^2/(2A)}, \qquad \operatorname{Re} A > 0.$$

The square completion for positive real $A$ and real $B$ extends to these complex parameters by analyticity, with the square root positive for $A > 0$. In the first $U$ coordinate take $A = 1 - 2it$, $B = 2it\sqrt{\lambda}$. Multiplying by $e^{it\lambda}$ leaves the exponent $it\lambda - 2t^2\lambda/(1-2it) = it\lambda/(1-2it)$. The other three coordinates have $B = 0$. Hence the four-dimensional factor is $(1-2it)^{-2} \exp(it\lambda/(1-2it))$. For $V$ replace $t, \lambda$ by $-t, \mu$. After multiplication by $e^{-iS\delta t}$ the exponent is

$$\frac{i\lambda t}{1-2it} - \frac{i\mu t}{1+2it} - iS\delta t = -\frac{2St^2}{1+4t^2} - i\frac{4S\delta t^3}{1+4t^2},$$

using $\lambda + \mu = S$ and $\lambda - \mu = S\delta$. Thus

$$\left\langle e^{itQ_{S,\delta}} \right\rangle_G = (1+4t^2)^{-2} \exp\left(-\frac{2St^2}{1+4t^2}\right) \exp\left(-i\frac{4S\delta t^3}{1+4t^2}\right). \tag{3.28}$$

The elementary sign identity

$$\operatorname{sign} x = \frac{2}{\pi} \lim_{\varepsilon \downarrow 0} \int_0^\infty e^{-\varepsilon t} \frac{\sin(tx)}{t}\, dt$$

follows because the integral vanishes at $x = 0$ and its $x$ derivative is $\int_0^\infty e^{-\varepsilon t} \cos(tx)\, dt = \varepsilon/(\varepsilon^2 + x^2)$. The integral is therefore $\arctan(x/\varepsilon)$. For fixed $\varepsilon > 0$, it can be averaged under the Gaussian integral by Fubini: for $0 < t < 1$ use $|\sin(tQ^{}_{S,\delta})|/t \le |Q^{}_{S,\delta}|$, which has a finite Gaussian integral, and for $t \ge 1$ use the exponential damping. Bounded convergence for $\arctan(Q^{}_{S,\delta}/\varepsilon)$ then gives the sign average as $\varepsilon \downarrow 0$. After inserting (3.28), the integrand is $O^{}_{S,\delta}(t^2)$ near zero and $O(t^{-5})$ at infinity. Thus dominated convergence also removes the damping in this formula, and we obtain

$$B(S,\delta) = -\frac{2}{\pi} \int_0^\infty \frac{e^{-2St^2/(1+4t^2)}}{t(1+4t^2)^2} \sin\left(\frac{4S\delta t^3}{1+4t^2}\right) dt.$$

Since $\sin v \le v$ for $v \ge 0$,

$$B(S,\delta) \ge -\frac{8S\delta}{\pi} \int_0^\infty \frac{t^2}{(1+4t^2)^3} \exp\left(-\frac{2St^2}{1+4t^2}\right) dt. \tag{3.29}$$

The same positive expression bounds $|B(S,\delta)|$, using $|\sin v| \le v$. Its $S^{\alpha-1}$ integral is finite, as the following computation shows. Tonelli's theorem for this positive majorant and the gamma integral give

$$\begin{aligned}\int_0^\infty S^{\alpha-1}B(S,\delta)\,dS &\ge -\delta\,\frac{8}{\pi}\Gamma(1+\alpha)2^{-1-\alpha}\int_0^\infty t^{-2\alpha}(1+4t^2)^{\alpha-2}\,dt\\ &= -\delta\,\frac{2^{\alpha-1}\Gamma(1+\alpha)\Gamma(\frac12-\alpha)}{\sqrt{\pi}\,\Gamma(2-\alpha)}.\end{aligned}$$

Indeed, the substitution $v = 4t^2$ gives

$$\int_0^\infty t^{-2\alpha}(1+4t^2)^{\alpha-2}\,dt = 2^{2\alpha-2}\int_0^\infty v^{-1/2-\alpha}(1+v)^{\alpha-2}\,dv = 2^{2\alpha-2}\mathrm{B}(\tfrac12-\alpha,\tfrac32).$$

Both beta parameters are positive. Combining the coefficient above with the one in (3.27) gives

$$\frac{\pi^4}{2^\alpha\Gamma(4+\alpha)}\,\frac{2^{\alpha-1}\Gamma(1+\alpha)\Gamma(\frac12-\alpha)}{\sqrt{\pi}\,\Gamma(2-\alpha)} = \mathcal{A}_\alpha.$$

The remaining factor is $\delta a^{-\alpha} = h_0 a^{-1-\alpha}$, which proves (3.26). □

3.4. **The nonlinear gain and a transport of strips.** Keep the notation of (3.23) and put $m = 5+2\alpha = 4+\sigma$. In (3.24) the cap and the quadratic comparison set have signs $\operatorname{sign}(z-h(r))$ and $\operatorname{sign}(z-h_0)$. Their difference is 2 on the strip $h(r) < z < h_0$ when $r > R$, and $-2$ on $h_0 < z < h(r)$ when $r < R$. Thus

$$-\mathcal{H}_s[E](P) = -\mathcal{H}_s[E^{\mathrm{quad}}_{h_0}](P) + \frac{\pi^2}{16}\mathcal{G}, \tag{3.30}$$

where the vertical integral is oriented and

$$\mathcal{G} = 2\,\mathrm{PV}\int_{\mathbb{R}^4}\int_{h(r)}^{h_0} M_\alpha(a,b)d^{-m}F_\alpha\left(1-\frac{d^2}{(a+b)^2}\right)dz\,dY. \tag{3.31}$$

Omitting the hypergeometric factor gives the principal gain

$$\mathcal{G}_M = 2\,\mathrm{PV}\int_{\mathbb{R}^4}\int_{h(r)}^{h_0} M_\alpha(a,b)d^{-m}\,dz\,dY. \tag{3.32}$$

The linear integral that will enter the fractional Laplacian is

$$I_\beta = \mathrm{PV}\int_{\mathbb{R}^4}\frac{(|Y|^2+h_0^2)^{-\beta/2}(h_0-h(|Y|))}{|Y-X_0|^{4+\sigma}}\,dY. \tag{3.33}$$

In these strip integrals we use the horizontal truncation $|Y-X_0| > \varepsilon$. To obtain (3.30) with this truncation, apply the local cancellation (3.25) to the difference of the two epigraphs with their common kernel coefficient. The difference of the frozen tangent halfspace signs is odd under $(Y-X_0, z-h_0) \mapsto (X_0-Y, h_0-z)$, and both the spherical and horizontal truncations are invariant under this map. The local remainder is absolutely integrable. This also applies with $F_\alpha$ omitted and with the coefficient replaced by a smooth horizontal weight. For $I_\beta$, the Taylor expansion of its numerator at $Y = X_0$ has an odd linear part and a remainder $O(D^2)$, where $D = |Y-X_0|$. The latter gives the radial bound $CD^{3+2-m}\,dD = CD^{-s}\,dD$. At infinity the strip height is bounded by one, $b \asymp r$, $d \asymp r$, and $M_\alpha(a,b) = O(r^\alpha)$, so the strip integrals converge by comparison with $\int_1^\infty r^{-2-\alpha}\,dr$. The linear integral also converges there because its numerator is bounded.

**Proposition 3.6** (Quantitative principal gain)**.** *One has*

$$\mathcal{G}_M \ge 2^{2+\alpha}a^{(1+\alpha)/2}c_L(\alpha)I_\beta. \tag{3.34}$$

*Proof.* For $u = b/a > 0$, use $\beta + (1+\alpha)/2 = 1$ to write

$$\mathcal{R}_\alpha(u) = \frac{(1+u)^{1+\alpha}}{u} - 2^{1+\alpha}u^{-\beta} = u^{-\beta}\left[(\sqrt{u} + u^{-1/2})^{1+\alpha} - 2^{1+\alpha}\right]. \tag{3.35}$$

The bracket is nonnegative by $\sqrt{u} + u^{-1/2} \geq 2$. Since $\alpha + \beta = (1+\alpha)/2$, this gives

$$M_\alpha(a,b) = 2^{1+\alpha}a^{(1+\alpha)/2}b^{-\beta} + a^\alpha \mathcal{R}_\alpha(b/a). \tag{3.36}$$

The contribution of the residual. Suppose $R > 0$. Map the inner strip into the outer strip by

$$\begin{gathered}(Y,z) = (tR\omega, h_0 + p) \longmapsto (Y',z') = (R\omega/t, h_0 - t^q p),\\ 0 < t < 1, \qquad 0 \leq p \leq h(tR) - h_0.\end{gathered} \tag{3.37}$$

Here $\omega \in S^3$, and (3.12) gives $0 \leq t^q p \leq h_0 - h(R/t)$, so the image belongs to the outer strip. The map is one-to-one because the horizontal inversion $Y' = R^2 Y/|Y|^2$ recovers $Y$ and the height then recovers $p$. Its horizontal differential has eigenvalue $-t^{-2}$ in the radial direction and $t^{-2}$ in the three perpendicular directions. The full differential is block triangular, with vertical derivative $-t^q$. Its absolute Jacobian is therefore

$$t^{-8}t^q = t^{q-8}. \tag{3.38}$$

Write $b_-$ for the radius $b$ at the inner point and $b_+$ for that at its image. The bounds

$$tR \leq R \leq R/t, \qquad h_0 \leq z \leq h(tR), \qquad h(R/t) \leq z' \leq h_0$$

and the increase of $r^2 + h(r)^2$ imply $b_- \leq a \leq b_+$. Also $b_-^2 \geq t^2R^2 + h_0^2 \geq t^2a^2$ and $b_+^2 \leq R^2/t^2 + h_0^2 \leq a^2/t^2$. The product has a further useful bound. With $t$ fixed,

$$b_-^2 = t^2R^2 + (h_0+p)^2, \qquad b_+^2 = R^2/t^2 + (h_0 - t^q p)^2, \qquad \frac{d}{dp}(b_-^2 b_+^2) = 2(zb_+^2 - t^q z' b_-^2) \geq 0.$$

The sign uses $z \geq z' \geq 0$, $t^q \leq 1$, and $b_+ \geq b_-$. At $p = 0$ the product is $a^4 + h_0^2R^2(t - t^{-1})^2$, so $b_-b_+ \geq a^2$.

The bracket in (3.35) increases with $|\log u|$, because $\sqrt{u} + u^{-1/2} = 2\cosh((\log u)/2)$. Since $b_+/a \geq a/b_- \geq 1$, its value at $b_+/a$ is at least its value at $b_-/a$. Also $b_+/b_- \leq t^{-2}$. It follows that

$$\mathcal{R}_\alpha(b_+/a) \geq (b_+/b_-)^{-\beta}\mathcal{R}_\alpha(b_-/a) \geq t^{1-\alpha}\mathcal{R}_\alpha(b_-/a).$$

Writing $X_0 = Re_1$, direct expansion of the horizontal distances gives

$$|X_0 - Y'|^2 = R^2(1 + t^{-2} - 2t^{-1}e_1 \cdot \omega) = t^{-2}|X_0 - Y|^2.$$

Thus the full distance $d'$ at the image satisfies $(d')^2 = D^2/t^2 + t^{2q}p^2 \leq d^2/t^2$. The outer residual density after transport is at least

$$t^{q-8}\, t^{5+2\alpha}\, t^{1-\alpha} = t^{q+\alpha-2} \geq 1$$

times the inner density, where $q + \alpha < 2$. The residual strip integrals are absolutely convergent. Near $Y = X_0$ the strip height is $O(D)$, $b - a = O(D)$, and $\mathcal{R}_\alpha(b/a) = O(D^2)$ because its bracket vanishes quadratically at $u = 1$. The resulting radial bound is $CD^{1-s}\,dD$. The tails converge as for the original strip integral, since $a^\alpha \mathcal{R}_\alpha(b/a) \leq M_\alpha(a,b)$. We can therefore compare the integrals by the change of variables. The contribution of the residual is nonnegative, including the positive outer strip left outside the image.

The principal power weight. Put $b_0(r) = \sqrt{r^2 + h_0^2}$. On the positive strip $z \leq h_0$, hence $b^{-\beta} \geq b_0(r)^{-\beta}$. On the negative strip $z \geq h_0$, hence $b^{-\beta} \leq b_0(r)^{-\beta}$. Replacing the power by $b_0(r)^{-\beta}$ thus gives a lower bound for the signed gain for every horizontal cutoff. The difference of the two weights is $O(D)$ near the evaluation point, so its integral over the strip has the radial bound $CD^{-s}\,dD$. It remains to prove

$$\mathrm{PV}\int_{\mathbb{R}^4} b_0(r)^{-\beta} J_{|X_0 - Y|}(h_0 - h(r))\,dY \geq c_L(\alpha) I_\beta, \tag{3.39}$$

where $J_D(v) = \int_0^v (D^2 + w^2)^{-m/2}\,dw$ for either sign of $v$.

For $R > 0$ pair $Y = tR\omega$ with $Y' = R\omega/t$ by the same horizontal inversion. Write $d_- = h(tR) - h_0$ and $d_+ = h_0 - h(R/t)$. The substitution $w = v/t$ gives $J_{D/t}(d_+) = t^{m-1}J_D(td_+)$. Including the horizontal Jacobian $t^{-8}$, the paired integrands for the left side of (3.39) and for $I_\beta$ are respectively

$$b_0(tR)^{-\beta}[\Lambda J_D(td_+) - J_D(d_-)], \qquad b_0(tR)^{-\beta}D^{-m}(\Lambda td_+ - d_-), \qquad \Lambda = t^{2\alpha-4}\left(\frac{b_0(tR)}{b_0(R/t)}\right)^{\beta}.$$

These pairings respect the principal values. Under inversion the cutoff on the outer integral is $D > t\varepsilon$, so the extra term is confined to $t\varepsilon < D \le \varepsilon$. There $t \ge 1 - \varepsilon/R$, and the resulting annulus in $\mathbb{R}^4$ has volume $O_R(\varepsilon^5)$. The pulled-back outer density for either integral is $O(D^{1-m})$ because the height difference is $O(D)$ and the factors involving $t$ and $b_0$ are bounded. The extra integral is therefore $O_R(\varepsilon^{1-s})$.

Since $b_0(tR)/b_0(R/t) \ge t^2$, we have $\Lambda \ge t^{\alpha-3} > 1$. By (3.12) and $q < 2 - \alpha$,

$$\Lambda td_+ \ge t^{\alpha-2}d_+ \ge t^{-q}d_+ \ge d_-.$$

The Lipschitz bound in (3.12) gives $d_- \le \ell\,|R - tR| \le \ell D$ and $td_+ \le t\ell\,|R/t - R| \le \ell D$. On $[0, \ell D]$ the function $J_D$ is increasing and concave, with

$$J_D'(v) = (D^2 + v^2)^{-m/2} \ge c_L(\alpha)D^{-m}.$$

For $u = td_+ \ge v = d_-$, use

$$\Lambda J_D(u) - J_D(v) = (\Lambda - 1)J_D(v) + \Lambda(J_D(u) - J_D(v)).$$

The derivative bound on both terms gives a lower bound $c_L(\alpha)D^{-m}(\Lambda u - v)$. For $u < v$, concavity gives $J_D(u) \ge (u/v)J_D(v)$. Since $\Lambda u/v - 1 \ge 0$ and $J_D(v) \ge c_L(\alpha)D^{-m}v$, we obtain the same lower bound. Integration over the common truncated region and passage to the principal values prove (3.39).

When $R = 0$, there is only a positive strip. The residual is nonnegative, the weight freezing gives the same lower bound, and $0 \le h_0 - h(r) \le \ell D$ gives (3.39) directly by integrating $J_D'$. Combining the bounds with the prefactor in (3.36) and the factor two in (3.32) gives (3.34). □

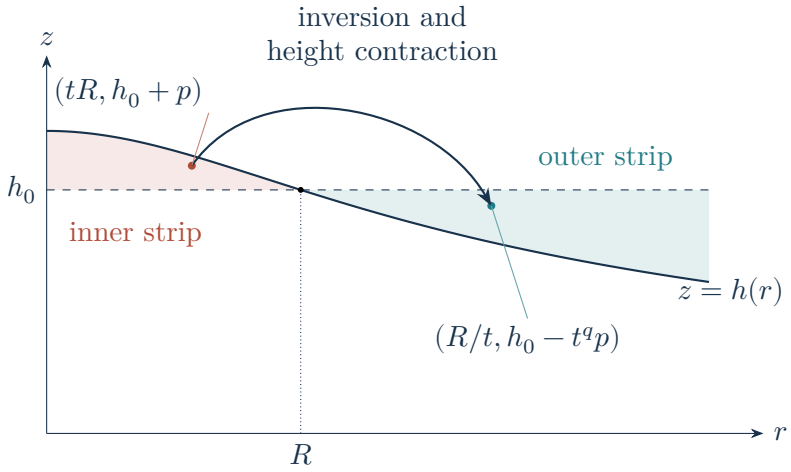



FIGURE 2. The strip transport in the $(r, z)$ plane of the Hopf quotient. The angular variable $\omega \in \mathbb{S}^3$ is kept fixed. The image of the inner strip is contained in the outer strip.

3.5. **The normalized fractional Laplacian in four dimensions.** We use the Fourier transform convention

$$\hat{f}(\zeta) = \int_{\mathbb{R}^4} e^{-iY\cdot\zeta} f(Y)\, dY, \qquad f(Y) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{iY\cdot\zeta} \hat{f}(\zeta)\, d\zeta.$$

For $0 < \sigma < 2$, write $L_\sigma = (-\Delta)^{\sigma/2}$, with Fourier multiplier $|\zeta|^\sigma$. Its singular-integral normalization is

$$L_\sigma f(X) = \gamma_{4,\sigma}\, \mathrm{PV} \int_{\mathbb{R}^4} \frac{f(X) - f(Y)}{|X-Y|^{4+\sigma}}\, dY, \qquad \gamma_{4,\sigma} = \frac{2^\sigma \Gamma(2+\sigma/2)}{\pi^2 |\Gamma(-\sigma/2)|}. \tag{3.40}$$

For the parameters in (3.2), this becomes

$$\gamma_{4,\sigma} = \frac{2^{2\alpha}(1+2\alpha)\Gamma(\frac{5}{2}+\alpha)}{\pi^2 \Gamma(\frac{1}{2}-\alpha)}. \tag{3.41}$$

**Lemma 3.7** (Powers and radial bubbles)**.** *If $0 < j < 4 - \sigma$, then, away from the origin,*

$$L_\sigma |Y|^{-j} = \mathfrak{c}_\sigma(j) |Y|^{-j-\sigma}. \tag{3.42}$$

*For $d > 0$, define*

$$K_\sigma(j) = 2^\sigma \Gamma(2+\theta) \frac{\Gamma(j/2+\theta)}{\Gamma(j/2)}, \tag{3.43}$$

$$\mathsf{F}_j(z) = {}_2F_1(-\theta, j/2+\theta; 2; z), \qquad \phi_j(z) = K_\sigma(j) \mathsf{F}_j(z). \tag{3.44}$$

*Then*

$$L_\sigma (|Y|^2 + d)^{-j/2} = (|Y|^2+d)^{-(j+\sigma)/2} \phi_j\left(\frac{|Y|^2}{|Y|^2+d}\right). \tag{3.45}$$

*Moreover,*

$$\phi_j(1) = \mathfrak{c}_\sigma(j). \tag{3.46}$$

*For $0 < \theta \le 1$ and $0 < j < 4 - 2\theta$, $\mathsf{F}_j$ is positive, decreasing, and concave on $[0,1]$.*

*Proof.* First, the normalization in (3.40) follows by computing the multiplier of the difference integral. The imaginary term cancels by oddness on each truncated domain. The Laplace representation of $|Y|^{-4-2\theta}$ and Gaussian integration give

$$\begin{aligned} \int_{\mathbb{R}^4} \frac{1-\cos(\zeta\cdot Y)}{|Y|^{4+2\theta}}\, dY &= \frac{\pi^2}{\Gamma(2+\theta)} \int_0^\infty t^{\theta-1} \left(1 - e^{-|\zeta|^2/(4t)}\right) dt \\ &= \frac{\pi^2 |\Gamma(-\theta)|}{4^\theta \Gamma(2+\theta)} |\zeta|^{2\theta}. \end{aligned}$$

In the second line, substitute $u = |\zeta|^2/(4t)$ and integrate by parts to obtain

$$\int_0^\infty (1-e^{-u}) u^{-1-\theta}\, du = \frac{\Gamma(1-\theta)}{\theta} = |\Gamma(-\theta)|.$$

The boundary terms vanish since $0 < \theta < 1$. The reciprocal coefficient of $|\zeta|^{2\theta}$ is $\gamma_{4,\sigma}$. Writing $|\Gamma(-\theta)| = \Gamma(1-\theta)/\theta$ and substituting $\theta = 1/2 + \alpha$ gives (3.41).

The identity

$$|Y|^{-j} = \frac{1}{\Gamma(j/2)} \int_0^\infty t^{j/2-1} e^{-t|Y|^2}\, dt$$

and the Fourier transform of a Gaussian yield, as tempered distributions,

$$\begin{aligned} \widehat{|Y|^{-j}}(\zeta) &= \frac{\pi^2}{\Gamma(j/2)} \int_0^\infty t^{j/2-3} e^{-|\zeta|^2/(4t)}\, dt \\ &= 2^{4-j} \pi^2 \frac{\Gamma((4-j)/2)}{\Gamma(j/2)} |\zeta|^{j-4}. \end{aligned}$$

Here $\widehat{e^{-t|Y|^2}} = (\pi/t)^2 e^{-|\zeta|^2/(4t)}$ and the second line follows by $u = |\zeta|^2/(4t)$. For the distributional justification, truncate the $t$ integral at both ends and pair with a Schwartz test function.

On the physical side, the truncations are dominated by $|Y|^{-j}$. On the Fourier side the transformed Gaussian integrals are positive and bounded by a constant times $|\zeta|^{j-4}$, which is locally integrable. These bounds allow passage to the limit for $0 < j < 4$. Multiplying the Fourier density by $|\zeta|^{\sigma}$ and comparing with the formula for $j + \sigma < 4$ gives the coefficient in (3.4).

To identify the distribution with the pointwise integral, first truncate the difference integral in (3.40) to shifts $|h| > \varepsilon$. For a Schwartz test function $\varphi$,

$$\int_{\mathbb{R}^4} |Y-h|^{-j}|\varphi(Y)|\,dY \le \|\varphi\|_{\infty}\int_{B_1}|Y|^{-j}\,dY + \|\varphi\|_{L^1},$$

uniformly in $h$, by splitting at $|Y-h| = 1$. The truncated shift kernel is integrable, so Fubini applies to this pairing. On the Fourier side the corresponding pairing is dominated by $C|\zeta|^{j-4}|\widehat{\varphi}(-\zeta)|$ times the truncated shift kernel. Thus the translation formula for the Fourier transform gives the truncated multiplier. By symmetry it is a nonnegative integral of $1-\cos(\zeta\cdot h)$, bounded by $|\zeta|^{\sigma}$, and tends to that value as the truncation is removed. The product with the Fourier density converges in tempered distributions since $|\zeta|^{j+\sigma-4}|\widehat{\varphi}(-\zeta)|$ is integrable near zero by $j+\sigma>0$ and at infinity by Schwartz decay. In physical space the truncated integrals converge locally uniformly away from the origin. To see this, subtract the odd linear Taylor term in a ball around the evaluation point of radius less than half its distance to the origin. The $O(|X-Y|^2)$ remainder is integrable since $\sigma < 2$. At the origin, $|Y|^{-j}$ is locally integrable because $j < 4$, and at infinity the kernel has integrable decay. This proves (3.42) at each nonzero point.

For the bubble at scale one, apply [9, Corollary 2, arXiv v1] in dimension four with the constant solid harmonic $V = 1$, operator order $\sigma$, upper parameters $j/2, 2$, and lower parameter 2. The balance condition on the smallest upper parameter is $j > -\sigma$, which holds here. Their regularized hypergeometric function agrees with the ordinary one because $\Gamma(2) = 1$. By the binomial identity ${}_2F_1(j/2, 2; 2; -|Y|^2) = (1+|Y|^2)^{-j/2}$, the corollary gives

$$L_{\sigma}(1+|Y|^2)^{-j/2} = K_{\sigma}(j)\,{}_2F_1(2+\theta, j/2+\theta; 2; -|Y|^2)$$

with the same Fourier multiplier $|\zeta|^{\sigma}$. Dilating $Y$ to $Y/\sqrt{d}$ multiplies the operator by $d^{-\sigma/2}$. Since the bubble itself has the factor $d^{-j/2}$, the formula at scale $d$ has prefactor $K_{\sigma}(j)d^{-(j+\sigma)/2}$ and argument $-|Y|^2/d$.

Put $b_j = j/2 + \theta \in (0,2)$ and $w = |Y|^2/d$. Pfaff's transformation (DLMF, Eq. 15.8.1) gives

$${}_2F_1(2+\theta, b_j; 2; -w) = (1+w)^{-b_j}\,{}_2F_1(-\theta, b_j; 2; w/(1+w)).$$

Here $1 + w > 0$ and the lower parameter is two, so the same normalization applies as in the proof of Lemma 3.3. Since $0 < b_j < 2$, the Euler integral for the last factor is

$$\mathsf{F}_j(z) = \frac{1}{\mathrm{B}(b_j, 2-b_j)}\int_0^1 t^{b_j-1}(1-t)^{1-b_j}(1-zt)^{\theta}\,dt. \tag{3.47}$$

Since $d^{-b_j}(1+w)^{-b_j} = (|Y|^2+d)^{-b_j}$, this gives (3.45). The factor $(1-zt)^{\theta}$ is positive, decreasing, and concave in $z \in [0,1]$ for each $0 < t < 1$ and $0 < \theta \le 1$. The beta weight is integrable, so integration preserves these properties and gives continuity at both endpoints. At $z = 1$, evaluation of the beta integral gives

$$\mathsf{F}_j(1) = \frac{\mathrm{B}(b_j, 2-b_j+\theta)}{\mathrm{B}(b_j, 2-b_j)} = \frac{\Gamma(2-j/2)}{\Gamma(2+\theta)\Gamma(2-j/2-\theta)}.$$

Multiplying by $K_{\sigma}(j)$ proves (3.46). □

3.6. **Two endpoint estimates for the weighted bubble.** The next two elementary estimates will allow us to control the weighted bubble at every radius, including both $R = 0$ and $R \to \infty$.

**Lemma 3.8** (An endpoint gamma bound)**.** *For $1/2 \le \theta \le 1$, set*

$$u = \frac{3}{2} - \frac{\theta}{2}, \qquad k(\theta) = \mathsf{F}_u(1) = \frac{\Gamma(\frac{5}{4} + \frac{\theta}{4})}{\Gamma(2+\theta)\Gamma(\frac{5}{4} - \frac{3\theta}{4})}.$$

*Then*

$$k(\theta) < \left(\frac{2}{5}\right)^{\theta}. \tag{3.48}$$

*Proof.* We use the same formula to define $k$ for $0 \le \theta \le 1$. The logarithmic derivative of $\Gamma$ is denoted by $\psi$. Its derivative has the convergent series

$$\psi'(x) = \sum_{n=0}^{\infty} (n+x)^{-2}, \qquad x > 0.$$

Thus $\psi'$ is positive and strictly decreasing. Differentiating twice gives

$$(\log k)'' = \frac{1}{16}\psi'\left(\frac{5}{4} + \frac{\theta}{4}\right) - \psi'(2+\theta) - \frac{9}{16}\psi'\left(\frac{5}{4} - \frac{3\theta}{4}\right) < 0.$$

Indeed, the first trigamma argument is at least the last, and its positive coefficient is smaller than the last negative coefficient. Also $k(0) = 1$.

At $\theta = 1/2$, gamma recurrence gives

$$k(1/2) = \frac{\Gamma(11/8)}{\Gamma(5/2)\Gamma(7/8)} = \frac{\Gamma(3/8)}{2\sqrt{\pi}\,\Gamma(7/8)} = \frac{3}{4\mathfrak{k}}, \qquad \mathfrak{k} = \frac{3\sqrt{\pi}}{2}\frac{\Gamma(7/8)}{\Gamma(3/8)}.$$

We claim that $\mathfrak{k} > 6/5$. Seven applications of the recurrence $\Gamma(x+1) = x\Gamma(x)$ give

$$\frac{\Gamma(7/8)}{\Gamma(3/8)} = \prod_{n=0}^{6} \frac{8n+3}{8n+7}\frac{\Gamma(63/8)}{\Gamma(59/8)} = \frac{375003}{2178215}\frac{\Gamma(63/8)}{\Gamma(59/8)}.$$

For $x > 0$, log-convexity of $\Gamma$ implies

$$\Gamma(x+1)^2 \le \Gamma(x+1/2)\Gamma(x+3/2) = (x+1/2)\Gamma(x+1/2)^2.$$

Therefore

$$\frac{\Gamma(x+1/2)}{\Gamma(x)} \ge \frac{x}{\sqrt{x+1/2}}.$$

Apply this at $x = 59/8$ and use $\pi > 157/50$ to obtain

$$\mathfrak{k}^2 > \frac{9}{4}\frac{157}{50}\left(\frac{375003}{2178215}\right)^2 \frac{(59/8)^2}{63/8} = \frac{36}{25} + \frac{333941978531853}{53139750565720000}.$$

The bound on $\pi$ follows from Machin's identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239)$$

and the alternating series:

$$\pi > 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3}\right) - \frac{4}{239} = \frac{1184}{375} - \frac{4}{239} > \frac{157}{50}.$$

To verify Machin's identity, put $A = \arctan(1/5)$ and $B = \arctan(1/239)$. The double-angle formula gives $\tan(2A) = 5/12$ and $\tan(4A) = 120/119$. Since $2A < \pi/4$ and $B < A$, we have $0 < 4A - B < \pi/2$. The tangent subtraction formula gives $\tan(4A - B) = 1$, so $4A - B = \pi/4$. This proves the claim and hence $k(1/2) < 5/8$.

Since $\log k$ is concave and $\log k(0) = 0$, the quotient $\log k(\theta)/\theta$ is decreasing for $\theta > 0$. For $\theta \ge 1/2$ it follows that

$$k(\theta) \le k(1/2)^{2\theta} < \left(\frac{25}{64}\right)^{\theta} < \left(\frac{2}{5}\right)^{\theta}. \qquad \square$$

**Lemma 3.9** (A geometric bubble inequality)**.** *For $0 \le w \le 1$, put*

$$b_*(w) = w + \frac{2}{5}(1-w)^{7/4}.$$

*For $1/2 \le \theta \le 1$, and $u = 3/2 - \theta/2$, one has*

$$b_*(w)^\theta \mathsf{F}_u(w) \ge \mathsf{F}_u(1). \tag{3.49}$$

*Proof.* We first establish a purely algebraic comparison:

$$1 - z + \frac{2}{5}z^{7/4} \ge \left(1 + \left(\sqrt{\frac{5}{2}} - 1\right) z\right)^{-2}, \qquad 0 \le z \le 1. \tag{3.50}$$

Write $d_* = \sqrt{5/2} - 1$ and subtract the right side from the left to obtain $f(z)$. Then $f(0) = f(1) = 0$,

$$f'(0) = 2\sqrt{5/2} - 3 > 0, \qquad f'(1) = -\frac{3}{10} + \frac{2d_*}{(1+d_*)^3} < 0.$$

The last inequality is equivalent to $\sqrt{5/2} < 8/5$. Also

$$f''(z) = \frac{21}{40} z^{-1/4} - 6d_*^2(1 + d_* z)^{-4}.$$

The ratio of the two positive terms in this difference has logarithmic derivative

$$-\frac{1}{4z} + \frac{4d_*}{1 + d_* z} = \frac{-1 + 15 d_* z}{4z(1 + d_* z)}.$$

The ratio starts at infinity and first decreases and then increases. Thus $f''$ can be negative only on a single interval, with $f'' \ge 0$ on either side. The endpoint signs of $f'$ force $f'$ to cross zero once on the interval where $f'' < 0$. It is positive on the initial increasing part and negative throughout any final increasing part since $f'(1) < 0$. Hence $f$ first increases and then decreases between its zero endpoint values. This proves $f \ge 0$ and (3.50).

Put $z = 1 - w$ and $k = \mathsf{F}_u(1)$. The concavity from Lemma 3.7 gives

$$\mathsf{F}_u(w) \ge (1 - z)k + z = k[1 + (k^{-1} - 1)z].$$

By Lemma 3.8, $k \le c^{-\theta}$. Since $2\theta \ge 1$, convexity of $x^{2\theta}$ gives

$$[1 + (\sqrt{c} - 1)z]^{2\theta} \le 1 + (c^\theta - 1)z.$$

Together with (3.50), this implies

$$b_*(w)^\theta \ge [1 + (c^\theta - 1)z]^{-1}.$$

Multiplying the last two inequalities and using $k^{-1} - 1 \ge c^\theta - 1$ proves (3.49). □

## 3.7. **The weighted fractional-Laplacian lower bound.**

**Proposition 3.10** (The weighted bubble estimate)**.** *For $I_\beta$ in (3.33),*

$$\gamma_{4,\sigma} I_\beta \ge h_0 a^{-\beta-\sigma}\big[\mathfrak{c}_\sigma(q+\beta) - \mathfrak{c}_\sigma(\beta)\big]. \tag{3.51}$$

*Proof.* Set

$$w_0(Y) = (|Y|^2 + h_0^2)^{-\beta/2}, \qquad u = q + \beta.$$

The parameters obey $\beta = 3/4 - \theta/2 > 0$ and $u + \sigma = 3/2 + 3\theta/2 \le 3$. Thus both $\beta$ and $u$ are in the range of Lemma 3.7. The singular-integral definition gives the exact identity

$$\gamma_{4,\sigma} I_\beta = L_\sigma(h w_0)(X_0) - h_0 L_\sigma w_0(X_0). \tag{3.52}$$

Indeed, the terms containing $w_0(X_0)$ cancel inside the same principal-value truncation. The surviving numerator is precisely $w_0(Y)(h_0 - h(|Y|))$.

We will represent the first term by beta averaging of bubbles. For positive $A, B$ and positive exponents $a_1, a_2$, the Feynman identity is

$$A^{-a_1} B^{-a_2} = \frac{1}{\mathrm{B}(a_1, a_2)} \int_0^1 \frac{t^{a_1 - 1}(1-t)^{a_2 - 1}}{[tA + (1-t)B]^{a_1 + a_2}}\, dt. \tag{3.53}$$

To verify it, multiply the two gamma-integral representations and use the change of variables $(v_1, v_2) = (\lambda t, \lambda(1-t))$. The Jacobian is $\lambda$. Integration of $\lambda^{a_1+a_2-1}e^{-\lambda[tA+(1-t)B]}$ in $\lambda$ gives $\Gamma(a_1+a_2)[tA+(1-t)B]^{-a_1-a_2}$. The remaining normalizing factor $\Gamma(a_1+a_2)/[\Gamma(a_1)\Gamma(a_2)]$ is $1/\mathrm{B}(a_1, a_2)$.

Apply it with $a_1 = q/2$, $a_2 = \beta/2$, $A = |Y|^2 + c$, and $B = |Y|^2 + h_0^2$. Since $h(Y) = c^{q/2}(|Y|^2+c)^{-q/2}$, the product $hw_0$ is $c^{q/2}$ times a beta average of bubbles with exponent $u$ and scale $tc + (1-t)h_0^2$. For fixed $X_0$ the scales lie in $[h_0^2, c] \subset (0, \infty)$. The bubbles have uniformly bounded Hessians near $X_0$ and are uniformly bounded globally. Pairing $X_0 + U$ with $X_0 - U$ gives an $O(|U|^2)$ local difference, uniformly in the scale. This and the integrable tail of the kernel justify applying $L_\sigma$ under the beta integral.

Introduce

$$w = \frac{R^2}{c+R^2}, \qquad b_* = \frac{a^2}{c+R^2} = w + \frac{2}{5}(1-w)^{7/4}, \qquad v = \frac{R^2}{a^2} = \frac{w}{b_*}. \tag{3.54}$$

Here $h_0^2 = (1-w)^q$ and $a^2 = R^2 + h_0^2$, which gives the displayed expression for $b_*$. Since $0 < h_0^2 \le 1 < c$, we have $0 < b_* < 1$ and $0 \le v < 1$. Using (3.45) in (3.52) gives

$$T := \frac{a^{\beta+\sigma}}{h_0}\gamma_{4,\sigma} I_\beta = \mathbb{E}\big[\mathcal{B}^\theta \phi_u(v\mathcal{B})\big] - \phi_\beta(v), \tag{3.55}$$

where

$$\mathcal{B} = b_* + (1-b_*)U, \qquad U \text{ has the beta density with parameters } \beta/2,\ q/2. \tag{3.56}$$

We verify this change of probability measure explicitly. In the original beta integral, let

$$M = \frac{R^2 + tc + (1-t)h_0^2}{a^2} = 1 + (b_*^{-1} - 1)t, \qquad \mathcal{B} = M^{-1}.$$

In (3.55) the scalar prefactor from (3.45) is

$$\frac{a^{\beta+\sigma}c^{q/2}}{h_0}(a^2 M)^{-(u+\sigma)/2} = b_*^{-q/2}\mathcal{B}^{u/2}\mathcal{B}^\theta.$$

The argument of $\phi_u$ is $R^2/(a^2M) = v\mathcal{B}$. The factor $b_*^{-q/2}\mathcal{B}^{u/2}$ is absorbed into the beta density, leaving $\mathcal{B}^\theta\phi_u(v\mathcal{B})$. The inverse map and its Jacobian are

$$t = \frac{b_*(1-\mathcal{B})}{(1-b_*)\mathcal{B}}, \qquad 1 - t = \frac{\mathcal{B} - b_*}{(1-b_*)\mathcal{B}}, \qquad \left|\frac{dt}{d\mathcal{B}}\right| = \frac{b_*}{(1-b_*)\mathcal{B}^2}.$$

Substitution into $t^{q/2-1}(1-t)^{\beta/2-1}b_*^{-q/2}\mathcal{B}^{u/2}\,dt$ cancels the standalone powers of $b_*$ and $\mathcal{B}$. The resulting normalized density is

$$\frac{(\mathcal{B}-b_*)^{\beta/2-1}(1-\mathcal{B})^{q/2-1}}{(1-b_*)^{u/2-1}\mathrm{B}(\beta/2, q/2)}\,d\mathcal{B}, \qquad b_* < \mathcal{B} < 1.$$

This is exactly (3.56).

For fixed $v$, the function $\mathcal{B} \mapsto \mathcal{B}^\theta\phi_u(v\mathcal{B})$ is concave on $[b_*, 1]$. In fact, Euler's representation (3.47) writes it as a positive multiple of an average of

$$[\mathcal{B}(1 - v\mathcal{B}t)]^\theta, \qquad 0 \le t \le 1.$$

The quadratic inside the brackets is nonnegative and concave. Composing it with the increasing, concave function $x^\theta$ preserves concavity, as does averaging. We can therefore bound the function below by its endpoint chord. The mean is $\mathbb{E}U = \mathrm{B}(\beta/2+1, q/2)/\mathrm{B}(\beta/2, q/2) = \beta/u$. Using $vb_* = w$ in the chord bound, (3.55) gives

$$T \ge P(v) + \frac{q}{u}\,b_*^\theta\phi_u(w), \qquad P(v) = \frac{\beta}{u}\phi_u(v) - \phi_\beta(v). \tag{3.57}$$

To compare $P(v)$ with $P(1)$, expand in powers of $v$. For $n \ge 1$, the coefficient of $v^n$ in $\phi_j$ is

$$\frac{2^\sigma\Gamma(2+\theta)}{(2)_n n!}(-\theta)_n\frac{\Gamma(j/2+\theta+n)}{\Gamma(j/2)}.$$

Here $(a)_n = a(a+1)\cdots(a+n-1)$. The factor $(-\theta)_n$ is negative for $0<\theta<1$. At $\theta=1$ it is negative for $n=1$ and zero thereafter. After multiplication of the $j=u$ coefficient by $\beta/u$, the difference to compare is

$$\frac{\beta}{2}\left[\frac{\Gamma(u/2+\theta+n)}{\Gamma(u/2+1)}-\frac{\Gamma(\beta/2+\theta+n)}{\Gamma(\beta/2+1)}\right]\geq 0.$$

The inequality follows because $x\mapsto\Gamma(x+\theta+n)/\Gamma(x+1)$ is increasing: its logarithmic derivative is $\psi(x+\theta+n)-\psi(x+1)\geq 0$. Thus all nonconstant coefficients of $P$ are nonpositive, and $P$ is decreasing for $0\leq v<1$. Continuity at 1 follows from (3.47) and gives $P(v)\geq P(1)$.

Finally, $u=q+\beta=3/2-\theta/2$. Lemma 3.9 gives $b_*^\theta\phi_u(w)\geq\phi_u(1)$. Apply this and the monotonicity of $P$ to (3.57):

$$\begin{aligned}T&\geq\frac{\beta}{u}\phi_u(1)-\phi_\beta(1)+\frac{q}{u}\phi_u(1)\\&=\mathfrak{c}_\sigma(u)-\mathfrak{c}_\sigma(\beta).\end{aligned}$$

This is (3.51). □

3.8. **An explicit bound for the hypergeometric remainder.** The principal gain uses the constant $F_\alpha(1)$ in place of the hypergeometric factor. We estimate the loss caused by this replacement.

**Lemma 3.11** (The remainder estimate)**.** *With the notation of* (3.31) *and* (3.32)*,*

$$\mathcal{G}\geq F_\alpha(1)\mathcal{G}_M-2\pi^2F_\alpha'(1)C_\alpha\,h_0a^{-1-\alpha}. \tag{3.58}$$

*Proof.* Subtract the gains with a common horizontal cutoff. By (3.20), the factor $F_\alpha(1)-F_\alpha(v)$ is nonnegative and is bounded by $F_\alpha'(1)d^2/(a+b)^2$. The difference $F_\alpha(1)\mathcal{G}_M-\mathcal{G}$ thus has a nonpositive contribution from the negatively oriented strip $r<R$. The resulting outer-strip bound is

$$F_\alpha(1)\mathcal{G}_M-\mathcal{G}\leq 2F_\alpha'(1)\int_{|Y|\geq R}\int_{h(|Y|)}^{h_0}\frac{(a+b)^{\alpha-1}}{b\,d^{3+2\alpha}}\,dz\,dY. \tag{3.59}$$

The integral on the right is finite. Near $Y=X_0$, the factor $(a+b)^{\alpha-1}/b$ is bounded, $d\geq|Y-X_0|$, and the Lipschitz bound on $h$ gives a strip of thickness at most $C|Y-X_0|$. The local radial integral in four dimensions is therefore bounded by $C\int_0^1 t^{1-2\alpha}\,dt$. For large $r$, use $b\geq r$, $d\geq r/2$, and a strip thickness at most $h_0$. The tail is bounded by $C\int_1^\infty r^{-2-\alpha}\,dr$. These bounds allow us to remove the cutoff.

On the outer strip, $z\geq h(r)>0$, so

$$b\geq\sqrt{r^2+h(r)^2}\geq\sqrt{R^2+h_0^2}=a,\qquad b\geq r.$$

The middle inequality follows from Lemma 3.2. Both factors $(a+b)^{\alpha-1}$ and $b^{-1}$ decrease in $b>0$. Define

$$\rho_*=\frac{R}{a},\qquad\rho=\frac{r}{a},\qquad q_\alpha(\rho)=\frac{(1+\max(1,\rho))^{\alpha-1}}{\max(1,\rho)}.$$

Since $b/a\geq\max(1,\rho)$, the weight in (3.59) is bounded by

$$a^{\alpha-2}q_\alpha(\rho)|Y-X_0|^{-3-2\alpha}.$$

Here $0\leq\rho_*<1$ and $q_\alpha$ is nonincreasing.

We also bound the strip thickness. The profile formula gives

$$\frac{h(a\rho)}{h_0}=\left(1+\frac{a^2}{c+R^2}(\rho^2-\rho_*^2)\right)^{-q/2},\qquad\frac{a^2}{c+R^2}=\frac{R^2+h_0^2}{R^2+c}\leq 1$$

because $h_0^2\leq 1<c$. The tangent line at zero to the convex function $x\mapsto(1+x)^{-q/2}$ gives $1-(1+x)^{-q/2}\leq(q/2)x$ for $x\geq 0$. Together with the bound by one, this yields

$$0\leq\frac{h_0-h(a\rho)}{h_0}\leq\min\left(1,\frac{q}{2}(\rho^2-\rho_*^2)\right),\qquad\rho\geq\rho_*. \tag{3.60}$$

For the angular integral we use

$$\frac{1}{|\mathbb{S}^3|}\int_{\mathbb{S}^3} |\rho\omega-\rho_* e_1|^{-3-2\alpha}\,d\omega \le \rho^{-2}(\rho^2-\rho_*^2)^{-1/2-\alpha}, \qquad \rho>\rho_*. \tag{3.61}$$

To see this, put $t=\rho_*/\rho$, $p=3/2+\alpha$, and $D(t,\omega)=1+t^2-2t\omega_1$. Denote normalized spherical averages by $\mathrm{Av}_\omega$. The function $Y\mapsto |e_1-Y|^{-2}$ is harmonic in the unit ball in $\mathbb{R}^4$, as follows from $(r^{-2})''+3r^{-1}(r^{-2})'=0$ for $r>0$. The mean-value property at the origin gives $\mathrm{Av}_\omega\, D(t,\omega)^{-1}=1$. Differentiating this identity for $t<1$ is legitimate because $D(t,\omega)\ge(1-t)^2$. It gives $\mathrm{Av}_\omega\,\omega_1 D^{-2}=t\,\mathrm{Av}_\omega\, D^{-2}$. Hence

$$1=\mathrm{Av}_\omega(DD^{-2})=(1-t^2)\,\mathrm{Av}_\omega\, D^{-2}.$$

Since $1<p<2$, Hölder's inequality applied to $D^{-p}=(D^{-1})^{2-p}(D^{-2})^{p-1}$ gives

$$\mathrm{Av}_\omega\, D^{-p}\le\left(\mathrm{Av}_\omega\, D^{-1}\right)^{2-p}\left(\mathrm{Av}_\omega\, D^{-2}\right)^{p-1}=(1-t^2)^{1-p}.$$

Multiplication by $\rho^{-2p}$ gives $\rho^{-2p}(1-t^2)^{1-p}=\rho^{-2}(\rho^2-\rho_*^2)^{-1/2-\alpha}$, proving (3.61).

In (3.59) put $Y=a\rho\omega$ and apply (3.60) and (3.61). The powers of $a$ are $a^{\alpha-2}$ from the weight, $a^4$ from $dY$, and $a^{-3-2\alpha}$ from the horizontal distance. Their product is $a^{-1-\alpha}$. Since $|\mathbb{S}^3|=2\pi^2$, the bound becomes

$$\begin{aligned}F_\alpha(1)\mathcal{G}_M-\mathcal{G}\le{}& 4\pi^2F_\alpha'(1)h_0a^{-1-\alpha}\\ &\times\int_{\rho_*}^\infty \min(1,c_0(\rho^2-\rho_*^2))q_\alpha(\rho)\rho(\rho^2-\rho_*^2)^{-1/2-\alpha}\,d\rho,\end{aligned}$$

where $c_0=q/2=3/8$. With $u=\rho^2-\rho_*^2$, the factor $\rho\,d\rho=du/2$ gives the coefficient $2\pi^2F_\alpha'(1)h_0a^{-1-\alpha}$ times

$$\mathcal{I}=\int_0^\infty \min(1,c_0u)\,q_\alpha(\sqrt{u+\rho_*^2})u^{-1/2-\alpha}\,du. \tag{3.62}$$

For $0<u\le1$, we have $q_\alpha(\sqrt{u+\rho_*^2})\le 2^{\alpha-1}$. For $u\ge1$, monotonicity and $\alpha<1$ give

$$q_\alpha(\sqrt{u+\rho_*^2})\le\frac{(1+\sqrt{u})^{\alpha-1}}{\sqrt{u}}\le u^{\alpha/2-1}.$$

Since $1/c_0>1$, splitting at 1 and $1/c_0$ gives

$$\begin{aligned}\mathcal{I}&\le c_0\,2^{\alpha-1}\int_0^1 u^{1/2-\alpha}\,du+c_0\int_1^{1/c_0}u^{-(1+\alpha)/2}\,du+\int_{1/c_0}^\infty u^{-(3+\alpha)/2}\,du\\ &=\frac{c_0\,2^{\alpha-1}}{3/2-\alpha}+\frac{c_0^{(1+\alpha)/2}-c_0}{(1-\alpha)/2}+\frac{c_0^{(1+\alpha)/2}}{(1+\alpha)/2}=C_\alpha.\end{aligned}$$

The displayed exponents give convergence at zero and infinity for $0<\alpha<1/2$, and the estimate also covers $R=0$. This proves (3.58). □

### 3.9. **Assembly of the curvature estimate.**

*Proof of Proposition 3.1.* At $P=(x,y)\in\partial E$, the defining equality gives $|y|^2-|x|^2=h_0>0$, so $P\in\partial E^{\mathrm{quad}}_{h_0}$. We can therefore insert the quadratic-plane bound (3.26) and the loss bound (3.58) into (3.30). Using the principal gain (3.34) and the weighted bubble bound (3.51), and writing $\Delta_\alpha=\mathfrak{c}_\sigma(q+\beta)-\mathfrak{c}_\sigma(\beta)$, we obtain

$$\begin{aligned}-\mathcal{H}_s[E](P)\ge{}&-\mathcal{A}_\alpha h_0a^{-1-\alpha}\\ &+\frac{\pi^2}{16}\frac{2^{2+\alpha}F_\alpha(1)}{\gamma_{4,\sigma}}\,c_L(\alpha)\,h_0a^{(1+\alpha)/2-\beta-\sigma}\Delta_\alpha\\ &-\frac{\pi^2}{16}2\pi^2F_\alpha'(1)C_\alpha h_0a^{-1-\alpha}.\end{aligned}$$

The middle term has the same power of $a$ as the other two, since

$$\frac{1+\alpha}{2}-\beta-\sigma=\frac{1+\alpha}{2}-\frac{1-\alpha}{2}-(1+2\alpha)=-1-\alpha.$$

To compare the constants, put

$$B_\alpha=\frac{16}{\pi^2}\mathcal{A}_\alpha=\frac{8\pi^{3/2}\Gamma(1+\alpha)\Gamma(\frac12-\alpha)}{\Gamma(4+\alpha)\Gamma(2-\alpha)}.$$

In the formula (3.21) for $F_\alpha(1)$, the duplication formula gives $\Gamma(2+\alpha/2)\Gamma(5/2+\alpha/2)=2^{-3-\alpha}\sqrt{\pi}\,\Gamma(4+\alpha)$. Thus

$$\frac{F_\alpha(1)}{B_\alpha}=\frac{2^\alpha\Gamma(\frac52+\alpha)\Gamma(2-\alpha)}{\pi^2\Gamma(1+\alpha)\Gamma(\frac12-\alpha)}.$$

Dividing by $\gamma_{4,\sigma}$ in (3.41) now gives

$$\frac{2^{2+\alpha}F_\alpha(1)}{\gamma_{4,\sigma}B_\alpha}=\frac{4\Gamma(2-\alpha)}{(1+2\alpha)\Gamma(1+\alpha)}. \tag{3.63}$$

For the loss term, (3.22) gives the first equality below. In the second, we use $\Gamma(\frac52+\alpha)=(\frac32+\alpha)(\frac12+\alpha)\Gamma(\frac12+\alpha)$, $6+4\alpha=4(\frac32+\alpha)$, and $\Gamma(2-\alpha)=(1-\alpha)\Gamma(1-\alpha)$.

$$\begin{aligned}\frac{2\pi^2F'_\alpha(1)C_\alpha}{B_\alpha}&=\frac{2^{1+\alpha}\alpha(1+\alpha)}{6+4\alpha}\,\frac{\Gamma(\frac52+\alpha)\Gamma(2-\alpha)}{\Gamma(1+\alpha)\Gamma(\frac12-\alpha)}\,C_\alpha\\&=2^{\alpha-1}\alpha(1-\alpha^2)\left(\tfrac12+\alpha\right)R_\alpha C_\alpha=E_\alpha.\end{aligned} \tag{3.64}$$

Factoring out $\mathcal{A}_\alpha h_0 a^{-1-\alpha}$ leaves $-1$ from the quadratic plane, $c_L(\alpha)Q_\alpha$ from (3.63) and (3.6), and $-E_\alpha$ from the loss. This is (3.11). □

## 4. An analytic estimate for the curvature margin

The scalar inequality in this section supplies the strict sign in Proposition 3.1. We use the parameters in (3.2), with $c_0=q/2=3/8$, and the quantities $Q_\alpha, C_\alpha, R_\alpha, E_\alpha, c_L(\alpha)$ from (3.6), (3.8)–(3.10), and (3.7).

For the scalar comparison we extend $\alpha$ to $[0,1/2]$. The arguments $\beta$ and $q+\beta$ of $\mathfrak{c}_\sigma$ stay in $(0,4-\sigma)$, since $\beta\ge1/4$ and $q+\beta\le5/4<2\le4-\sigma$. At $\alpha=1/2$, use the Gamma recurrence to extend (3.9) as

$$R_\alpha=\left(\frac12-\alpha\right)\frac{\Gamma(1/2+\alpha)\Gamma(1-\alpha)}{\Gamma(3/2-\alpha)\Gamma(1+\alpha)},\qquad R_{1/2}=0. \tag{4.1}$$

All the scalar factors are then continuous on the closed interval. The geometric argument uses $0<s<1$.

**Theorem 4.1** (Uniform scalar margin)**.** *For every* $0\le\alpha\le1/2$*,*

$$c_L(\alpha)Q_\alpha-E_\alpha-1>\frac{1}{2000}. \tag{4.2}$$

### 4.1. A positive radial representation for the gain.

Put

$$t=\frac12-\alpha,\qquad \nu=\frac{\sigma}{2}=1-t,\qquad m=2-\nu=1+t,\qquad x_t=\frac{m}{2},\qquad A_t=\frac{2+\nu}{2}.$$

The power coefficient in (3.4) has a convenient form when the power is written as $m-2z$. In fact, its numerator has Gamma arguments $A_t-z, A_t+z$, and its denominator has arguments $x_t-z, x_t+z$. Thus, for $|z|<x_t$, define

$$G_t(z)=\mathfrak{c}_{2\nu}(m-2z)=4^{1-t}\frac{\Gamma(A_t+z)\Gamma(A_t-z)}{\Gamma(x_t+z)\Gamma(x_t-z)},\qquad P(t)=\frac{2\Gamma(3/2+t)}{(1-t)\Gamma(3/2-t)}. \tag{4.3}$$

In particular $G_t$ is even. The powers in (3.6) are $m-2z_0=\beta$ and $m-2z_1=\beta+q$ with

$$z_0=\frac38+\frac{t}{4},\qquad z_1=\frac{t}{4},\qquad 0\le z_1<z_0<x_t.$$

Replacing $\alpha$ by $1/2-t$ in the prefactor of (3.6) gives $P(t)$, so

$$Q_\alpha = P(t)(G_t(z_1) - G_t(z_0)). \tag{4.4}$$

With $\psi = \Gamma'/\Gamma$, put

$$D(t) = \psi'(x_t) - \psi'(A_t), \qquad W(t) = \frac{P(t)G_t(0)D(t)}{8}. \tag{4.5}$$

**Lemma 4.2.** *For $0 \le t \le 1/2$,*

$$Q_\alpha \ge \left(\frac{9}{8} + \frac{3}{2}t\right) W(t). \tag{4.6}$$

*Proof.* Suppose first that $t > 0$. For $|z| \le z_0$, all three powers $m$ and $m \pm 2z$ are in $(0, 4-2\nu)$ since $4-2\nu = 2m$ and $2z_0 < m$. Apply (3.40) and (3.42) at a unit vector $e \in \mathbb{R}^4$, using $r = |Y|$. In the principal-value expression for $G_t(z)$ we may average the power formulas at $z$ and $-z$ by evenness. Subtracting this average from $G_t(0)$ with the same centered deletion leaves the numerator

$$\tfrac{1}{2}(r^{-m+2z} + r^{-m-2z}) - r^{-m} = r^{-m}(\cosh(2z\log r) - 1).$$

It is $O(|Y-e|^2)$ near $e$, since $\log|Y| = O(|Y-e|)$. Its integral against the kernel is bounded there by $C\int_0^1 d^{1-2\nu}\,dd$, where $d = |Y-e|$. Near $Y = 0$ the radial bound is $Cr^{3-m-2z_0}\,dr$, and at infinity it is $Cr^{-1-2\nu-m+2z_0}\,dr$. The first power is greater than $-1$ because $m + 2z_0 < 4$, and the second is less than $-1$ because $2z_0 < m$. Thus the centered deletion can be removed before inversion.

Write

$$K_\nu(r) = \frac{1}{|S^3|}\int_{S^3} |e - r\omega|^{-4-2\nu}\,d\omega.$$

Since $|e - \omega/r| = |e - r\omega|/r$, the part with radial coordinate larger than one has the same integral as the part with $0 < r < 1$ after inversion. Indeed the cosh is invariant under $r \mapsto 1/r$, and the radial factors become $r^{-(3-m)}r^{-2}r^{4+2\nu} = r^{1+\nu} = r^{3-m}$. We obtain

$$G_t(0) - G_t(z) = 2\gamma_{4,2\nu}|S^3|\int_0^1 K_\nu(r)r^{1+\nu}(\cosh(2z\log r) - 1)\,dr. \tag{4.7}$$

Let $A = 2|\log r|$. For $z \ge 0$, the power series of the cosh shows that $(\cosh(Az) - 1)/z^2$ is nondecreasing for $z > 0$ and tends to $A^2/2$ at zero. Hence dominated convergence in (4.7), with the integrable majorant at $z = z_0$ divided by $z_0^2$, identifies the integral of $A^2/2$ against its radial measure with $-G_t''(0)/2$. Also $\sinh(Az) \ge Az$ on $z \ge 0$. Integrating $A\sinh(Az)$ between $z_1$ and $z_0$ gives

$$\cosh(Az_0) - \cosh(Az_1) \ge \tfrac{1}{2}A^2(z_0^2 - z_1^2).$$

Subtracting the two identities (4.7) now yields

$$G_t(z_1) - G_t(z_0) \ge (z_0^2 - z_1^2)\left(-\frac{1}{2}G_t''(0)\right).$$

The first derivative of $\log G_t$ at zero is zero, and its second derivative there is $2\psi'(A_t) - 2\psi'(x_t) = -2D(t)$. Thus $-G_t''(0)/2 = G_t(0)D(t)$. Since $8(z_0^2 - z_1^2) = 9/8 + 3t/2$, (4.4) and (4.5) prove the bound.

At $t = 0$, Gamma recurrence gives $G_0(z) = 1 - 4z^2$ and $P(0) = 2$. Here $-G_0''(0)/2 = 4 = G_0(0)D(0)$, so $W(0) = 1$, and (4.4) gives $Q_{1/2} = 9/8$. □

### 4.2. **A lower bound for the centered Gamma factor.**

**Lemma 4.3.** *For $0 \le t \le 1/2$,*

$$W(t) \ge 1 - \frac{3}{5}t. \tag{4.8}$$

*Proof.* Put $\mathcal{B}(t) = P(t)G_t(0)/8$, so that $W = \mathcal{B}D$. The recurrence $\Gamma((3-t)/2) = \frac{1-t}{2}\Gamma((1-t)/2)$ reduces the prefactor $4^{1-t}/(1-t)$ in $4\mathcal{B}$ to $(1-t)4^{-t}$. Thus

$$\begin{aligned}\log(4\mathcal{B}(t)) = {} & \log(1-t) - t\log 4 \\ & + \log\Gamma(3/2+t) - \log\Gamma(3/2-t) \\ & + 2\log\Gamma(1/2-t/2) - 2\log\Gamma(1/2+t/2). \end{aligned} \tag{4.9}$$

For the expansion of these Gamma terms, use the series for $\psi'$ from Lemma 3.8. For $x > 0$ and $|u| \le r < x$, expanding each $(x+k+u)^{-2}$ as a power series in $u/(x+k)$ gives an absolutely and uniformly convergent double series. Its absolute sum is bounded by

$$\sum_{k=0}^{\infty}(x+k)^{-2}\sum_{j=0}^{\infty}(j+1)(r/x)^j < \infty.$$

Integrating twice gives the Taylor series of $\log\Gamma(x+u)$ on the same interval. We apply this with $(x,r) = (3/2, 1/2)$ and $(x,r) = (1/2, 1/4)$. Differentiating the series for $\psi'$ gives, for $n \ge 2$,

$$\psi^{(n-1)}(1/2) = (-1)^n(n-1)!\sum_{k=0}^{\infty}(k+1/2)^{-n} = (-1)^n(n-1)!(2^n-1)\zeta(n).$$

The recurrence $\psi(x+1) = \psi(x) + 1/x$ then gives

$$\psi^{(n-1)}(3/2) = \psi^{(n-1)}(1/2) + (-1)^{n-1}(n-1)!2^n.$$

For odd $n \ge 3$, the coefficient of $t^n$ in (4.9) is therefore

$$\frac{a_n}{n} = -\frac{1}{n} + \frac{2\psi^{(n-1)}(3/2) - 2^{2-n}\psi^{(n-1)}(1/2)}{n!},$$

where

$$a_n = 2^{n+1} - 1 - (2^{n+1} - 6 + 2^{2-n})\zeta(n). \tag{4.10}$$

The Gamma differences have only odd powers. Their linear term is $2(\psi(3/2) - \psi(1/2))t = 4t$, and the even part of $\log(1-t)$ is $\frac{1}{2}\log(1-t^2)$. Consequently,

$$\log(4\mathcal{B}(t)) = (3 - 2\log 2)t + \frac{1}{2}\log(1-t^2) + \sum_{\substack{n\ge 3\\ n\ \text{odd}}}\frac{a_n}{n}t^n. \tag{4.11}$$

The preceding uniform convergence also justifies this formula at $t = 1/2$.

The rational bounds we need are

$$\log 2 < \frac{25}{36}, \qquad \zeta(3) < \frac{101}{84}. \tag{4.12}$$

Indeed, integrating the geometric series for $(1-u^2)^{-1}$ from zero to $1/3$ gives

$$\log 2 = 2\sum_{j=0}^{\infty}\frac{3^{-2j-1}}{2j+1} < \frac{56}{81} + \frac{2}{5}\sum_{j=2}^{\infty}3^{-2j-1} = \frac{1123}{1620} < \frac{25}{36}.$$

For each integer $k \ge 5$, strict convexity of $r^{-3}$ gives $k^{-3} < \int_{k-1/2}^{k+1/2} r^{-3}\,dr$ by the midpoint inequality. Hence

$$\zeta(3) < 1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \int_{9/2}^{\infty} r^{-3}\,dr = \frac{6233}{5184} < \frac{101}{84}.$$

In particular $a_3 = 15 - \frac{21}{2}\zeta(3) > 19/8$. For $n \ge 5$, bounding the terms from $k = 4$ onward by $\int_3^{\infty} r^{-n}\,dr$ gives

$$\zeta(n) \le 1 + 2^{-n} + \left(1 + \frac{3}{n-1}\right)3^{-n}.$$

With $b_n = 2^{n+1} - 6 + 2^{2-n}$ we have

$$2^{n+1} - 1 - b_n(1 + 2^{-n}) = 3 + 2^{1-n} - 2^{2-2n} \ge 3.$$

Since $0 < b_n < 2^{n+1}$, this yields

$$a_n \geq 3 - 2\left(1 + \frac{3}{n-1}\right)(2/3)^n \geq 3 - \frac{112}{243} > \frac{19}{8},$$

where the second inequality uses $n - 1 \geq 4$ and $n \geq 5$. Inserting these bounds into (4.11) and summing the odd powers of $t$ gives

$$4\mathcal{B}(t) \geq \sqrt{1-t^2}\,\exp\left(\frac{29}{18}t + \frac{19}{8}(\operatorname{arctanh} t - t)\right). \tag{4.13}$$

For $D(t)$, the same series for $\psi'$ gives

$$\begin{aligned}\frac{D(t)}{4} &= \frac{1}{(1+t)^2} + \sum_{\substack{j\geq 3\\ j \text{ odd}}} \left((j+t)^{-2} - (j-t)^{-2}\right)\\ &= \frac{1}{(1+t)^2} - 4t \sum_{\substack{j\geq 3\\ j \text{ odd}}} \frac{j}{(j^2-t^2)^2}.\end{aligned} \tag{4.14}$$

The sum of $j^{-3}$ over odd $j \geq 3$ is $\frac{7}{8}\zeta(3) - 1 < 5/96$, by (4.12). Also $j/(j^2-t^2)^2 = j^{-3}(1-t^2/j^2)^{-2}$, and the last factor is bounded by $(1 - t^2/9)^{-2}$ when $j \geq 3$. Thus

$$\frac{D(t)}{4} \geq \frac{1}{(1+t)^2} - \frac{5t}{24(1-t^2/9)^2}. \tag{4.15}$$

Multiplying the right side by $(1+t)^2$ gives $N(t)/(8(9-t^2)^2)$, where

$$N(t) = 8(9-t^2)^2 - 135t(1+t)^2 = 648 - 135t - 414t^2 - 135t^3 + 8t^4 \geq \frac{3681}{8} > 0.$$

The last bound follows by putting $t = 1/2$ in the negative terms and dropping $8t^4$. We may now multiply the lower bounds in (4.13) and (4.15). Using $2\operatorname{arctanh} t = \log((1+t)/(1-t))$, we obtain

$$W(t) \geq (1-t)^{-11/16}(1+t)^{-5/16}e^{-55t/72}\frac{N(t)}{8(9-t^2)^2} =: W_{\mathrm{lo}}(t). \tag{4.16}$$

To compare this with $1 - 3t/5$, define $\ell(t) = \log(W_{\mathrm{lo}}(t)/(1-3t/5))$. Both terms in the ratio are positive and $\ell(0) = 0$. Logarithmic differentiation gives

$$\ell'(t) = \frac{3+8t}{8(1-t^2)} - \frac{55}{72} + \frac{N'(t)}{N(t)} + \frac{4t}{9-t^2} + \frac{3}{5-3t}.$$

Write $S(t) = (1-t^2)(9-t^2)(5-3t)$. Multiplying by $72S(t)N(t)$ gives

$$\begin{aligned}72SN\ell' = 72SN' + \big[&9(3+8t)(9-t^2)(5-3t) - 55S\\ &+ 288t(1-t^2)(5-3t) + 216(1-t^2)(9-t^2)\big]N.\end{aligned}$$

Expanding these products and grouping powers gives

$$72S(t)N(t)\ell'(t) = 5832 + tA_3(t) + t^5B_4(t), \tag{4.17}$$

where

$$\begin{aligned}A_3(t) &= 1009908 - 1758348t - 628101t^2 + 920889t^3,\\ B_4(t) &= 466074 + 93190t - 134097t^2 - 21019t^3 + 1320t^4.\end{aligned}$$

For $0 \leq t \leq 1/2$ we have

$$A_3'(t) \leq -1758348 + \frac{3\cdot 920889}{4} = -\frac{4270725}{4} < 0, \qquad A_3(t) \geq A_3(1/2) = \frac{710559}{8} > 0,$$

and

$$B_4(t) \geq 466074 - \frac{134097}{4} - \frac{21019}{8} = \frac{3439379}{8} > 0.$$

Since $S(t)N(t) > 0$, (4.17) gives $\ell'(t) > 0$. This proves $W(t) \geq W_{\mathrm{lo}}(t) \geq 1 - 3t/5$. □

**Corollary 4.4.** *For $t = 1/2 - \alpha \in [0, 1/2]$,*

$$Q_\alpha \geq \left(\frac{9}{8} + \frac{3}{2}t\right)\left(1 - \frac{3}{5}t\right) = \frac{9}{8} + \frac{33}{40}t - \frac{9}{10}t^2. \tag{4.18}$$

*Proof.* Substitute (4.8) into (4.6), whose multiplier of $W(t)$ is positive. □

4.3. **Two analytic bounds for the error.**

**Lemma 4.5.** *For $0 \leq \alpha \leq 1/2$, $R_\alpha \leq 1 - 2\alpha$.*

*Proof.* Let $f(x) = \log\Gamma(x+1/2) - \log\Gamma(x)$ for $x > 0$. The trigamma series used in Lemma 3.8 shows that $\psi'$ is strictly decreasing. Thus

$$f''(x) = \psi'(x + 1/2) - \psi'(x) < 0.$$

The sum $S(\alpha) = f(1/2 + \alpha) + f(1 - \alpha)$ is concave. At both endpoints it equals

$$f(1/2) + f(1) = \log\frac{\Gamma(3/2)}{\Gamma(1/2)} = \log(1/2).$$

The chord joining these endpoint values gives $S(\alpha) \geq \log(1/2)$. By (4.1),

$$R_\alpha = \frac{1 - 2\alpha}{2}\exp(-S(\alpha)) \leq 1 - 2\alpha.$$

This expression also gives $R_{1/2} = 0$. □

The next convexity observation controls the product $2^\alpha C_\alpha$ in the error.

**Lemma 4.6.** *The function $J(\alpha) = 2^\alpha C_\alpha$ is convex on $[0, 1/2]$. It satisfies*

$$J(\alpha) < \frac{15}{8} \qquad (0 \leq \alpha \leq 1/2), \tag{4.19}$$

$$J(\alpha) < \frac{9}{5} \qquad (1/10 \leq \alpha \leq 7/20). \tag{4.20}$$

*Proof.* The majorant evaluated after (3.62) gives the integral representation

$$C_\alpha = L(\alpha) + T(\alpha), \qquad L(\alpha) = \frac{c_0\, 2^{\alpha-1}}{3/2 - \alpha}, \qquad T(\alpha) = \int_1^\infty \min(1, c_0 u) u^{-3/2-\alpha/2}\, du.$$

Here $T$ combines the integrals over $[1, 1/c_0]$ and $[1/c_0, \infty)$ from that calculation. The representation holds at both endpoints by continuity, since the tail integrand is dominated by $u^{-3/2}$. The first summand of $J$ is convex because

$$\frac{d^2}{d\alpha^2}\log\left(\frac{c_0\, 2^{2\alpha-1}}{3/2 - \alpha}\right) = \frac{1}{(3/2 - \alpha)^2} > 0$$

and a positive log-convex function is convex. The second summand is

$$2^\alpha T(\alpha) = \int_1^\infty \min(1, c_0 u) u^{-3/2} e^{\alpha(\log 2 - \frac{1}{2}\log u)}\, du.$$

For each $u$, the integrand is convex in $\alpha$. Integrating its convexity inequality proves convexity of this summand. All integrals are finite, since the integrand is bounded by $\sqrt{2}\, u^{-3/2}$.

For the endpoint evaluations, combining the last two terms in (3.8) gives

$$T(\alpha) = \frac{4c_0^{(1+\alpha)/2}}{1 - \alpha^2} - \frac{2c_0}{1 - \alpha}.$$

At zero, with $c_0 = 3/8$, this yields

$$J(0) = \frac{c_0}{3} + 4\sqrt{c_0} - 2c_0 = \sqrt{6} - \frac{5}{8} < \frac{15}{8},$$

where $\sqrt{6} < 5/2$. At the other endpoint,

$$C_{1/2} = \frac{3}{8\sqrt{2}} + \frac{16}{3}(3/8)^{3/4} - \frac{3}{2} < \frac{17}{64} + \frac{64}{25} - \frac{3}{2} = \frac{2121}{1600}.$$

The bounds on the first two terms follow from $288 < 289$ after squaring and $27 \cdot 25^4 < 512 \cdot 12^4$ after taking fourth powers, respectively. Since $2 \cdot 70^2 < 99^2$, we obtain

$$J(1/2) = \sqrt{2}\, C_{1/2} < \frac{99}{70}\frac{2121}{1600} = \frac{209979}{112000} < \frac{15}{8}.$$

Convexity then proves (4.19).

For (4.20), the required strict rational upper bounds are

| $\alpha$ | $2^{2\alpha-1}$ | $2^{\alpha}$ | $(3/8)^{(1+\alpha)/2}$ |
|---|---|---|---|
| $1/10$ | $23/40$ | $43/40$ | $7/12$ |
| $7/20$ | $13/16$ | $51/40$ | $129/250.$ |

Raise each comparison to the power that clears the exponent denominator. After cancellation, the two rows give respectively

$$16 \cdot 23^5 > 40^5, \qquad 43^{10} > 2 \cdot 40^{10}, \qquad 7^{20} > 2^7\, 3^{31},$$
$$8 \cdot 13^{10} > 16^{10}, \qquad 51^{20} > 2^7\, 40^{20}, \qquad 2^{41} 3^{13} 43^{40} > 5^{120}.$$

For example, $(7/12)^{20} > (3/8)^{11}$ is equivalent to $7^{20} > 2^7\, 3^{31}$. Substitution into $J(\alpha) = 2^{\alpha} L(\alpha) + 2^{\alpha} T(\alpha)$ gives

$$J(1/10) < \frac{15}{56}\frac{23}{40} + \frac{43}{40}\left(\frac{400}{99}\frac{7}{12} - \frac{5}{6}\right) = \frac{238417}{133056} < \frac{9}{5},$$
$$J(7/20) < \frac{15}{46}\frac{13}{16} + \frac{51}{40}\left(\frac{1600}{351}\frac{129}{250} - \frac{15}{13}\right) = \frac{1286489}{717600} < \frac{9}{5}.$$

In each row we first bound $T(\alpha)$, which is positive by its integral representation, and then bound its multiplier $2^{\alpha}$. The endpoint bounds imply (4.20) by convexity. □

**Lemma 4.7.** *For $t = 1/2 - \alpha$,*

$$E_{\alpha} \le \min\left\{\frac{45}{64}t, \frac{2}{25}\right\}. \tag{4.21}$$

*Proof.* Set $M(\alpha) = \alpha(1-\alpha^2)(1/2+\alpha)$. Its derivative is

$$M'(\alpha) = \frac{1}{2} + 2\alpha - \frac{3}{2}\alpha^2 - 4\alpha^3 \ge \frac{1}{2} + 2\alpha - \frac{3}{4}\alpha - \alpha = \frac{1}{2} + \frac{\alpha}{4}.$$

The inequality uses $\alpha^2 \le \alpha/2$ and $\alpha^3 \le \alpha/4$ on $[0, 1/2]$. Hence $M(\alpha) \le M(1/2) = 3/8$. By (3.10), Lemma 4.5, and (4.19),

$$E_{\alpha} = \tfrac{1}{2} J(\alpha) M(\alpha) R_{\alpha} \le t J(\alpha) M(\alpha) \le \frac{15}{8}\frac{3}{8}\, t = \frac{45}{64} t.$$

For the uniform bound, write

$$P_e(\alpha) = M(\alpha)(1-2\alpha) = \frac{\alpha}{2}(1-\alpha^2)(1-4\alpha^2) = \frac{\alpha}{2} - \frac{5}{2}\alpha^3 + 2\alpha^5.$$

Then $E_{\alpha} \le \frac{1}{2} J(\alpha) P_e(\alpha)$. The derivative $P_e'(\alpha) = \frac{1}{2}(1 - 15\alpha^2 + 20\alpha^4)$ vanishes where $\alpha^2 = (15 \pm \sqrt{145})/40$. Since $12 < \sqrt{145} < 13$, the larger root is above $1/4$, whereas the smaller root $y = (15 - \sqrt{145})/40$ lies between $1/20$ and $3/40$. Thus the derivative changes sign from positive to negative at $\alpha_* = \sqrt{y}$, the only critical point in $[0, 1/2]$, and $1/10 < \alpha_* < 7/20$.

The relation $20y^2 = 15y - 1$ gives $P_e(\alpha_*) = \alpha_*(2/5 - y)$ and

$$P_e(\alpha_*)^2 = y^3 - \frac{4}{5}y^2 + \frac{4}{25}y = \frac{1+29y}{400} = \frac{475 - 29\sqrt{145}}{16000}.$$

Since $301^2 < 145 \cdot 25^2$, this is less than $1573/200000$. The difference $16/2025 - 1573/200000$ equals $587/16200000 > 0$, so $P_e(\alpha) < 4/45$ throughout $[0, 1/2]$. In the middle interval we obtain

$$E_{\alpha} \le \tfrac{1}{2} J(\alpha) P_e(\alpha) \le \tfrac{1}{2} \cdot \tfrac{9}{5} \cdot \tfrac{4}{45} = \frac{2}{25} \qquad (1/10 \le \alpha \le 7/20).$$

On the other two intervals, (4.19) gives $\frac{1}{2}J(\alpha) < 1$. By the derivative sign, $P_e$ is bounded there by its values at the nearest endpoint of the middle interval. These are

$$P_e(1/10) = \frac{297}{6250} < \frac{2}{25}, \qquad P_e(7/20) = \frac{125307}{1600000} < \frac{2}{25}.$$

Thus $E_\alpha \le 2/25$ on the full interval. □

4.4. **Completion of the scalar comparison.**

*Proof of Theorem 4.1.* The slope estimate (3.12) gives $\ell^2 < 3/80$. Since $1 + \ell^2 \ge 1$ and $(5 + 2\alpha)/2 \le 3$, the attenuation satisfies

$$c_L(\alpha) \ge (1 + \ell^2)^{-3} > \left(\frac{80}{83}\right)^3 = \frac{512000}{571787} =: c_*.$$

Write $t = 1/2 - \alpha$ and

$$p(t) = \frac{9}{8} + \frac{33}{40}t - \frac{9}{10}t^2.$$

This polynomial is positive on $[0, 1/2]$ because $p(t) - 9/8 = t(33/40 - 9t/10) \ge 3t/8$. Thus Corollary 4.4 and Lemma 4.7 give

$$c_L(\alpha)Q_\alpha - E_\alpha - 1 \ge c_* p(t) - 1 - \min\left\{\frac{45}{64}t, \frac{2}{25}\right\} =: F(t).$$

The two error bounds agree at $t_* = (2/25)(64/45) = 128/1125$. The minimum equals $45t/64$ for $t \le t_*$ and $2/25$ for $t \ge t_*$. On each of the two intervals $F$ is a quadratic with second derivative $-9c_*/5 < 0$, so it lies above the chord between the corresponding endpoint values. It suffices to check $F$ at $0, t_*, 1/2$.

At the switching point the gain polynomial is

$$p(t_*) = \frac{9}{8} + \frac{176}{1875} - \frac{8192}{703125} = \frac{6790589}{5625000}.$$

Substitution gives the three comparisons

| $t$ | $F(t)$ | $2000\,(\text{numerator}) - (\text{denominator})$ |
|---|---|---|
| 0 | $c_*\frac{9}{8} - 1 = \frac{4213}{571787}$ | 7854213 |
| $t_*$ | $c_*\frac{6790589}{5625000} - \frac{27}{25} = \frac{3175543}{3216301875}$ | 3134784125 |
| 1/2 | $c_*\frac{21}{16} - \frac{27}{25} = \frac{1361751}{14294675}$ | 2709207325. |

All three differences in the last column are positive. Hence the endpoint values exceed $1/2000$, and the chord bound gives (4.2) on the full interval. □

*Remark* 4.8. The scalar margin is uniform in $\alpha$. Its application to curvature uses $0 < s < 1$, where the dimensional factor $\mathcal{A}_{s/2}$ is finite and positive.

**Theorem 4.9** (The strict sign of the explicit cap)**.** *Let $0 < s < 1$ and take $E$ as in (3.3). With the exterior-minus-interior convention (3.1), every $P \in \partial E$ satisfies*

$$-\mathcal{H}_s[E](P) > \frac{\mathcal{A}_{s/2}}{2000} \frac{h(2|x||y|)}{(|x|^2 + |y|^2)^{1+s/2}} > 0, \qquad P = (x, y).$$

*The same strict sign holds for every dilated set $bE$, $b > 0$.*

*Proof.* In (3.11) the prefactor $\mathcal{A}_\alpha h_0 a^{-1-\alpha}$ is positive, and Theorem 4.1 bounds the bracket strictly from below by $1/2000$. This proves the estimate.

For a dilation, put $X = bZ$ in the truncated integral for $\mathcal{H}_s[bE](bP)$ in (3.1). The sign function becomes that of $E$, the measure contributes $b^8$, and the kernel contributes $b^{-8-s}$. The deleted ball becomes $|Z - P| \le \varepsilon/b$, so taking the principal value gives

$$\mathcal{H}_s[bE](bP) = b^{-s}\mathcal{H}_s[E](P).$$

The factor $b^{-s}$ is positive, so the strict sign persists. □

## 5. Global signed critical profiles

We keep $s \in (0,1)$ fixed and use the normal coordinates of (2.1). The intermediate exponent is $\kappa = 5/2$. For a radial function, $g = O_j(\rho^{-a})$ means

$$|g^{(i)}(\rho)| \le C_i \rho^{-a-i}, \qquad 0 \le i \le j.$$

The normal fields are measured using the outward normal of $D$. A cap contained in $D$ therefore has negative displacement.

### 5.1. Pointwise curvature estimates for exterior normal graphs.

**Lemma 5.1** (A weighted Jacobi estimate)**.** *Let $0 < a < 7$. If a radial function satisfies*

$$|w^{(j)}(\rho)| \le L\rho^{-a-j}, \qquad j = 0,1,2,$$

*then*

$$|J_s w(P_R)| \le C_{s,a} L R^{-a-1-s}, \qquad |P_R| = R. \tag{5.1}$$

*The function may vanish near the vertex or have the integrable singularity allowed by the displayed bounds.*

*Proof.* For $L > 0$ put $u(\rho) = R^a w(R\rho)/L$. The cone scaling in (2.8) gives

$$J_s w(P_R) = L R^{-a-1-s} J_s u(P_1), \qquad P_1 = P_R/R.$$

The bounds on $u^{(j)}$ are $\rho^{-a-j}$. In a fixed tangent chart at $P_1$ the radius is bounded away from zero, so the composition of $u$ with the chart has bounded $C^2$ norm. The cancellation following (2.8) then bounds the local principal value by $C_{s,a}\int_0^1 r^{-s}\,dr$. In particular the Euclidean cutoff and the tangent cutoff have the same limit, since the shell between them contributes $O(\varepsilon^{2-s})$. Away from the chart the integral over comparable radii is bounded. The vertex and distant terms are bounded respectively by

$$C_{s,a}\int_0^{1/2} \rho^{6-a}\,d\rho, \qquad C_{s,a}\int_2^{\infty} (\rho^{-a}+1)\rho^{-2-s}\,d\rho.$$

Here the powers of $\rho$ in the surface measure come from (2.3). These bounds and the finite potential $A_s$ prove (5.1). For $L = 0$ the estimate holds with both sides zero. □

**Lemma 5.2** (Quadratic curvature remainder)**.** *Let $0 < a < 3$. Suppose that $w$ is radial, vanishes near the vertex, and satisfies*

$$|w^{(j)}(\rho)| \le L\rho^{-a-j} \quad (0 \le j \le 5), \qquad |w(\rho)| \le \rho/10. \tag{5.2}$$

*Let $E_w$ be the set $\tau < w(\rho)$ in normal coordinates. For all sufficiently large $R$,*

$$|\mathcal{H}_s[E_w](P_R + w(R)\nu(P_R)) + 2J_s w(P_R)| \le C_{s,a} L^2 R^{-2a-2-s}. \tag{5.3}$$

*The threshold for $R$ depends on $L$ and $a$. The estimate is uniform over all functions with these bounds.*

*Proof.* First assume $w$ is smooth, and consider $E_{\theta w}$ at $p_\theta = P_R + \theta w(R)\nu(P_R)$, for $0 \le \theta \le 1$. To estimate its curvature locally, dilate by $1/R$ and set

$$w^\sharp(\rho) = R^{-1} w(R\rho), \qquad \epsilon = LR^{-a-1}.$$

On any fixed annulus about the unit sphere, $|(w^\sharp)^{(j)}| \le C_a\epsilon$ for $0 \le j \le 5$. Use a fixed tangent frame at $P_1 = P_R/R$, with vertical direction $\nu(P_1)$. In these coordinates the cone is the graph $q(\xi) = (\xi, g_0(\xi))$, where $g_0(0) = Dg_0(0) = 0$ and $\xi \in \mathbb{R}^7$. Write its normal as $(\nu_h, \nu_v)$, and let $\widehat{w}(\xi)$ be $w^\sharp$ evaluated at the cone radius of $q(\xi)$. The deformed horizontal and vertical coordinates are

$$T_\theta(\xi) = \xi + \theta v(\xi), \qquad U_\theta(\xi) = g_0(\xi) + \theta\widehat{w}(\xi)\nu_v(\xi), \qquad v = \widehat{w}\nu_h.$$

Since $v(0) = 0$ and $\|Dv\|_\infty \le C_a\epsilon$, for small enough $\epsilon$ the horizontal map has an inverse $S_\theta$ on a fixed smaller ball. Indeed, $\xi = \eta - \theta v(\xi)$ is a contraction of a fixed closed chart ball into itself for $\eta$ in the smaller ball. The derivatives of $S_\theta$ through order three are uniformly bounded by

differentiating $DS_\theta = (I+\theta Dv(S_\theta))^{-1}$. The graph describes the full boundary near the moving point. To see this, any deformed point near that point has undeformed cone radius comparable to one, since its norm is $(\rho^2+\theta^2 w^\sharp(\rho)^2)^{1/2}$. Its displacement is then $O_a(\epsilon)$, so its cone point belongs to the original chart if the neighborhood and $\epsilon$ are sufficiently small.

In coordinates centered at the moving point the deformed graph has height

$$g_\theta(\eta) = U_\theta(S_\theta(\eta)) - \theta w^\sharp(1), \qquad g_\theta(0) = 0.$$

Put $M_\theta = (I+\theta Dv(S_\theta))^{-1}$. Differentiating $S_\theta + \theta v(S_\theta) = \eta$ gives

$$\dot S_\theta = -M_\theta v(S_\theta), \qquad \ddot S_\theta = -M_\theta(2Dv(S_\theta)\dot S_\theta + \theta D^2 v(S_\theta)[\dot S_\theta, \dot S_\theta]).$$

The uniform bounds on $S_\theta$ and the $C^5$ bounds on the field show that these derivatives are $O_{C^2}(\epsilon)$ and $O_{C^2}(\epsilon^2)$, respectively. For example, the second derivative of $g_\theta$ is

$$2D(\widehat{w}\nu_v)(S_\theta)\dot S_\theta + D^2U_\theta(S_\theta)[\dot S_\theta, \dot S_\theta] + DU_\theta(S_\theta)\ddot S_\theta.$$

Taking up to two spatial derivatives in these identities gives

$$\|g_\theta\|_{C^2} \le C_a, \qquad \|\dot g_\theta\|_{C^2} \le C_a\epsilon, \qquad \|\ddot g_\theta\|_{C^2} \le C_a\epsilon^2. \tag{5.4}$$

In this chart $E_{\theta w}$ lies below the deformed graph. Indeed, $\tau - \theta w^\sharp(\rho)$ has positive derivative in the vertical direction near the moving point, by the small $C^1$ norm and the choice of $\nu(P_1)$ as the vertical direction. Choose a smooth radial cutoff $\chi$ equal to one near zero, supported in a fixed scaled ball on which the graph description holds. Write $K_s^{\mathrm{loc}}(\eta,z) = \chi((|\eta|^2+z^2)^{1/2})(|\eta|^2 + z^2)^{-(8+s)/2}$. The tangent plane through the moving point has height $\ell_\theta(\eta) = Dg_\theta(0)\eta$. Its halfspace has zero curvature under the radial truncations of Section 2. Subtraction leaves the absolutely integrable local contribution

$$\mathcal{H}_s^{\mathrm{loc}}(\theta) = -2\int_{\mathbb{R}^7}\int_{\ell_\theta(\eta)}^{g_\theta(\eta)} K_s^{\mathrm{loc}}(\eta,z)\,dz\,d\eta. \tag{5.5}$$

Here the inner integral is oriented, and the cutoff restricts the formula to the chart. In particular this formula gives the same value as the Euclidean principal value.

Set $b_\theta = g_\theta - \ell_\theta$ and parametrize the strip by $z_\theta = \ell_\theta + ub_\theta$, $0 \le u \le 1$. If $r = |\eta|$, the quantities $b_\theta, \dot b_\theta, \ddot b_\theta$ are bounded by $C_a r^2, C_a\epsilon r^2, C_a\epsilon^2 r^2$ by spatial Taylor's formula. The derivatives $\dot z_\theta, \ddot z_\theta$ are bounded by $C_a\epsilon r, C_a\epsilon^2 r$. For $j = 0,1,2$, $\left|\partial_z^j K_s^{\mathrm{loc}}(\eta,z)\right| \le C_s r^{-8-s-j}$. With $K = K_s^{\mathrm{loc}}(\eta, z_\theta)$, the first two parameter derivatives of $b_\theta K$ are

$$\dot b_\theta K + b_\theta K_z \dot z_\theta, \qquad \ddot b_\theta K + 2\dot b_\theta K_z \dot z_\theta + b_\theta(K_{zz}\dot z_\theta^2 + K_z \ddot z_\theta).$$

Their absolute bounds are $C_{s,a}\epsilon r^{-6-s}$ and $C_{s,a}\epsilon^2 r^{-6-s}$, respectively. The factor $r^6$ in $d\eta$ gives an integrable radial majorant. These estimates justify differentiation under the integral. Taylor's formula and the curvature scaling factor $R^{-s}$ give a local remainder bounded by $C_{s,a}R^{-s}\theta^2\epsilon^2$.

In the unscaled variables the distant kernel is

$$W_\theta(Y) = (1-\chi(|Y-p_\theta|/R))|Y-p_\theta|^{-8-s}.$$

By (2.3) the distant curvature equals

$$\begin{aligned}\mathcal{H}_s^{\mathrm{far}}(\theta) = &\int_{\mathbb{R}^8} \mathrm{sign}(\tau(Y))W_\theta(Y)\,dY \\ &- 2\int_\Sigma\int_0^{\theta w(\rho)} W_\theta(Z+t\nu(Z))\,\mathcal{J}(\rho,t)\,dt\,d\sigma(Z), \qquad \mathcal{J}(\rho,t) = (1-t^2/\rho^2)^3.\end{aligned}$$

The sign changes by $-2$ on a positive oriented strip. All integrals here are absolute. As a function of $Y - p_\theta$, $W_\theta$ and its first two derivatives have bounds $C_{s,j}(R+|Y-p_\theta|)^{-8-s-j}$. Integrating the second derivative bound over $Y \in \mathbb{R}^8$ gives $C_s R^{-2-s}$, so the first integral has Taylor remainder bounded by $C_s\theta^2|w(R)|^2R^{-2-s}$.

On the strip, $|Y| = |Z + t\nu(Z)| \geq \rho$ and $|p_\theta| \leq 2R$, so $R + \rho \leq 3(R + |Y - p_\theta|)$. This comparison applies throughout the source and center segments, including when $Y = Z$. The mean-value theorem and $|\mathcal{J}(\rho, t) - 1| \leq C|t|/\rho$ thus give

$$\begin{aligned}&|W_\theta(Z + t\nu(Z))\mathcal{J}(\rho, t) - W_0(Z)|\\ &\qquad \leq C_s\left[(\theta|w(R)| + |t|)(R + \rho)^{-9-s} + \frac{|t|}{\rho}(R + \rho)^{-8-s}\right].\end{aligned}$$

Since $R + \rho \geq \rho$, the $|t|$-term with power $-9 - s$ is bounded by the last term. Integrating from 0 to $\theta w(\rho)$ after subtracting the linear term $\theta w(\rho)W_0(Z)$ bounds the strip remainder by

$$C_s\theta^2 \int_\Sigma \left[|w(R)w(\rho)|(R + \rho)^{-9-s} + \frac{|w(\rho)|^2}{\rho}(R + \rho)^{-8-s}\right] d\sigma(Z).$$

The weighted bounds and $\rho = Ru$ bound this by

$$C_s\theta^2 L^2 R^{-2a-2-s}\left[\int_0^\infty u^{6-a}(1 + u)^{-9-s}\, du + \int_0^\infty u^{5-2a}(1 + u)^{-8-s}\, du\right].$$

Both integrals are finite at infinity, and $a < 3$ gives integrability at zero.

The constant term of the combined local and distant expansions is zero by (2.4). Their linear coefficient is the derivative of the full curvature at $\theta = 0$, which is $-2J_s w(P_R)$ by (2.10). The hypotheses of that formula hold for the present smooth field by the vanishing near the vertex and the weighted bounds. Combining the remainders and taking $\theta = 1$ proves the estimate for smooth fields.

For a field with the stated five derivatives, take a smooth probability density $\eta_\delta$ supported on $[-\delta, \delta]$ and set

$$w_\delta(\rho) = \int_{\mathbb{R}} e^{-u} w(e^u \rho)\eta_\delta(u)\, du.$$

In logarithmic coordinates $w_\delta(e^x)/e^x$ is a smooth convolution of $w(e^x)/e^x$. If $w$ vanishes on $(0, \rho_*)$, then $w_\delta$ vanishes on $(0, e^{-\delta}\rho_*)$. Also $|w_\delta(\rho)| \leq \rho/10$, and the $j$-th derivative integrand is bounded by $Le^{-(a+1)u}\rho^{-a-j}$, so $L_\delta = Le^{(a+1)\delta}$ works in the weighted bounds. The approximants converge in $C^2$ on compact radial intervals. Choose the radius threshold for $2L$ so it applies for all sufficiently small $\delta$. The horizontal graph coordinates above give convergence in $C^2$ in the scaled chart. Formula (5.5) has a common integrable bound because the tangent-subtracted heights are $O(r^2)$. For the distant contribution the moving points converge and the indicators converge almost everywhere. The cutoff kernels have a common integrable bound by the distance estimate above, for each fixed $R$. The Jacobi integrals converge by the tangent subtraction in (2.8) and the bounds of Lemma 5.1. Passing to the limit with $L_\delta \to L$ proves (5.3). □

**Corollary 5.3** (Expansion with an arbitrary fixed cap)**.** *Suppose a set agrees, outside a fixed normal radius $R_0 \geq 1$, with the sublevel set of*

$$f(\rho) = a_0\rho^{-a_1} + b_0\rho^{-a_2} + g(\rho), \qquad a_1, a_2 \in (0, 3), \qquad g = O_5(\rho^{-b}),$$

*where $\min(a_1, a_2) < b < 7$ and $|f(\rho)| \leq \rho/10$. Inside the fixed radius the set may be arbitrary. Put $a = \min(a_1, a_2)$. At its exterior points,*

$$\begin{aligned}\mathcal{H}_s[E](P_R + f(R)\nu(P_R)) = {}& 2a_0(H_s(a_1) - A_s)R^{-a_1-1-s} + 2b_0(H_s(a_2) - A_s)R^{-a_2-1-s}\\ &+ O_s(R^{-2a-2-s} + R^{-b-1-s} + R^{-8-s}).\end{aligned} \tag{5.6}$$

*The $R^{-b-1-s}$ term is omitted when $g = 0$. For fixed $s, a_1, a_2, b$, the constants and exterior threshold are uniform for $1 \leq R_0 \leq R_{\max}$ and common upper bounds on the coefficients and the weighted $C^5$ norms.*

*Proof.* Let $\zeta(\rho) = \zeta_0(\rho/R_0)$, with $\zeta_0$ smooth, valued in $[0, 1]$, zero on $[0, 1]$, and one on $[2, \infty)$. Define $w = \zeta f$ for $\rho > R_0$ and extend it by zero inside. The flat cutoff at $R_0$ makes the extension $C^5$. On $[R_0, 2R_0]$ the cutoff derivatives satisfy $|\zeta^{(j)}| \leq C_j R_0^{-j} \leq 2^j C_j \rho^{-j}$. The product rule

gives the bounds (5.2) with exponent $a$, since $\rho \geq 1$ and $a \leq a_i, b$. It also gives the weighted bounds for $\zeta g$ with exponent $b$.

The sets $E$ and $E_w$ differ only on $\rho \leq 2R_0$. By the normal coordinate formula this region lies in $|Y| \leq 2\sqrt{2}R_0$. For $R \geq 4R_0$ they share the evaluation point $p = P_R + f(R)\nu(P_R)$ with $|p| \geq R$. The distance to the region of disagreement is at least $(1 - 1/\sqrt{2})R$. Subtracting their curvature integrals directly therefore gives

$$|\mathcal{H}_s[E](p) - \mathcal{H}_s[E_w](p)| \leq C_s R^{-8-s} R_0^8. \tag{5.7}$$

Apply Lemma 5.2. With $p_0(\rho) = a_0\rho^{-a_1} + b_0\rho^{-a_2}$, linearity gives $J_s w = J_s p_0 + J_s(\zeta g) + J_s((\zeta - 1)p_0)$. The first term is (2.9), and the second is $O_s(R^{-b-1-s})$ by Lemma 5.1. For the last term use the fact that for any $v$ supported in $\rho \leq 2R_0$, both $v(P_R)$ and the potential contribution vanish, so

$$|J_s v(P_R)| \leq C_s R^{-8-s} \int_{\{\rho \leq 2R_0\}} |v| \, d\sigma. \tag{5.8}$$

Indeed, the source lies at least $R/2$ from $P_R$. For $v = (\zeta - 1)p_0$ the integral is bounded by

$$|\Lambda| \int_0^{2R_0} (|a_0|\rho^{6-a_1} + |b_0|\rho^{6-a_2}) \, d\rho = |\Lambda| \left( \frac{|a_0|(2R_0)^{7-a_1}}{7-a_1} + \frac{|b_0|(2R_0)^{7-a_2}}{7-a_2} \right).$$

Here $|\Lambda|$ denotes the surface measure of the link. These bounds prove the expansion and its stated uniformity. □

5.2. **Geometry of the explicit cap and the critical root.** Write $\phi$ for the profile $h$ in (3.3) and $E_*$ for its cap, which lies in $D$. We use the names $C_0 = c = 5/2$ and $\vartheta = q = 3/4$ for its two parameters.

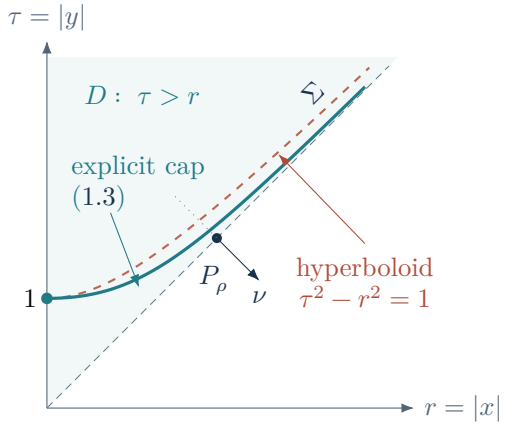



FIGURE 3. Exact radial traces of the explicit cap and the unit hyperboloid. The cap has negative exterior-minus-interior curvature. The hyperboloid has positive curvature. The cone normal $\nu$ points out of $D$. A negative normal displacement moves a point into $D$.

**Lemma 5.4** (Geometry and expansion of the explicit cap)**.** *The boundary of $E_*$ is $|y| = h_*(|x|)$, where $h_*$ is smooth and even at zero and satisfies*

$$h_*(0) = 1, \qquad h_*(v) > v, \qquad h_*''(v) > 0, \qquad 0 < h_*'(v) < 1 \quad (v > 0), \qquad h_*(v) - v h_*'(v) > 0.$$

*Its normal graph is defined for $\rho \geq \rho_{\mathrm{ax}} := 1/\sqrt{2}$, with $f_0(\rho_{\mathrm{ax}}) = -\rho_{\mathrm{ax}}$. The endpoint is the smooth factor axis $x = 0$, and the set $E_*$ is empty in normal coordinates with $\rho < \rho_{\mathrm{ax}}$. With $B = C_0^{3/8}/2$, the normal field has the expansion*

$$f_0(\rho) = -B\rho^{-\kappa} + \frac{3C_0 B}{8}\rho^{-\kappa-4} + O_5(\rho^{-\kappa-7}). \tag{5.9}$$

*Moreover, there is $c_s > 0$ such that, for $\rho \geq \rho_{\mathrm{ax}}$,*

$$\mathcal{H}_s[E_*](P_\rho + f_0(\rho)\nu(P_\rho)) \leq -c_s(1+\rho)^{-\kappa-1-s}. \tag{5.10}$$

*At the axis this expression is understood through the smooth boundary; the physical point does not depend on the collapsed angular coordinate in the $x$-factor.*

*Proof.* For a fixed $v = |x|$, the equation for $u = |y|$ is

$$u^2 - v^2 = \phi(2uv).$$

The difference $u^2 - v^2 - \phi(2uv)$ is negative for $u \leq v$, tends to infinity with $u$, and has derivative $2(u - v\phi'(2uv)) > 0$ for $u > v$. Thus it has one zero $u = h_*(v) > v$, and the cap inequality is equivalent to $u > h_*(v)$. At $v = 0$, the zero is $u = 1$. The defining function is smooth and even in $v$, with nonzero derivative in $u$. The implicit function theorem therefore proves smoothness and evenness, including at zero.

Along $u = h_*(v)$, put $r = 2uv$, $a = u^2 + v^2 = \sqrt{r^2 + \phi^2}$, and

$$A = u - v\phi'(r) > 0, \qquad B_1 = v + u\phi'(r).$$

Implicit differentiation gives $u' = B_1/A$, and

$$u + vu' = \frac{uA + vB_1}{A} = \frac{a}{A}, \qquad r' = \frac{2a}{A}.$$

Differentiating $Au' = B_1$ therefore gives

$$Au'' = 1 + 2\phi' u' - (u')^2 + (u + vu')\phi'' r', \qquad A^2 + 2\phi' A B_1 - B_1^2 = (\phi - r\phi')(1 + \phi'^2).$$

Thus

$$\begin{aligned} h_*'(v) &= \frac{v + u\phi'(r)}{u - v\phi'(r)}, \\ h_*''(v) &= \frac{(\phi - r\phi')(1+\phi'^2) + 2(r^2+\phi^2)\phi''}{(u - v\phi')^3}, \\ h_*(v) - vh_*'(v) &= \frac{uA - vB_1}{A} = \frac{\phi - r\phi'}{u - v\phi'} > 0. \end{aligned} \tag{5.11}$$

The last sign follows from $\phi > 0$ and $\phi' \leq 0$.

To check the sign of $h_*''$, put $t = r^2/C_0$ and $k = \phi^2/C_0$. If $N$ denotes its numerator, the derivatives in (3.13) give

$$\frac{(1+t)^3 N}{\phi} = \big[1 + (1+\vartheta)t\big]\big[(1+t)^2 + \vartheta^2 tk\big] + 2\vartheta(t+k)\big[(1+\vartheta)t - 1\big](1+t).$$

Collecting the terms independent of $k$ and those linear in $k$ yields

$$\begin{aligned} \frac{(1+t)^3 N}{\phi} &= [1 + (2-\vartheta)t + (1 + 3\vartheta + 2\vartheta^2)t^2](1+t) \\ &\quad + \vartheta k[-2 + 3\vartheta t + (\vartheta+1)(\vartheta+2)t^2]. \end{aligned} \tag{5.12}$$

Since $t \geq 0$, $\vartheta = 3/4$, and $0 < k \leq 1/C_0$, this expression is at least $1 - 2\vartheta k \geq 2/5$. Hence $h_*'' > 0$. Evenness gives $h_*'(0) = 0$, so $h_*' > 0$ for positive arguments. The first formula in (5.11) also gives $h_*' < 1$, because $(u - v) - (u + v)\phi' > 0$. The normal radius along this graph is $\rho(v) = (v + h_*(v))/\sqrt{2}$. It is strictly increasing, starts at $1/\sqrt{2}$, and tends to infinity. Its normal displacement is $f_0(\rho(v)) = (v - h_*(v))/\sqrt{2}$. At fixed $\rho$, increasing $\tau$ increases $|x|$ and decreases $|y|$. Since $h_*' \geq 0$, the quantity $|y| - h_*(|x|)$ is strictly decreasing in $\tau$. Thus on this normal slice $E_*$ is exactly the sublevel set $\tau < f_0(\rho)$. If a point of $E_*$ had normal radius below $1/\sqrt{2}$, its factor radii would satisfy both $|y| > h_*(|x|) \geq 1$ and $|x| + |y| < 1$, a contradiction. This proves the normal-domain assertions as well as the geometric claims.

In normal coordinates, the same equation reads

$$-2\rho f_0 = \phi(\rho^2 - f_0^2).$$

Let $w = -f_0 > 0$. Since $\phi \le 1$, we first obtain $w \le 1/(2\rho)$. The equation becomes

$$2\rho w = C_0^{3/8}\left[(\rho^2 - w^2)^2 + C_0\right]^{-3/8}.$$

After writing $w = B\rho^{-5/2}v$, the equation is

$$v = \left[1 + C_0\rho^{-4} - 2B^2\rho^{-7}v^2 + B^4\rho^{-14}v^4\right]^{-3/8}.$$

In the equation for $w$, the expression inside the power is $\rho^4[(1-(w/\rho)^2)^2 + C_0\rho^{-4}]$. Since $w/\rho = O(\rho^{-2})$, dividing out the leading power gives $v \to 1$. To differentiate the expansion, put $U = \rho^{-4}$ and $W = \rho^{-7}$ and regard them as independent. The derivative in $v$ of the left side minus the right side of the rescaled equation is one at $(v, U, W) = (1, 0, 0)$. The implicit function theorem gives a unique smooth branch near this point, which agrees with the cap branch for large $\rho$ by $v \to 1$. At $W = 0$ the branch equals $(1 + C_0U)^{-3/8}$. Taylor expansion in $U$ there and integration of the $W$-derivative give

$$v = 1 - \frac{3C_0}{8}U + U^2F_1(U) + WF_2(U, W)$$

with $F_1, F_2$ smooth near zero. If $\mathcal{D} = \rho\, d/d\rho$, then on this branch $\mathcal{D} = -4U\partial_U - 7W\partial_W$. Each application preserves the factors $U^2$ and $W$, so the remainder and its first five $\mathcal{D}$-derivatives are $O(\rho^{-7})$. The identity $\rho^j(d/d\rho)^j = \mathcal{D}(\mathcal{D}-1)\cdots(\mathcal{D}-j+1)$ gives the corresponding weighted ordinary derivatives. Substitution into $f_0 = -B\rho^{-5/2}v$ proves (5.9).

Finally, Theorem 4.9 gives a negative curvature bound by a positive constant times $\phi(r)a^{-1-s/2}$. Since $a = \rho^2 + f_0^2$ and $\phi(r) = -2\rho f_0$, the expansion gives $\phi(r)a^{-1-s/2} \sim 2B\rho^{1-\kappa}\rho^{-2-s}$. The ratio of this positive quantity to $(1+\rho)^{-\kappa-1-s}$ is continuous and strictly positive up to the axis on bounded radial intervals. This proves (5.10). □

**Lemma 5.5** (The critical exponent)**.** *For every $0 < s < 1$,*

$$\Delta_s := H_s(\kappa) - A_s > 0. \tag{5.13}$$

*The smaller radial root $\gamma = \gamma_-(s)$ is defined and satisfies*

$$1 < \gamma < \kappa = \frac{5}{2} < \frac{6-s}{2}, \qquad J_s(\rho^{-\gamma}) = 0.$$

*With*

$$p = 1 + \gamma, \qquad \eta = \frac{\kappa - \gamma}{1 + \gamma}, \tag{5.14}$$

*we have*

$$0 < \eta < 1, \qquad 2\gamma + 1 > \kappa, \qquad 2\gamma + 1 < 7, \qquad p(1+\eta) = \kappa + 1 = \frac{7}{2} < 8. \tag{5.15}$$

*Proof.* In Corollary 5.3, take the term $-B\rho^{-\kappa}$ as the sole explicit mode and the remaining terms in (5.9) as $g = O_5(\rho^{-\kappa-4})$. The exponents satisfy $\kappa < 3$ and $\kappa + 4 < 7$. The errors $R^{-2\kappa-2-s}$, $R^{-\kappa-5-s}$, and $R^{-8-s}$ are all $o(R^{-\kappa-1-s})$, so the corollary gives

$$\mathcal{H}_s[E_*](P_R + f_0(R)\nu(P_R)) = -2B(H_s(\kappa) - A_s)R^{-\kappa-1-s} + o(R^{-\kappa-1-s}).$$

Multiply this identity and (5.10) by $R^{\kappa+1+s}$, and let $R \to \infty$. We obtain

$$-2B(H_s(\kappa) - A_s) \le -c_s < 0.$$

Since $B > 0$, this gives the quantitative conclusion $\Delta_s \ge c_s/(2B) > 0$. Lemma 2.3 now supplies the root and its asserted range. Since $\gamma > 1$, $\kappa - \gamma < 1 + \gamma$ and $2\gamma + 1 > 3 > \kappa$. Since $\gamma < 5/2$, $2\gamma + 1 < 6 < 7$. The remaining identity is the definition of $\eta$. □

5.3. **The exact negative critical cap.** We first record the curvature continuity argument used below and for the rescaled graph barriers.

**Lemma 5.6** (Continuity on a compact part of a surface)**.** *Let $E_j, E \subset \mathbb{R}^d$ be measurable sets, $d \geq 2$. Near a fixed compact part of $\partial E$ and the corresponding parts of $\partial E_j$, suppose the sets are given by one side of local $C^2$ boundary graphs. Assume the graphs converge in $C^2$ with uniform chart bounds and that $\chi_{E_j} \to \chi_E$ in $L^1$ on every bounded set. Then their curvatures at corresponding points converge uniformly on that compact part.*

*Proof.* For corresponding points $p_j \to p$ on the compact family, truncate the kernel smoothly at distance $\delta$ from its evaluation point. After subtracting the tangent halfspace, the local difference is confined to a slab of thickness $Cr^2$ at tangential distance $r$. Thus the removed local integrals are bounded by $C\int_0^{2\delta} r^{d-2}r^2 r^{-d-s}\,dr = C\int_0^{2\delta} r^{-s}\,dr$, uniformly by the common chart bound. For fixed $\delta$, let $K_\delta$ denote the remaining kernel. On the ball $B_M$, the difference of the truncated integrals is bounded by

$$2\|K_\delta\|_\infty \|\chi_{E_j} - \chi_E\|_{L^1(B_M)} + |B_M|\|\nabla K_\delta\|_\infty\, |p_j - p|.$$

This tends to zero uniformly, by the assumed local convergence and the uniform convergence of the corresponding points in their charts. Since the centers stay bounded, the integral outside $B_M$ is at most $C_s M^{-s}$ for $M$ large, uniformly for both surfaces. Choosing $\delta$ small and $M$ large before taking $j \to \infty$ proves the assertion. □

**Proposition 5.7** (A negative cap with an exact critical tail)**.** *There is a smooth even vertical profile $h_-$, with*

$$h_- > r, \qquad h_-'' > 0, \qquad 0 < h_-' < 1 \quad (r > 0), \qquad h_- - r h_-' > 0,$$

*whose superlevel set $E_-^0 = \{|y| > h_-(|x|)\}$ has strictly negative curvature everywhere. Outside a finite normal radius, its normal field is exactly*

$$f_-(\rho) = -A_-\rho^{-\gamma} - B_-\rho^{-\kappa}, \qquad A_-, B_- > 0. \tag{5.16}$$

*It has the global bound*

$$\mathcal{H}_s[E_-^0](re_1, h_-(r)e_1) \leq -c_s(1+r)^{-\kappa-1-s}. \tag{5.17}$$

*Proof.* All deformations keep the original cap near its axis, with empty slices for $\rho < \rho_{\mathrm{ax}}$. The relations $r = (\rho + f(\rho))/\sqrt{2}$ and $h = (\rho - f(\rho))/\sqrt{2}$ convert the normal field $f$ to a vertical profile whenever $1 + f' > 0$. Applying $d/dr = \sqrt{2}(1+f')^{-1}d/d\rho$ gives

$$h' = \frac{1-f'}{1+f'}, \qquad h'' = -\frac{2\sqrt{2}f''}{(1+f')^3}, \qquad h - rh' = \frac{\sqrt{2}(\rho f' - f)}{1+f'}. \tag{5.18}$$

The signs $f < 0$, $0 < f' < 1$, and $f'' < 0$ thus suffice for the desired geometry away from the axis. If $f(\rho_{\mathrm{ax}}) = -\rho_{\mathrm{ax}}$, the map $r_f(\rho) = (\rho + f(\rho))/\sqrt{2}$ increases from zero to infinity. The profile defined by $h(r_f(\rho)) = (\rho - f(\rho))/\sqrt{2}$ is increasing. For $\rho \geq \rho_{\mathrm{ax}}$, the condition $\tau < f(\rho)$ is equivalent to $r < r_f(\rho)$, hence to $\rho > r_f^{-1}(r)$. Since $|y| = \sqrt{2}\rho - r$, this is $|y| > h(r)$. For $\rho < \rho_{\mathrm{ax}}$ both sets are empty, as $|y| \leq \sqrt{2}\rho < h(0) = \sqrt{2}\rho_{\mathrm{ax}}$. This identifies the set determined by the normal field with the vertical superlevel set. Conversely, on the original cap $f_0 < 0$ by $h_* > r$. The inequalities $0 < h_*' < 1$ and $h_*'' > 0$ give $0 < f_0' < 1$ and $f_0'' < 0$ by the first two identities in (5.18).

Let $g = f_0 + B\rho^{-\kappa}$. For a smooth cutoff $\chi$ taking values in $[0,1]$, zero on $[0,1]$ and one on $[2,\infty)$, put

$$f_R = f_0 + v_R, \qquad v_R = -\chi(\rho/R)g.$$

With $b = \kappa + 4 = 13/2 < 7$, Leibniz's rule gives

$$v_R = 0 \quad (\rho \leq R), \qquad |v_R^{(j)}(\rho)| \leq C_j\rho^{-b-j} \quad (0 \leq j \leq 5), \tag{5.19}$$

with constants independent of $R$. Indeed $g = O_5(\rho^{-b})$ by (5.9), and every positive-order derivative of $\chi(\rho/R)$ has bound $C_j R^{-j}$ and is supported where $\rho \asymp R$. The derivative bounds also hold for $g_R = (1-\chi(\rho/R))g$, so $f_R = -B\rho^{-\kappa} + g_R$. For all sufficiently large $R$ this gives

$$f_R = -B\rho^{-\kappa} + O(\rho^{-b}), \qquad f_R' = \kappa B\rho^{-\kappa-1} + O(\rho^{-b-1}), \qquad f_R'' = -\kappa(\kappa+1)B\rho^{-\kappa-2} + O(\rho^{-b-2})$$

for $\rho \geq R$, uniformly in $R$. The leading terms give $f_R < 0$, $0 < f_R' < 1$, and $f_R'' < 0$ there, while $f_R = f_0$ for $\rho \leq R$.

Choose a second cutoff $\chi_0$ equal to zero for $\rho \leq L$ and one for $\rho \geq 2L$, where $L > \rho_{\mathrm{ax}}$ is fixed. For $R \geq 4L$, consider

$$f_{R,\varepsilon} = f_R - \varepsilon\chi_0(\rho)\rho^{-\gamma}, \qquad 0 \leq \varepsilon \leq \varepsilon_0.$$

Choose $L$ large enough that $R \geq 4L$ lies in the range of the preceding sign estimates and that $f_R' \leq 1/2$ for all $\rho \geq 2L$ and $R \geq 4L$, using the uniform decay of $f_R'$ there. On the fixed transition $[L, 2L]$, the unperturbed field is $f_0$. By its strict signs there, we may choose $\varepsilon_0 > 0$ independent of $R$ so that $f_{R,\varepsilon} < 0$, $0 < f_{R,\varepsilon}' < 1$, and $f_{R,\varepsilon}'' < 0$ throughout the transition. On $\rho \geq 2L$, the term $-\varepsilon\rho^{-\gamma}$ preserves the signs of $f_R, f_R', f_R''$. Its derivative is bounded by $\varepsilon_0\gamma(2L)^{-\gamma-1}$, which can be made less than $1/2$, so $f_{R,\varepsilon}' < 1$ there as well. The axis region is unchanged. The normal-to-vertical conversion above therefore gives the claimed geometry and identifies the two descriptions of the set.

On $\rho \geq 2L$ the fields have the form

$$f_{R,\varepsilon} = -\varepsilon\rho^{-\gamma} - B\rho^{-\kappa} + g_R.$$

Choose one exterior radius $R_0 \geq \max(2L, 1)$ such that $|f_{R,\varepsilon}| \leq \rho/10$ for $\rho \geq R_0$, using the common $O(\rho^{-\gamma})$ bound. All weighted bounds and coefficients are uniform in $R \geq 4L$ and $0 \leq \varepsilon \leq \varepsilon_0$. Corollary 5.3 applies with this fixed $R_0$, and the critical mode has zero Jacobi coefficient. It gives

$$\mathcal{H}_s[E_{f_{R,\varepsilon}}] = -2B\Delta_s\rho^{-\kappa-1-s} + O_s(\rho^{-2\gamma-2-s} + \rho^{-b-1-s} + \rho^{-8-s}) \tag{5.20}$$

at exterior boundary points, uniformly in both parameters. Relative to $\rho^{-\kappa-1-s}$, the three errors decay with positive powers $2\gamma+1-\kappa$, $b-\kappa$, and $7-\kappa$. We can therefore fix one radius $S$ beyond which the curvature is at most $-B\Delta_s\rho^{-\kappa-1-s}$.

To treat $\rho \leq S$, first set $\varepsilon = 0$ and take $R \geq \max(4L, 4S)$. The surfaces for $f_R$ and $f_0$ agree near each evaluation point $p$ of normal radius $\rho \leq S$. Their sets can differ only on slices of radii $r \geq R$, where the difference has width $|v_R(r)|$. Every strip point $Y$ has $|Y| \geq r$, whereas $|p| \leq \sqrt{2}S$, so $|p - Y| \geq r/2$. The volume formula (2.3) gives the absolutely convergent bound

$$|\mathcal{H}_s[E_{f_R}](p) - \mathcal{H}_s[E_{f_0}](p)| \leq C_s \int_R^\infty r^6 |v_R(r)| r^{-8-s}\, dr \leq C_s R^{-b-1-s}. \tag{5.21}$$

By (5.10), the curvature of $E_{f_0}$ is at most $-c_s(1+S)^{-\kappa-1-s}$ on this compact portion. Fix $R$ so large that the error above is less than half this margin.

Only now choose a positive $\varepsilon$ small enough for the compact portion $\rho \leq S$. As $\varepsilon \downarrow 0$, the fields $f_{R,\varepsilon}$ converge smoothly on bounded radial intervals to $f_R$. The change of variables $r = (\rho + f_{R,\varepsilon})/\sqrt{2}$ has derivative at least $1/\sqrt{2}$, so its inverse converges smoothly on bounded intervals of $r$. The axis region is unchanged. The vertical profiles therefore converge smoothly on compact intervals, and their indicators converge in local $L^1$ by dominated convergence away from the limiting boundary. Lemma 5.6 shows that the curvatures converge uniformly on the compact portion with $\rho \leq S$. There the straightened cap has a strictly negative maximum, so taking $\varepsilon > 0$ small preserves a fixed negative bound on that portion. For $\rho \geq 2R$, both cutoffs are one, and the field is exactly the tail in (5.16) with $A_- = \varepsilon$ and $B_- = B$. Finally, $r < h_-(r) \leq r + h_-(0)$ follows from $0 < h_-' < 1$ and the profile's position above the cone. Thus $1 + \rho = 1 + (r + h_-(r))/\sqrt{2}$ is comparable to $1 + r$, and the compact and exterior bounds give (5.17). □

5.4. **The positive critical cap.** For this construction, distinguish the spectral midpoint from the chosen second exponent:

$$d_{\mathrm{mid}} = \frac{6-s}{2} - \gamma, \qquad d = \kappa - \gamma, \qquad q = \gamma - 1, \qquad b = q/d. \tag{5.22}$$

By Lemma 5.5, $0 < d < d_{\mathrm{mid}}$, $q > 0$, and $q + d = \kappa - 1 = 3/2$.

Write $w(\rho) = \rho^{-\gamma} v(\log \rho)$ and $t = \log \rho$. Subtract $v(t) J_s(\rho^{-\gamma})(P_\rho) = 0$ from $J_s w(P_\rho)$, using the same cutoff in both principal values. The potential and evaluation-point terms cancel, leaving the source numerator $|Z|^{-\gamma}[v(\log|Z|) - v(t)]$. After scaling $Z$ by $\rho$, the radial measure is $r^6\,dr$ and the overall prefactor is $\rho^{-\gamma-1-s}$. Putting $r = e^u$ gives the logarithmic kernel

$$k_\gamma(u) = e^{(7-\gamma)u} \mathcal{A}_s(e^u).$$

The angular inversion identity of Lemma 2.1 gives

$$k_\gamma(-u) = e^{(1+s+\gamma)u} \mathcal{A}_s(e^u),$$

hence

$$k_\gamma(u) = e^{2d_{\mathrm{mid}} u} k_\gamma(-u). \tag{5.23}$$

Pairing the positive and negative logarithms now yields

$$J_s w(P_\rho) = \rho^{-\gamma-1-s} \int_0^\infty k_\gamma(-u) \big[ e^{2d_{\mathrm{mid}} u} (v(t+u) - v(t)) - (v(t) - v(t-u)) \big]\, du, \qquad t = \log \rho. \tag{5.24}$$

The pairing uses precisely the inversion from Lemma 2.1. In scaled variables the two deletions on $0 < r < 1$ are $|P_1 - r\omega| > \varepsilon$ and $|P_1 - r\omega| > r\varepsilon$. For fixed $t$ the unpaired numerator, including its smooth radial prefactors near $r = 1$, is $O(|r-1|)$ by smoothness of $v$. The shell estimate in that lemma therefore gives an error $O(\varepsilon^{1-s})$. The bracket in (5.24) is

$$(v''(t) + 2d_{\mathrm{mid}} v'(t)) u^2 + O(u^3) \qquad (u \to 0).$$

The bound for $\mathcal{A}_s$ in the same lemma gives $k_\gamma(-u) = O(u^{-2-s})$ there, so the paired integrand is $O(u^{-s})$. For bounded $v$, the two unpaired tails are integrable by the radial bounds $Cr^{6-\gamma}$ at zero and $Cr^{-2-s-\gamma}$ at infinity. In particular the formula applies to all the functions constructed below.

**Lemma 5.8** (A logarithmic sign criterion)**.** *Suppose $v' > 0$ and*

$$-d \le (\log v')' \le q.$$

*Whenever (5.24) is defined,*

$$J_s(\rho^{-\gamma} v(\log \rho))(P_\rho) \ge c_s \rho^{-\gamma-1-s} v'(\log \rho) > 0.$$

*Proof.* Integrating the lower bound for $(\log v')'$ over an interval of length $u$ gives $v'(t+\xi) \ge e^{-du} v'(t-u+\xi)$ for $0 \le \xi \le u$. Integration in $\xi$ yields

$$v(t+u) - v(t) \ge e^{-du}(v(t) - v(t-u)).$$

Thus the bracket in (5.24) is at least

$$(e^{(2d_{\mathrm{mid}} - d)u} - 1)(v(t) - v(t-u)) > 0.$$

For $1 \le u \le 2$, the upper bound for $(\log v')'$ also gives

$$v(t) - v(t-u) = \int_0^u v'(t-\xi)\, d\xi \ge v'(t) \int_0^1 e^{-q\xi}\, d\xi.$$

On this interval the factor $e^{(2d_{\mathrm{mid}} - d)u} - 1$ has a positive lower bound and $k_\gamma(-u)$ has a positive finite integral. Retaining this interval proves the quantitative bound. □

Start the interpolation with

$$v_0(t) = (1 + e^{-dt})^{-b}, \qquad a_0(t) = v_0'(t) = qe^{qt}(1 + e^{dt})^{-b-1}.$$

Its logarithmic derivative is

$$\ell_0(t) := (\log a_0)'(t) = -d + \frac{q+d}{1+e^{dt}}, \qquad -d < \ell_0 < q.$$

Since $\ell_0(t) \to -d$ and $a_0(t)/v_0(t) = q/(1+e^{dt}) \to 0$ as $t \to \infty$, choose $M$ with $\ell_0(M) < 0$ and

$$\frac{a_0(M)}{v_0(M)} < \frac{\gamma(\gamma+1)}{4(2\gamma+1+d)}. \tag{5.25}$$

Let $0 \le \theta \le 1$ be smooth, equal to zero for $t \le M$ and one for $t \ge M+1$. Set

$$a(t) = a_0(t) \exp\left( - \int_{-\infty}^{t} \theta(u)(\ell_0(u) + d)\, du \right), \qquad v(t) = \int_{-\infty}^{t} a(u)\, du.$$

The integral in the exponent is nonnegative and bounded by $\int_M^\infty (q+d)e^{-du}\, du$. Also $(\log a)' = (1-\theta)\ell_0 - \theta d$, so

$$c_s a_0 \le a \le a_0, \qquad c_s v_0 \le v \le v_0, \qquad -d \le (\log a)' \le q. \tag{5.26}$$

The comparison for $v$ follows by integration from $-\infty$, where $v_0$ vanishes. For $t \le M$, we have $v = v_0$. For $t \ge M+1$, we have $(\log a)' = -d$, whence

$$v(t) = A_0 - B_0 e^{-dt} \qquad (t \ge M+1), \tag{5.27}$$

with $A_0 = \int_{\mathbb{R}} a(u)\, du$ and $B_0 = d^{-1} e^{d(M+1)} a(M+1)$, both positive.

Set

$$w(\rho) = \rho^{-\gamma} v(\log \rho), \qquad w_R(\rho) = R^{-1} w(\rho/R), \qquad R \ge 1.$$

The factor multiplying $\rho^{-\gamma}$ in $w_R$ is $R^q v(t - \log R)$ with $t = \log \rho$. Its logarithmic derivative bounds for application of Lemma 5.8 are inherited from $v$ by translation and multiplication, and

$$R^q v'(t - \log R) \ge c_s R^q a_0(t - \log R) = c_s q\, \rho^q (1 + (\rho/R)^d)^{-b-1}.$$

Since $q = \gamma - 1$, the lemma gives

$$J_s w_R(P_\rho) \ge c_s \rho^{-2-s} (1 + (\rho/R)^d)^{-b-1} > 0. \tag{5.28}$$

We also have

$$0 < w_R \le \rho^{-1}, \qquad -\gamma w_R/\rho \le w_R' \le -w_R/\rho < 0, \qquad w_R'' > 0. \tag{5.29}$$

It suffices to check these claims for $w$, since they are preserved by the dilation defining $w_R$. The size bound follows from $v \le v_0 \le e^{qt}$. The inequality $(\log a)' \le q$ gives $a(u) \ge a(t) e^{-q(t-u)}$ for $u < t$, so $v(t) \ge a(t)/q$. Substituting $0 < a(t)/v(t) \le q$ into

$$w'(\rho) = \rho^{-\gamma-1} \big[ a(\log \rho) - \gamma v(\log \rho) \big]$$

gives the two derivative bounds. To see $w'' > 0$ for $\log \rho \le M$, use the formula $w(\rho) = \rho^{-1}(1+\rho^d)^{-b}$. Its logarithmic slope $\lambda = -\rho w'/w$ satisfies

$$\lambda = 1 + q \frac{\rho^d}{1+\rho^d}, \qquad \frac{d\lambda}{d \log \rho} = qd \frac{\rho^d}{(1+\rho^d)^2}, \qquad \frac{\rho^2 w''}{w} = \lambda(\lambda+1) - \frac{d\lambda}{d\log\rho}.$$

Here $\lambda \ge 1$ and $qd \le (q+d)^2/4 = 9/16$, so the last expression is positive. For $t \ge M$, the formula for $\ell_0$ gives $(\log a)' < 0$, while $v$ is increasing. Thus $a(t)/v(t) \le a_0(M)/v_0(M)$, and

$$w''(\rho) = \rho^{-\gamma-2} \big[ \gamma(\gamma+1)v - (2\gamma+1)a + a' \big]_{t = \log \rho} > 0.$$

The sign follows from $a' \ge -da$ and (5.25).

For every fixed derivative order we have

$$|w_R^{(j)}(\rho)| \le C_{s,j} \rho^{-1-j} \quad (R \ge 1). \tag{5.30}$$

Indeed, with $\mathcal{D} = \rho\, d/d\rho$, the small-$\rho$ formula is $\rho^{-1}H(\rho^d)$ for $H(z) = (1+z)^{-b}$. Applying powers of $\mathcal{D}$ amounts, after factoring out $\rho^{-1}$, to applying powers of $d\, z\, \frac{d}{dz} - 1$ to $H$ on $0 \le z \le e^{dM}$. These are bounded there. For $\log\rho \ge M+1$, the two powers in (5.27) give the same estimate because $\gamma, \kappa > 1$. The smooth intermediate region is compact. The identity $\rho^j(d/d\rho)^j = \mathcal{D}(\mathcal{D}-1)\cdots(\mathcal{D}-j+1)$ and the dilation defining $w_R$ give the stated bounds.

On any compact subinterval of $(0,\infty)$, for all sufficiently large $R$ we have $w_R(\rho) = \rho^{-1}(1 + (\rho/R)^d)^{-b}$. This converges to $\rho^{-1}$ with every fixed number of derivatives. Substituting $t = \log(\rho/R)$ in (5.27) gives the exact tail

$$w_R(\rho) = A_0 R^{\gamma-1}\rho^{-\gamma} - B_0 R^{\kappa-1}\rho^{-\kappa}, \qquad \rho \ge Re^{M+1}. \tag{5.31}$$

**Proposition 5.9** (A positive cap with an exact critical tail)**.** *There is a smooth even vertical profile $h_+$, with*

$$h_+(0) = 1, \quad h_+ > r, \quad h_+'' > 0, \quad 0 < h_+' < 1 \quad (r>0), \quad h_+ - rh_+' > 0,$$

*such that $E_+^0 = \{|y| > h_+(|x|)\}$ satisfies*

$$\mathcal{H}_s[E_+^0](re_1, h_+(r)e_1) \ge c_s(1+r)^{-\kappa-1-s}. \tag{5.32}$$

*Outside a finite normal radius its normal field is exactly*

$$f_+(\rho) = -A_+\rho^{-\gamma} + B_+\rho^{-\kappa}, \qquad A_+, B_+ > 0. \tag{5.33}$$

*Proof.* The unit hyperboloid has normal field $f_H(\rho) = -1/(2\rho)$ for $\rho \ge \rho_0 = 1/\sqrt{2}$, since $|y|^2 - |x|^2 = -2\rho\tau$. Its superlevel set is empty below $\rho_0$. Choose a fixed $L > \max\{\rho_0, \sqrt{5}\}$ with $\gamma/(2L^2) < 1/2$, and a smooth $0 \le \chi \le 1$ that vanishes on $\rho \le L$ and equals one on $\rho \ge 2L$. For $\rho \ge \rho_0$, put

$$f_R(\rho) = -\frac{1}{2}\big[(1-\chi(\rho))\rho^{-1} + \chi(\rho)w_R(\rho)\big].$$

The set in normal coordinates is $\{\tau < f_R(\rho)\}$ above $\rho_0$ and empty below it. The field equals $f_H$ through radius $L$. On $[L, 2L]$, it tends smoothly to $f_H$ as $R \to \infty$, so $0 < f_R' < 1$, $f_R'' < 0$, and $-\rho < f_R < 0$ on that interval for large $R$. These signs also hold on $(\rho_0, L]$ by the hyperboloid formula and on $[2L,\infty)$ by (5.29), which in particular gives $0 < -w_R'/2 \le \gamma/(2\rho^2) < 1/2$.

Apply the normal-to-vertical conversion in the proof of Proposition 5.7, with $\rho_0$ as the axis radius and $r_R(\rho) = (\rho + f_R(\rho))/\sqrt{2}$. It gives $E_{h_R} = \{|y| > h_R(|x|)\}$ and the geometric inequalities in (5.18). Near $r = 0$ the profile agrees with $\sqrt{1+r^2}$, so it is smooth and even there, with $h_R(0) = 1$.

For the exterior estimate, take the reference field $\tilde{f}_R = -\chi w_R/2$. By (5.30) and the product rule, its weighted $C^5$ bounds of exponent one are uniform in $R$. It vanishes near the vertex and obeys $|\tilde{f}_R(\rho)| \le 1/(2\rho) \le \rho/10$ on $\rho \ge L$. Lemma 5.2 then gives, at $X_R = P_\rho + f_R(\rho)\nu(P_\rho)$ for sufficiently large $\rho$,

$$\mathcal{H}_s[E_{\tilde{f}_R}](X_R) = J_s(\chi w_R)(P_\rho) + O_s(\rho^{-4-s}).$$

The actual and reference sets agree for normal radii at least $2L$, so (5.7) bounds their curvature difference by $C_s\rho^{-8-s}$ for $\rho \ge 4L$. The field $(\chi-1)w_R$ is supported on the cone within radius $2L$, and $w_R(u) \le u^{-1}$. Hence (5.8) bounds its Jacobi integral by $C_s\rho^{-8-s}\int_0^{2L} u^5\, du$. Consequently

$$\mathcal{H}_s[E_{h_R}](X_R) \ge J_s w_R(P_\rho) - C_s(\rho^{-4-s} + \rho^{-8-s}). \tag{5.34}$$

All constants and the exterior threshold are independent of $R$.

For $R \ge 1$ and $\rho \ge 1$, we have $1 + (\rho/R)^d \le 2\rho^d$ and $d(b+1) = \kappa - 1$. Thus (5.28) gives $J_s w_R(P_\rho) \ge c_s\rho^{-\kappa-1-s}$. The ratios of the two errors in (5.34) to this lower bound are bounded by $C_s\rho^{\kappa-3}$ and $C_s\rho^{\kappa-7}$. Both tend to zero, so one can choose $T_s \ge 4L$ independent of $R$ such that $\mathcal{H}_s[E_{h_R}](X_R) \ge c_s'\rho^{-\kappa-1-s}$ for $\rho \ge T_s$.

On the compact portion $\rho_0 \le \rho \le T_s$, the boundary charts converge in $C^2$ to those of the unit hyperboloid. Indeed, the profiles agree near the axis. Away from the axis, $f_R \to f_H$ smoothly on bounded intervals and the maps $r_R$ have derivatives bounded below by $1/\sqrt{2}$, so their inverses converge smoothly on compact intervals of $r$. The bound $0 < -f_R(\rho) \le 1/(2\rho)$ ensures that

bounded $r$ corresponds to bounded $\rho$ uniformly in $R$. It follows that $h_R \to \sqrt{1+r^2}$ on bounded intervals, giving local $L^1$ convergence of the indicators by dominated convergence. Lemma 5.6 gives uniform curvature convergence on the compact portion. The limiting curvature has a strictly positive minimum there by Proposition 2.2 and continuity. Choose $R$ large enough to preserve half this minimum. Since $T_s$ was chosen independently of $R$, this choice retains the exterior bound.

Fix this $R$ and set $h_+ = h_R$. Since $\rho = \sqrt{2}r - f_R(\rho) \le \sqrt{2}r + 1/(2\rho_0)$, the exterior lower bound and the compact minimum give (5.32). Finally, (5.31) gives (5.33) with $A_+ = \frac{1}{2}A_0 R^{\gamma-1}$ and $B_+ = \frac{1}{2}B_0 R^{\kappa-1}$. □

5.5. **Ordering the profiles and labeling their dilations.**

**Theorem 5.10** (Global signed critical profiles)**.** *For every $0 < s < 1$, let $\gamma$, $p$, and $\eta$ be as in Lemma 5.5. There are two smooth even profiles $h_\pm : [0,\infty) \to (0,\infty)$ such that*

$$h_- > h_+ > r, \qquad h_\pm'' > 0, \qquad 0 < h_\pm' < 1 \quad (r>0), \qquad h_\pm - r h_\pm' > 0. \tag{5.35}$$

*Their superlevel sets*

$$E_\pm^0 = \{(x,y) : |y| > h_\pm(|x|)\}$$

*obey*

$$\begin{aligned} \mathcal{H}_s[E_+^0](re_1, h_+(r)e_1) &\ge c_s(1+r)^{-\kappa-1-s}, \\ \mathcal{H}_s[E_-^0](re_1, h_-(r)e_1) &\le -c_s(1+r)^{-\kappa-1-s}. \end{aligned} \tag{5.36}$$

*For a finite common radius $R_*$, their normal fields are exactly*

$$f_\pm(\rho) = -A_\pm \rho^{-\gamma} \pm B_\pm \rho^{-\kappa} \qquad (\rho \ge R_*), \qquad A_\pm, B_\pm > 0. \tag{5.37}$$

*The dilations of either boundary are disjoint and fill $D$.*

*Proof.* Take the two profiles from Propositions 5.7 and 5.9. We first convert the normal expansions to the vertical coordinate. For either profile write $f(\rho) = -A\rho^{-\gamma} + D_0\rho^{-\kappa}$ on its tail. Put $\xi = \sqrt{2}r = \rho + f(\rho)$ and $u = f(\rho)/\xi$, so that $\rho = \xi(1-u)$. The exact equation is

$$u = -A\xi^{-1-\gamma}(1-u)^{-\gamma} + D_0\xi^{-1-\kappa}(1-u)^{-\kappa}.$$

Along the leaf, $u \to 0$, since $f(\rho) = O(\rho^{-\gamma})$ and $\xi/\rho \to 1$. Treat the powers $\alpha_1 = \xi^{-1-\gamma}$, $\alpha_2 = \xi^{-1-\kappa}$ as independent variables near zero. The derivative of the defining equation $u + A\alpha_1(1-u)^{-\gamma} - D_0\alpha_2(1-u)^{-\kappa} = 0$ with respect to $u$ at $(u,\alpha_1,\alpha_2) = (0,0,0)$ is one. The smooth solution has derivatives $-A, D_0$ in the two variables there. Its Taylor remainder is a sum of $\alpha_i\alpha_j$ times smooth bounded functions. The operator

$$\xi\frac{d}{d\xi} = -(1+\gamma)\alpha_1\partial_{\alpha_1} - (1+\kappa)\alpha_2\partial_{\alpha_2}$$

preserves these quadratic factors. As in Lemma 5.4, this gives the corresponding weighted ordinary derivative bounds. Since $h(r) - r = -\sqrt{2}f(\rho) = -\sqrt{2}\,\xi u$, we have

$$h(r) = r + ar^{-\gamma} + br^{-\kappa} + O_5(r^{-2\gamma-1}), \qquad a = A2^{(1-\gamma)/2} > 0, \quad b = -D_0\,2^{(1-\kappa)/2}. \tag{5.38}$$

In particular $h(r) - r \asymp_s (1+r)^{-\gamma}$, by the positive leading term at infinity and the strictly positive continuous difference on the compact part. Differentiating (5.38) also gives

$$h - rh' = (1+\gamma)ar^{-\gamma} + O(r^{-\kappa}), \qquad h'' = \gamma(1+\gamma)ar^{-\gamma-2} + O(r^{-\kappa-2}),$$

where $2\gamma + 1 > \kappa$ bounds the differentiated remainders. Together with the compact geometry, these yield

$$\begin{aligned} h(\tau) \asymp_s 1+\tau, \qquad h(\tau) - \tau \asymp_s (1+\tau)^{-\gamma}, \\ \mathfrak{q}(\tau) := h(\tau) - \tau h'(\tau) \asymp_s (1+\tau)^{-\gamma}, \qquad h''(\tau) \asymp_s (1+\tau)^{-\gamma-2}. \end{aligned} \tag{5.39}$$

Choose constants $m_- > 0$ and $M_+ < \infty$ from (5.39) such that

$$h_-(r) - r \ge m_-(1+r)^{-\gamma}, \qquad h_+(r) - r \le M_+(1+r)^{-\gamma}.$$

Dilating the negative profile by a fixed $\mu \geq 1$ gives $h_-^\mu(r) = \mu h_-(r/\mu)$, with

$$h_-^\mu(r) - r \geq \mu m_-(1 + r/\mu)^{-\gamma} \geq \mu m_-(1+r)^{-\gamma}.$$

Take $\mu m_- > M_+$. Then $h_-^\mu > h_+$ for all $r$. Its first derivative is $h_-'(r/\mu)$, its second is $\mu^{-1}h_-''(r/\mu)$, and $h_-^\mu - r(h_-^\mu)' = \mu\, \mathfrak{q}_-(r/\mu)$. Thus the geometric inequalities hold for the dilated profile. Its normal field is $\mu f_-(\rho/\mu)$, so the two coefficients are multiplied by $\mu^{1+\gamma}$ and $\mu^{1+\kappa}$, respectively. The same dilation preserves the form of (5.38) and (5.39). By the curvature scaling in Theorem 4.9, the negative margin for $h_-^\mu$ has magnitude at least $\mu^{-s}c_s(1+r/\mu)^{-\kappa-1-s} \geq \mu^{-s}c_s(1+r)^{-\kappa-1-s}$. Use $h_-^\mu$ as $h_-$ from now on and take a common tail radius $R_*$ for the two profiles.

For either ordered profile and fixed $r \geq 0$, the function $\lambda \mapsto \lambda h(r/\lambda)$ is strictly increasing, since

$$\frac{d}{d\lambda}[\lambda h(r/\lambda)] = \mathfrak{q}(r/\lambda) > 0.$$

If $r > 0$, then $\lambda h(r/\lambda) = r + O(\lambda^{1+\gamma} r^{-\gamma})$ as $\lambda \downarrow 0$, by (5.38). For $r = 0$, the function is $\lambda h(0)$. In both cases the limit at zero is $r$ and the limit at infinity is infinity, the latter because $h(r/\lambda) \to h(0) > 0$. Thus for every $t > r$ there is exactly one dilation with height $t$ at radial coordinate $r$. □

Define their labels by

$$t = \lambda_\pm(r,t)\, h_\pm(r/\lambda_\pm(r,t)), \qquad V_\pm(x,y) = \lambda_\pm(|x|,|y|)^p \quad \text{when } t = |y| > r = |x|. \tag{5.40}$$

Extend $V_\pm$ by zero on $\Sigma$ and by oddness under factor interchange: $V_\pm(y,x) = -V_\pm(x,y)$.

**Lemma 5.11** (Bounds and regularity of the foliation labels)**.** *The labels are smooth in $D$, including its factor axis. They are homogeneous of degree $p$, are locally $C^{1,\eta}$ on $\mathbb{R}^8$, and satisfy $0 < V_- < V_+$ in $D$. Write $\varrho = |X|$, $r = |x|$, and $t = |y|$. For $X \in D$,*

$$V_\pm(X) \asymp_s \varrho^\gamma(t-r), \qquad |DV_\pm(X)| \asymp_s \varrho^\gamma, \tag{5.41}$$

$$|D^2 V_\pm(X)| \leq C_s \varrho^{3\gamma-\kappa} V_\pm(X)^{\eta-1}. \tag{5.42}$$

*For $v = V_\pm(X) > 0$, the superlevel set $\{V_\pm > v\}$ has the corresponding signed curvature bound*

$$\begin{aligned} \mathcal{H}_s[\{V_+ > v\}](X) &\geq c_s v^{1+\eta} \varrho^{-\kappa-1-s}, \\ \mathcal{H}_s[\{V_- > v\}](X) &\leq -c_s v^{1+\eta} \varrho^{-\kappa-1-s}. \end{aligned} \tag{5.43}$$

*Finally, wherever $\rho/v^{1/p} \geq R_*$, its normal field is exactly*

$$f_{\pm,v}(\rho) = -A_\pm v \rho^{-\gamma} \pm B_\pm v^{1+\eta} \rho^{-\kappa}. \tag{5.44}$$

*Proof.* The implicit function theorem applies to $t = \lambda h(r/\lambda)$ because the $\lambda$ derivative is $\mathfrak{q}(r/\lambda) > 0$. For smooth even $h$, the function $x \mapsto h(|x|)$ is smooth at $x = 0$. Taylor polynomials of $h$ have only even powers, and the remainder after degree $2N$ has derivatives of order $k$ bounded by $C_N r^{2N+2-k}$ for $k \leq 2N+1$. After composition with $r = |x|$, the same orders follow from $|D^j(|x|)| \leq C_j |x|^{1-j}$ away from the origin. Taking $N$ arbitrarily large gives smoothness at the axis. Evaluating this radial function at $x/\lambda$ and using $y \neq 0$ in $D$ proves smoothness of the defining equation there. Uniqueness in (5.40) gives $\lambda_\pm(\mu X) = \mu \lambda_\pm(X)$ for $\mu > 0$. Also, for any fixed $\lambda$ and $r$ the dilation of $h_-$ has larger height than that of $h_+$. At a fixed point this gives $\lambda_- < \lambda_+$, and hence $0 < V_- < V_+$.

Write $\tau = r/\lambda$, omitting the sign of the profile. On the leaf $r = \lambda\tau$, $t = \lambda h(\tau)$, and

$$\varrho = \lambda\sqrt{\tau^2 + h(\tau)^2} \asymp_s \lambda(1+\tau).$$

Since $t - r = \lambda(h(\tau) - \tau)$, (5.39) gives $\varrho^\gamma(t-r) \asymp_s \lambda^{1+\gamma} = V$. Implicit differentiation with respect to $r$ and $t$ gives

$$\lambda_r = -\frac{h'}{\mathfrak{q}}, \qquad \lambda_t = \frac{1}{\mathfrak{q}}, \qquad V_r = -\frac{p\lambda^{p-1}h'}{\mathfrak{q}}, \qquad V_t = \frac{p\lambda^{p-1}}{\mathfrak{q}}.$$

For $r > 0$, the two radial gradients in $x$ and $y$ are orthogonal as vectors in $\mathbb{R}^8$, so $|DV| = (V_r^2 + V_t^2)^{1/2}$. The formula extends to the axis by smoothness. As $0 \le h' < 1$, the bounds on $\mathfrak{q}$ give $|DV| \asymp_s \lambda^\gamma (1+\tau)^\gamma \asymp_s \varrho^\gamma$.

The derivatives of the profile parameter and of the dilation derivative are

$$\tau_r = \frac{\mathfrak{q} + \tau h'}{\lambda \mathfrak{q}} = \frac{h}{\lambda \mathfrak{q}}, \qquad \tau_t = -\frac{\tau}{\lambda \mathfrak{q}}, \qquad \mathfrak{q}' = -\tau h''.$$

For example, differentiating $V_t$ in $t$ yields

$$V_{tt} = p\gamma \lambda^{p-2} \frac{\lambda_t}{\mathfrak{q}} - p\lambda^{p-1} \frac{\mathfrak{q}' \tau_t}{\mathfrak{q}^2} = \frac{p\lambda^{p-2}}{\mathfrak{q}^3} (\gamma \mathfrak{q} - \tau^2 h'').$$

The same differentiation, using $(h'/\mathfrak{q})' = hh''/\mathfrak{q}^2$, gives

$$V_{rr} = \frac{p\lambda^{p-2}}{\mathfrak{q}^3} (\gamma h'^2 \mathfrak{q} - h^2 h''), \qquad V_{rt} = \frac{p\lambda^{p-2}}{\mathfrak{q}^3} (-\gamma h' \mathfrak{q} + \tau h h'').$$

Inserting the differentiated powers from (5.38) into the $tt$ bracket cancels the term of order $\tau^{-\gamma}$ and gives

$$\gamma \mathfrak{q} - \tau^2 h'' = (\kappa+1)(\gamma - \kappa) b \tau^{-\kappa} + O(\tau^{-2\gamma-1}).$$

The $rr$ bracket differs from the $tt$ bracket by $\gamma(h'^2 - 1)\mathfrak{q} - (h^2 - \tau^2)h''$. The $rt$ bracket plus the $tt$ bracket equals $\gamma(1-h')\mathfrak{q} + \tau(h-\tau)h''$. In these expressions $h' - 1 = O(\tau^{-\gamma-1})$, $h - \tau = O(\tau^{-\gamma})$, and $h'' = O(\tau^{-\gamma-2})$, so all the extra terms are $O(\tau^{-2\gamma-1})$. Thus each bracket is bounded by $C_s(1+\tau)^{-\kappa}$, using $2\gamma + 1 > \kappa$ at infinity and smoothness on the compact part.

In the angular directions perpendicular to $x$ and $y$ the Hessian acts by $V_r/r$ and $V_t/t$. This follows by differentiating the radial unit vectors, whose derivatives in orthogonal directions are $1/r$ and $1/t$ times the direction. At the first axis the quotient is interpreted by smoothness. Indeed $h'(\tau) = O(\tau)$ near zero and $h' < 1$ everywhere, so $|h'(\tau)|/\tau \le C_s/(1+\tau)$. Also $1/h(\tau) \le C_s/(1+\tau)$. Using the bounds on $\mathfrak{q}$, the angular entries are therefore at most $C_s \lambda^{\gamma-1}(1+\tau)^{\gamma-1}$ in absolute value. The radial entries are at most $C_s \lambda^{\gamma-1}(1+\tau)^{3\gamma-\kappa}$. The latter also bounds the angular entries since $2\gamma + 1 > \kappa$. Finally,

$$\varrho^{3\gamma-\kappa} V^{\eta-1} \asymp_s \lambda^{3\gamma-\kappa+p(\eta-1)} (1+\tau)^{3\gamma-\kappa} = \lambda^{\gamma-1}(1+\tau)^{3\gamma-\kappa}.$$

Here $p(\eta - 1) = \kappa - 2\gamma - 1$, by (5.14). This proves (5.42).

For $v > 0$, the inequality $V_\pm(X) > v$ is equivalent to $t > v^{1/p} h_\pm(r/v^{1/p})$, by the strict increase of the height with the dilation parameter. The odd extension contributes no points from the opposite chamber, so $\{V_\pm > v\} = v^{1/p} E_\pm^0$. The bounds in (5.36) apply in all angular directions by rotational invariance. With $\lambda = v^{1/p}$ at the evaluation point, scaling gives the lower bound for the magnitude of the corresponding signed curvature

$$c_s \lambda^{-s} (1+\tau)^{-\kappa-1-s} \ge c_s' \lambda^{\kappa+1} \varrho^{-\kappa-1-s} = c_s' v^{1+\eta} \varrho^{-\kappa-1-s}.$$

This gives (5.43). In normal coordinates the scaled field is $\lambda f_\pm(\rho/\lambda)$. Substituting (5.37) and $p(1+\eta) = 1 + \kappa$ gives (5.44).

Near a nonzero cone point use the smooth variables

$$\rho = (r+t)/\sqrt{2}, \qquad \delta = \frac{t-r}{r+t}, \qquad \zeta = \frac{V}{\rho^p}.$$

Since $\rho \le \varrho \le \sqrt{2}\,\rho$, the first estimate in (5.41) gives $\zeta \asymp_s \delta$ for $\delta > 0$. Then $\lambda/\rho = \zeta^{1/p} \to 0$ as $\delta \downarrow 0$, so the exact normal field (5.44) applies near the cone. The signed normal displacement at $X$ equals $(r-t)/\sqrt{2} = -\rho\delta$. Dividing the field equation by $-\rho$ gives

$$\delta = A\zeta - D_0 \zeta^{1+\eta}, \tag{5.45}$$

where $D_0 = \pm B_\pm$ is the coefficient on the ordered leaf. The derivative of the right side in $\zeta$ is at least $A/2$ near zero. Hence this equation determines a Lipschitz function of $\delta$ for small $\delta \ge 0$, independent of $\rho$ and of the angular directions, with

$$\zeta'(\delta) = \left[A - (1+\eta) D_0 \zeta(\delta)^\eta\right]^{-1} \quad (\delta > 0), \qquad \lim_{\delta \downarrow 0} \frac{\zeta(\delta)}{\delta} = \frac{1}{A}.$$

The derivative is $\eta$-Hölder up to zero because $|a^\eta - b^\eta| \le |a-b|^\eta$ for $a, b \ge 0$ and the denominator is bounded away from zero. Under factor interchange $\rho$ stays fixed and $\delta$ changes sign. Thus the odd extension of $\zeta$ has the same derivative $1/A$ at zero, and its derivative on either side equals $\zeta'(|\delta|)$. The Hölder estimate for opposite signs follows from $\big||\delta| - |\delta'|\big| \le |\delta - \delta'|$. It follows that $V = \rho^p \zeta(\delta)$ is $C^{1,\eta}$ near each nonzero cone point.

At the origin, $|V(X)| \le C_s |X|^p$ gives differentiability with $DV(0) = 0$, and $|DV(X)| \le C_s |X|^\gamma$ gives continuity of the gradient. These bounds extend from $D$ by oddness and continuity across the punctured cone. On a fixed closed annulus the gradient is $\eta$-Hölder, by a finite covering of the local estimates just proved and smoothness inside each chamber. Differentiating $V(\mu X) = \mu^p V(X)$ gives $DV(\mu X) = \mu^\gamma DV(X)$. To scale the bound, let $M = \max(|X|, |Y|) > 0$. If $\min(|X|, |Y|) \ge M/2$, the two scaled points lie in the annulus $1/2 \le |Z| \le 1$ and the gradient homogeneity gives

$$|DV(X) - DV(Y)| \le C_s M^{\gamma-\eta} |X - Y|^\eta.$$

If $\min(|X|, |Y|) < M/2$, then $|X - Y| \ge M/2$, and the size bound $C_s M^\gamma$ for the difference gives the same estimate. Since $\gamma > \eta$, this gives the local Hölder estimate at the origin as well. □

## 6. Global graph barriers

Fix $s \in (0,1)$ throughout this section. All constants and all smallness thresholds are allowed to depend on this fixed exponent. By Lemma 5.5 and the gap established above, we use

$$\kappa = \frac{5}{2}, \qquad 1 < \gamma < \kappa, \qquad p = 1 + \gamma, \qquad \eta = \frac{\kappa - \gamma}{p} > 0, \qquad \Delta_\kappa = H_s(\kappa) - A_s > 0. \tag{6.1}$$

In particular $2 < p < 7/2$, $0 < \eta < 1$, and $p(1+\eta) = 1 + \kappa$. Here $H_s(\lambda)$ denotes the radial Jacobi coefficient on the base cone, whereas $\mathcal{H}_s[E]$ denotes the curvature of a set. We first record the graph operator and its angular kernel ordering, then analyze the height variable, estimate the nonlinear error, and finally invert the two geometric foliations from Theorem 5.10.

The reason for retaining two tail modes is that they control different errors. The critical mode $\rho^{-\gamma}$ has zero base Jacobi curvature, but its height dependence provides a strict margin. The faster mode $\rho^{-\kappa}$ provides a second margin through $\Delta_\kappa > 0$. Near the profile core we use the full curvature sign of the smooth cap instead of a small-displacement expansion. The proof therefore covers the core, the exterior with height comparable to the base radius, and the exterior with larger height. Every point belongs to one of these regions, and their estimates will be combined only after the constants in each region have been fixed.

6.1. **The graph operator and its weak form.** Let $u : \mathbb{R}^8 \to \mathbb{R}$ be measurable and finite almost everywhere. For its subgraph $E_u = \{(X, z) : z < u(X)\}$, put

$$p_s = \frac{9+s}{2}, \qquad G_s(q) = \int_0^q (1+\tau^2)^{-p_s}\, d\tau, \qquad \mathcal{G}_s(q) = \int_0^q G_s(\tau)\, d\tau.$$

The function $G_s$ is odd, strictly increasing, and bounded. If $u$ is $C^{1,1}$ near $X$, then

$$\mathcal{H}_s[E_u](X, u(X)) = 2\,\mathrm{PV} \int_{\mathbb{R}^8} G_s\!\left(\frac{u(X) - u(Y)}{|X - Y|}\right) \frac{dY}{|X - Y|^{8+s}} =: \mathcal{H}_s u(X). \tag{6.2}$$

The right-hand principal value deletes a ball in the base variables. For $d = |X - Y| > 0$ and $b = u(X) - u(Y)$, write the vertical integration variable as $u(X) + \tau$. Subtracting the odd function $\mathrm{sign}(\tau)$ against the even vertical kernel gives

$$\int_{\mathbb{R}} \frac{\mathrm{sign}(\tau + b)}{(d^2 + \tau^2)^{p_s}}\, d\tau = 2 \int_0^b (d^2 + \tau^2)^{-p_s}\, d\tau = 2 d^{-8-s} G_s(b/d).$$

The integral from 0 to $b$ is oriented.

To pass from the ambient principal value to this vertical calculation, compare the ambient ball deletion with the cylinder deletion $|X - Y| < \varepsilon$. The sign function of the tangent halfspace integrates to zero over the difference of the two deleted regions by reflection about $(X, u(X))$. On every column in the cylinder, the discrepancy between the two sign functions is supported

between the graph and its affine tangent, in a vertical interval of length at most $Cd^2$. Its absolute kernel integral over these columns is at most

$$C\int_0^{\varepsilon} r^7 r^2 r^{-9-s}\,dr = \frac{C}{1-s}\varepsilon^{1-s}.$$

This proves the equivalence of the deletions. With the cylinder deleted the kernel is integrable, so Fubini applies to the vertical calculation.

The weak form is

$$\langle \mathcal{H}_s u, \psi\rangle = \iint_{\mathbb{R}^8\times\mathbb{R}^8} G_s\left(\frac{u(X)-u(Y)}{|X-Y|}\right)\frac{\psi(X)-\psi(Y)}{|X-Y|^{8+s}}\,dX\,dY. \tag{6.3}$$

It is defined for every $\psi \in W^{s,1}(\mathbb{R}^8)$, with

$$|\langle \mathcal{H}_s u, \psi\rangle| \le \|G_s\|_\infty [\psi]_{W^{s,1}(\mathbb{R}^8)}. \tag{6.4}$$

For $\psi \in C_c^\infty$ and $u$ of class $C^{1,1}$ near its support, the weak form equals $\int \mathcal{H}_s u(X)\psi(X)\,dX$. With $|X-Y|>\varepsilon$ on both sides this follows by swapping $X, Y$ and using the oddness of $G_s$. On a uniform small ball around each point in $\operatorname{supp}\psi$, subtract the flux $G_s(-Du(X)\cdot(Y-X)/|Y-X|)$ from the pointwise integrand. Its integral vanishes on centered annuli. Taylor's formula for $u$ and $|G_s'| \le 1$ bound the difference by $C|Y-X|$ before multiplication by the base kernel. Its radial integral is $C\int_0^1 r^{-s}\,dr$, and boundedness of $G_s$ controls the tails. This justifies the pointwise limit uniformly on $\operatorname{supp}\psi$. The weak limit follows from (6.4) and absolute integrability.

A weak subsolution satisfies $\langle \mathcal{H}_s u, \psi\rangle \le 0$ for every nonnegative $\psi \in C_c^\infty$ of the open set in question. For a weak supersolution the inequality is reversed.

6.2. **An averaged-kernel inequality.** Use the chamber $D$ and the factor interchange $T$ from Section 1. A function invariant under $O(4)\times O(4)$ is determined by the radii $r=|x|$, $t=|y|$. Here oddness on the base means $u(TX) = -u(X)$. Let

$$\mathcal{D} = \{(r,t) : 0 \le r < t\}, \qquad d\mu(r,t) = |S^3|^2 r^3 t^3\,dr\,dt.$$

For $a=(r,t)$, $b=(\rho,\sigma)$ in $\mathcal{D}$ and $h\in\mathbb{R}$, introduce the normalized angular averages

$$K_{\mathrm{d}}(a,b;h) = \frac{1}{|S^3|^2}\int_{S^3\times S^3}\left(r^2+t^2+\rho^2+\sigma^2-2r\rho\omega_1-2t\sigma\eta_1+h^2\right)^{-p_s}\,d\omega\,d\eta, \tag{6.5}$$

$$K_{\mathrm{c}}(a,b;h) = K_{\mathrm{d}}((r,t),(\sigma,\rho);h). \tag{6.6}$$

The letters d and c stand for direct and crossed interactions. For $h=0$ the direct denominator is bounded below by $(r-\rho)^2+(t-\sigma)^2$, so the direct average is finite off the radial diagonal. That diagonal has zero $d\mu(a)d\mu(b)$ measure. The crossed denominator is bounded below by $(r-\sigma)^2+(t-\rho)^2>0$ on $\mathcal{D}^2$. Both kernels are even in $h$ and decrease as $|h|$ increases.

**Lemma 6.1** (Direct interactions dominate crossed interactions)**.** *For $a,b\in\mathcal{D}$ and all $h\in\mathbb{R}$,*

$$K_{\mathrm{d}}(a,b;h) \ge K_{\mathrm{c}}(a,b;h) > 0$$

*whenever the quantities are finite. The first inequality is strict when $r<t$ and $\rho<\sigma$.*

*Proof.* Apply [10, Proposition 2.4 and (1.10), arXiv v2] to the radial kernel $k_h(r) = (r^2+h^2)^{-p_s}$ in dimension $2m=8$. Its required convexity follows from

$$\frac{d^2}{du^2}k_h(\sqrt{u}) = p_s(p_s+1)(u+h^2)^{-p_s-2} > 0 \qquad (u>0).$$

Normalize Haar measure on $O(4)^2$ to have mass one. Its image under $(Q_1,Q_2)\mapsto(Q_1e_1,Q_2e_1)$ is the product of the uniform probability measures on $S^3$, giving the angular average over $S^3\times S^3$. The average depends only on the two pairs of radii, so it can be evaluated with representatives $(re_1,te_1)$ and $(\rho e_1,\sigma e_1)$. The direct and reflected averages are exactly (6.5) and (6.6). Simultaneous interchange of the two factors identifies the two choices of chamber. The strict kernel comparison in the cited theorem gives the conclusion for the finite averages. □

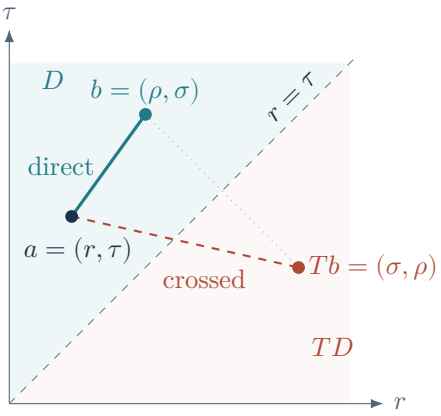



FIGURE 4. Direct and crossed interactions in the radial chamber. The comparison proof uses the angular averages of the geometric kernels. For nonnegative chamber values, those averages satisfy $K_{\mathrm{d}} \geq K_{\mathrm{c}} > 0$. This inequality permits comparison of odd functions even though their order reverses in the opposite chamber.

6.3. **The cylinder Jacobi operator and its height kernel.** The cylinder Jacobi operator separates into a base-cone term and an operator in the height variable. Put $\mathcal{C} = \Sigma \times \mathbb{R} \subset \mathbb{R}^9$ and keep the exterior normal $\nu$ of $D$ on the base cone. The normal to the cylinder is $(\nu, 0)$. At $(P, z) \in \mathcal{C}$, $P \neq 0$, define

$$\mathcal{J}_s f(P,z) = \mathrm{p.v.} \int_{\Sigma\times\mathbb{R}} \frac{f(Q,w) - f(P,z)}{(|P-Q|^2 + |w-z|^2)^{(9+s)/2}}\, d\sigma(Q)\, dw + \mathcal{A}_s(P) f(P,z),$$

where

$$\mathcal{A}_s(P) = \int_{\Sigma\times\mathbb{R}} \frac{1 - \nu(P)\cdot\nu(Q)}{(|P-Q|^2 + |w|^2)^{(9+s)/2}}\, d\sigma(Q)\, dw.$$

This uses the same Jacobi sign convention as for the base cone. For a smooth compactly supported normal displacement away from the singular line $\{0\} \times \mathbb{R}$, the first variation of exterior-minus-interior curvature is $-2\mathcal{J}_s f$. The calculation in Lemma 6.13 also identifies the linear term for the cutoff fields used below.

Set

$$c_s^{\mathrm{v}} = \int_{\mathbb{R}} (1+t^2)^{-(9+s)/2}\, dt.$$

Integrating in $w$ by $w = |P-Q|t$ gives the base potential kernel multiplied by $c_s^{\mathrm{v}}$. Thus the base scaling in (2.8) gives

$$\mathcal{A}_s(P) = c_s^{\mathrm{v}} A_s |P|^{-1-s}. \tag{6.7}$$

For $0 < \lambda < 7$, $\rho = |P|$, and $\tau \neq 0$, define

$$K_{\rho,\lambda,s}(\tau) = \int_{\Sigma} \frac{|Q|^{-\lambda}}{(|P-Q|^2 + \tau^2)^{(9+s)/2}}\, d\sigma(Q). \tag{6.8}$$

The group $O(4)\times O(4)$ acts transitively on the link of $\Sigma$ and preserves distance, surface measure, and $|Q|$. Consequently this positive even kernel depends on $P$ only through $\rho$.

**Lemma 6.2** (The height kernel at both scales)**.** *For the fixed $s \in (0,1)$ and every $0 < \lambda < 7$, there are positive constants $c, C$ such that*

$$c\rho^{-\lambda}|\tau|^{-2-s}(1+|\tau|/\rho)^{-\lambda} \leq K_{\rho,\lambda,s}(\tau) \leq C\rho^{-\lambda}|\tau|^{-2-s}(1+|\tau|/\rho)^{-\lambda}. \tag{6.9}$$

*Moreover*

$$K_{\rho,\lambda,s}(\tau) = \rho^{-\lambda-2-s} K_{1,\lambda,s}(\tau/\rho).$$

*Proof.* The scaling formula follows by writing $Q = \rho Q'$ in (6.8); the surface measure scales by $\rho^7$. It suffices to take $\rho = 1$.

For $0 < |\tau| \le 1$, use a fixed smooth tangent chart about $P$. On that chart $|Q|^{-\lambda}$ is bounded above and below, the surface measure is comparable to seven-dimensional Lebesgue measure, and $|P - Q|$ is comparable to the chart radius. The substitution $\xi = |\tau|\eta$ gives

$$\int_{|\xi|<c} (|\xi|^2 + \tau^2)^{-(9+s)/2}\, d\xi = |\tau|^{-2-s} \int_{|\eta|<c/|\tau|} (1 + |\eta|^2)^{-(9+s)/2}\, d\eta \asymp |\tau|^{-2-s}.$$

The last integral is bounded below by the integral over $|\eta| < c$ and above by the integral over $\mathbb{R}^7$. Off the chart the integral is bounded: the radial integral at the vertex is $\int_0^{1/2} r^{6-\lambda}\, dr$, and the integral at infinity is bounded by $C \int_2^\infty r^{-3-s-\lambda}\, dr$.

For $|\tau| \ge 1$, set $Q = |\tau|Q'$. The remaining integral is

$$\int_\Sigma |Q'|^{-\lambda} (|P/|\tau| - Q'|^2 + 1)^{-(9+s)/2}\, d\sigma(Q').$$

On $|Q'| \le 2$, the denominator is at least one and $|Q'|^{-\lambda}$ is integrable. On $|Q'| > 2$, $|P/|\tau| - Q'| \ge |Q'|/2$, since $|P| = 1$. This gives the uniform upper bound. Restricting the integral to $2 < |Q'| < 3$, where the denominator is bounded above, gives the uniform positive lower bound. The factor removed by the substitution is $|\tau|^{-2-s-\lambda}$. These two ranges prove the lemma. □

Whenever the paired integral converges, put

$$\mathcal{T}_{\rho,\lambda,s} g(z) = \int_0^\infty [g(z+\tau) + g(z-\tau) - 2g(z)] K_{\rho,\lambda,s}(\tau)\, d\tau.$$

This equals the one-dimensional principal value $\mathrm{p.v.} \int_{\mathbb{R}} [g(z+\tau) - g(z)] K_{\rho,\lambda,s}(\tau)\, d\tau$ under the next hypotheses.

**Theorem 6.3** (Exact separation in the height variable)**.** *Let $0 < \lambda < 7$ and $\rho = |P|$. Suppose $g$ is measurable and locally bounded on $\mathbb{R}$, is $C^2$ in a neighborhood of the evaluation height $z$, and satisfies $|g(w)| \le C(1 + |w|^a)$ for every $w \in \mathbb{R}$, for some $0 \le a < 1 + s + \lambda$. Then*

$$\begin{aligned} \mathcal{J}_s(|P|^{-\lambda} g(z)) = c_s^{\mathrm{v}} (A_s - H_s(\lambda)) \rho^{-\lambda-1-s} g(z) \\ + \mathcal{T}_{\rho,\lambda,s} g(z). \end{aligned} \tag{6.10}$$

*In particular, at the critical exponent $\lambda = \gamma$,*

$$\mathcal{J}_s(|P|^{-\gamma} g(z)) = \mathcal{T}_{\rho,\gamma,s} g(z). \tag{6.11}$$

*Proof.* In the cylinder principal value put $\tau = w - z$ and split the numerator as

$$|Q|^{-\lambda} g(z+\tau) - \rho^{-\lambda} g(z) = g(z)(|Q|^{-\lambda} - \rho^{-\lambda}) + |Q|^{-\lambda}[g(z+\tau) - g(z)].$$

The Euclidean deletion $|P - Q|^2 + \tau^2 \le \varepsilon^2$ is invariant under $\tau \mapsto -\tau$ at each $Q$, so the second term can be paired before taking the limit. Near $\tau = 0$ the paired numerator in the height variable is $O(\tau^2)$. After integration in $Q$, (6.9) gives the majorants $C\tau^{-s}$ near zero and $C(1 + \tau^a)\tau^{-2-s-\lambda}$ at infinity. They are integrable by the assumptions. Dominated convergence gives precisely $\mathcal{T}_{\rho,\lambda,s} g(z)$ for this term.

For the first term, we may replace the Euclidean deletion by $|P - Q| \le \varepsilon$. In fact, in the tangent chart used for (2.8), write $Q = P + \xi + b(\xi)\nu(P)$ with $b(\xi) = O(|\xi|^2)$. The area density is $1 + O(|\xi|^2)$ and $|Q|^{-\lambda} - \rho^{-\lambda} = L(\xi) + O(|\xi|^2)$, with $L$ linear. With the flat kernel $(|\xi|^2 + \tau^2)^{-(9+s)/2}$, the $L(\xi)$ term integrates to zero on a centered product chart under either flat deletion, by $\xi \mapsto -\xi$. The change in the kernel and area density has relative size $O(|\xi|^2)$. After subtracting the linear term the local integrals are therefore absolutely convergent, with radial bound $Cr^{-s}\, dr$ in the eight tangent variables.

The difference between the curved and flat product-ball deletions lies in a shell of thickness $O(\varepsilon^3)$, since $|P - Q|^2 = |\xi|^2 + O(|\xi|^4)$. The shell has volume $O(\varepsilon^{10})$, and the kernel times numerator is $O(\varepsilon^{-8-s})$ there. For the horizontal deletion the shell in the seven base variables has volume $O(\varepsilon^9)$. Integrating first in $\tau$ bounds its kernel by $C|\xi|^{-8-s}$. Both shell errors are $O(\varepsilon^{2-s})$.

Off the local product chart the first term is absolutely integrable. For $Q$ away from the base chart this follows by vertical integration and $\lambda < 7$ at the cone vertex. For $Q$ near $P$ and $\tau$ separated from zero it follows directly. Thus the two curved deletions have the same limit.

With the horizontal deletion in place, integration in $\tau$ gives $c_s^{\mathrm{v}}|P-Q|^{-8-s}$. By (6.7), the first term and the potential together give $c_s^{\mathrm{v}} g(z) J_s(\rho^{-\lambda})(P)$. Use (2.9) to obtain (6.10), and $H_s(\gamma) = A_s$ for (6.11). □

**Proposition 6.4** (Odd convex functions have the useful sign)**.** *Suppose $g$ is odd, $C^2$, nonnegative on $(0,\infty)$, and convex there, and satisfies the growth assumption in Theorem 6.3. Then, for $z > 0$,*

$$\mathcal{T}_{\rho,\lambda,s} g(z) \geq 0.$$

*If $g$ is strictly convex on $(0,\infty)$, the inequality is strict. If $g$ satisfies the growth hypothesis also at $\lambda = \gamma$, the normal displacement $-a|P|^{-\gamma} g(z)$, $a > 0$, has nonnegative linearized exterior-minus-interior curvature in the positive-height region.*

*Proof.* For $0 < \tau \leq z$ the paired difference in the definition of $\mathcal{T}$ is nonnegative by convexity, and is positive if $g$ is strictly convex. For $\tau > z$ oddness gives the expression $g(\tau+z) - g(\tau-z) - 2g(z)$. Since $g'$ is nondecreasing,

$$g(\tau+z) - g(\tau-z) = \int_0^{2z} g'(\tau - z + t)\, dt \geq g(2z) - g(0) \geq 2g(z).$$

The last inequality uses convexity and $g(0) = 0$, and is strict under strict convexity. Integrating against the positive kernel proves the sign. The final assertion follows from the linearization $-2\mathcal{J}_s$ and (6.11). □

6.4. **A monotonicity formula for the height kernel.**

**Theorem 6.5** (Monotonicity after removing the singular power)**.** *If*

$$1 < \lambda < 4 - s,$$

*then $\tau^{2+s} K_{\rho,\lambda,s}(\tau)$ is strictly decreasing as a function of $\tau > 0$, for each $\rho > 0$. In particular, this statement applies to both $\lambda = \gamma$ and $\lambda = \kappa = 5/2$.*

*Proof.* Use the Pochhammer notation from Section 3. The series ${}_3F_2(A, B, C; d, e; q)$ has coefficient $(A)_j (B)_j (C)_j / [(d)_j (e)_j j!]$ at $q^j$.

By the rotational symmetries, we may take

$$P = \frac{\rho}{\sqrt{2}}(e_1, e_1), \qquad Q = \frac{r}{\sqrt{2}}(\xi, \zeta), \qquad \xi, \zeta \in S^3.$$

Then

$$d\sigma(Q) = 2^{-3} r^6\, dr\, d\sigma(\xi)\, d\sigma(\zeta), \qquad |P-Q|^2 = r^2 + \rho^2 - \rho r(\xi_1 + \zeta_1).$$

Let $\mathbb{E}$ denote the average over the two unit spheres. The coordinate moments of a uniform point of $S^3$ are

$$\mathbb{E}\xi_1^{2j} = \frac{(1/2)_j}{(2)_j}, \qquad \mathbb{E}\xi_1^{2j+1} = 0.$$

These are the same moments used after (3.19). Expanding a power of $\xi_1 + \zeta_1$ gives

$$\frac{\mathbb{E}(\xi_1 + \zeta_1)^{2j}}{(2j)!} = 4^{-j} \sum_{k=0}^{j} \frac{1}{k!(k+1)!(j-k)!(j-k+1)!} = \frac{(3/2)_j}{j!(2)_j (3)_j}. \tag{6.12}$$

For the first equality, divide the binomial expansion by $(2j)!$ and use $\mathbb{E}\xi_1^{2k}/(2k)! = 4^{-k}/[k!(k+1)!]$. For the second equality, multiplying the sum by $j!(j+2)!$ gives $\sum_{k=0}^{j} \binom{j}{k}\binom{j+2}{j+1-k} = \binom{2j+2}{j+1}$, the coefficient of $t^{j+1}$ in $(1+t)^j (1+t)^{j+2}$. Now $4^{-j}\binom{2j+2}{j+1} = 2(3/2)_j/(j+1)!$ and $(3)_j = (j+2)!/2$, yielding the stated coefficient.

Set

$$A = \frac{7-\lambda}{2}, \qquad B = \frac{2+s+\lambda}{2}, \qquad c = \frac{2+s}{2}, \qquad q = \frac{\rho^2}{\rho^2 + \tau^2}.$$

Put $U=\rho^2+\tau^2$. The binomial expansion of the kernel in $\rho r(\xi_1+\zeta_1)/(r^2+U)$ is legitimate, since

$$\left|\frac{\rho r(\xi_1+\zeta_1)}{r^2+U}\right| \le \frac{2\rho r}{r^2+U} \le \frac{\rho}{\sqrt{U}} < 1.$$

Pairing its two signs removes the odd powers and leaves nonnegative terms, so Tonelli applies to the angular and radial integrals. The radial integral in the $2j$-th term is, by $r^2=Uv$,

$$\int_0^\infty \frac{r^{6-\lambda+2j}}{(r^2+U)^{p_s+2j}}\,dr = \frac{1}{2}U^{-B-j}\mathrm{B}(A+j,B+j).$$

Since $p_s=A+B$, $(p_s)_{2j}\mathrm{B}(A+j,B+j)=\mathrm{B}(A,B)(A)_j(B)_j$. Combining this identity with (6.12) and $\rho^{2j}U^{-j}=q^j$ gives

$$K_{\rho,\lambda,s}(\tau)=C_{\lambda,s}(\rho^2+\tau^2)^{-B}{}_3F_2(A,B,3/2;2,3;q), \qquad C_{\lambda,s}=\frac{|S^3|^2}{16}\,\mathrm{B}(A,B)>0. \tag{6.13}$$

The beta average of the ${}_2F_1$ series gives

$${}_3F_2(A,B,3/2;2,3;q)=\frac{1}{\mathrm{B}(3/2,1/2)}\int_0^1 t^{1/2}(1-t)^{-1/2}{}_2F_1(A,B;3;qt)\,dt.$$

Integrating the $j$-th term contributes $(3/2)_j/(2)_j$ by the beta integral. Applying Euler's transformation in the form used after (3.19) gives

$${}_2F_1(A,B;3;x)=(1-x)^{-c-1/2}{}_2F_1\left(\frac{\lambda-1}{2},\frac{4-s-\lambda}{2};3;x\right).$$

In the beta integral, make the substitution

$$t=\frac{1-v}{1-qv}.$$

Here $1-t=v(1-q)/(1-qv)$, $1-qt=(1-q)/(1-qv)$, and $|dt/dv|=(1-q)/(1-qv)^2$. Thus the transformed weight is $(1-q)^{-c}v^{-1/2}(1-v)^{1/2}(1-qv)^{c-3/2}$, and

$$\begin{aligned}(1-q)^c{}_3F_2(A,B,3/2;2,3;q)&\\ =\frac{1}{\mathrm{B}(3/2,1/2)}\int_0^1 v^{-1/2}(1-v)^{1/2}(1-qv)^{-(1-s)/2}&\\ \cdot{}_2F_1\left(\frac{\lambda-1}{2},\frac{4-s-\lambda}{2};3;\frac{q(1-v)}{1-qv}\right)dv.&\end{aligned} \tag{6.14}$$

Both transformed upper parameters are strictly positive under the hypotheses of the theorem. Each coefficient of the displayed ${}_2F_1$ is therefore positive. For each $v\in(0,1)$, both $(1-qv)^{-(1-s)/2}$ and $q(1-v)/(1-qv)$ increase with $q$. Consequently the right side of (6.14) is nondecreasing in $q$.

Finally, (6.13) gives

$$\tau^{2+s}K_{\rho,\lambda,s}(\tau)=C_{\lambda,s}\rho^{-\lambda}q^{\lambda/2}(1-q)^c{}_3F_2(A,B,3/2;2,3;q).$$

The factor $q^{\lambda/2}$ is strictly increasing, and the other positive factor has just been proved nondecreasing. Since $q$ strictly decreases with $\tau$, the required strict monotonicity follows.

Lemma 5.5 gives $1<\gamma<\kappa=5/2<4-s$ for $0<s<1$. Both exponents therefore satisfy the theorem's hypotheses. □

**Lemma 6.6** (A one-sided power has negative height curvature)**.** *Let $1<\lambda<4-s$, $0<\alpha<(1+s)/2$, and $v_a(w)=(w+a)_+^\alpha$. For $z+a>0$,*

$$\mathcal{T}_{\rho,\lambda,s}v_a(z)\le -c_{\alpha,\lambda,s}(z+a)^{\alpha-1-s}(\rho+z+a)^{-\lambda}<0.$$

*The definition of $\mathcal{T}$ uses the paired integral; the possible nonsmoothness at $w=-a$ is separated from the evaluation point and causes no principal-value issue.*

*Proof.* Put $x = z + a > 0$ and $q_0 = 1 + s \in (1, 2)$. The paired second difference is

$$D_x(\tau) = (x + \tau)^\alpha + (x - \tau)_+^\alpha - 2x^\alpha.$$

It is negative for $0 < \tau < \tau_* = (2^{1/\alpha} - 1)x$, and positive for $\tau > \tau_*$. Indeed, for $\tau \le x$ this follows from strict concavity, and for $\tau > x$ the bracket is $(x + \tau)^\alpha - 2x^\alpha$.

The unweighted fractional integral has a strictly negative constant:

$$I_{\alpha,q_0} = \int_0^\infty [(1 + t)^\alpha + (1 - t)_+^\alpha - 2]t^{-1-q_0}\, dt < 0.$$

Integrate by parts on $t > \varepsilon$, noting that the second difference is continuous at $t = 1$ with an integrable derivative there. In the first derivative term, $t^{-q_0}(1+t)^{\alpha-1}\, dt = u^{-q_0}(1-u)^{q_0-\alpha-1}\, du$ under $u = t/(1 + t)$. The derivative term from $(1 - t)_+^\alpha$ already runs over $(\varepsilon, 1)$. Taking the limit gives

$$I_{\alpha,q_0} = \frac{\alpha}{q_0} \int_0^1 u^{-q_0} \left[(1 - u)^{q_0-\alpha-1} - (1 - u)^{\alpha-1}\right] du. \tag{6.15}$$

The boundary terms vanish because the original second difference is $O(t^2)$ at zero and $O(t^\alpha + 1)$ at infinity. The lower cutoffs in the derivative terms are $\varepsilon/(1 + \varepsilon)$ and $\varepsilon$. They differ by $O(\varepsilon^2)$, which costs $O(\varepsilon^{2-q_0})$ because the derivative integrands are $O(\varepsilon^{-q_0})$ on this interval. This justifies combining the two individually divergent derivative integrals. The integrand in (6.15) is integrable: at zero its cancellation leaves $O(u^{1-q_0})$, and at one both exponents exceed $-1$. It is strictly negative because $q_0 - \alpha > \alpha$.

Let $m(\tau) = \tau^{2+s} K_{\rho,\lambda,s}(\tau)$. By Theorem 6.5, $m$ decreases. On each side of the single sign change of $D_x$, one has

$$D_x(\tau)m(\tau) \le D_x(\tau)m(\tau_*).$$

The integrals converge absolutely after pairing, so

$$\mathcal{T}_{\rho,\lambda,s} v_a(z) \le m(\tau_*) x^{\alpha-1-s} I_{\alpha,1+s}.$$

The lower kernel bound (6.9) gives $m(\tau_*) \ge c(\rho + \tau_*)^{-\lambda} \ge c_\alpha(\rho + x)^{-\lambda}$. Since the remaining constant is negative, this proves the stated upper bound. □

**Corollary 6.7** (The stronger bounded-height margin)**.** *Under the assumptions of Lemma 6.6, let* $v(w) = (w + 1)_+^\alpha$. *Uniformly for* $0 \le z \le 1$,

$$-\mathcal{T}_{\rho,\lambda,s} v(z) \ge c_{\alpha,\lambda,s} \rho^{-\lambda} \min\{\rho, 1\}^{1-s}, \qquad \rho > 0.$$

*Proof.* For $\rho \ge 1$, the preceding lemma gives the result directly, since $1 \le z + 1 \le 2$. For small $\rho$, Taylor's formula and the uniform bound $v''(w) \le -c_\alpha$ on $[-1/2, 3/2]$ give

$$v(z + \tau) + v(z - \tau) - 2v(z) \le -c_\alpha \tau^2 \qquad (0 < \tau < 1/2).$$

For $0 < \rho < 1/2$, the kernel bound gives $K_{\rho,\lambda,s}(\tau) \ge c\rho^{-\lambda}\tau^{-2-s}$ on $0 < \tau < \rho$. Integration of the negative difference there contributes at most $-c_{\alpha,\lambda,s}\rho^{1-s-\lambda}$. All positive second differences are confined to $\tau \ge 2^{1/\alpha} - 1 > 1$. Their total integral is bounded above by a constant independent of $0 < \rho \le 1$ and $0 \le z \le 1$, by the upper kernel bound and the growth $v(w) \le C(1 + w_+)^\alpha$. Because $1 - s - \lambda < 0$, the negative local term dominates this constant for $0 < \rho \le \rho_0$, with one fixed $\rho_0 > 0$. On the remaining compact interval $\rho_0 \le \rho \le 1$, Lemma 6.6 gives a uniform strictly negative bound. Decreasing the final constant proves the result on that interval as well. □

6.5. **Two concave powers with a small linear term.** Choose exponents satisfying

$$0 < \alpha_0 < \frac{1 + s}{2(1 + \eta)}, \qquad \max\{\alpha_0, 0, 1 - \gamma\eta\} < \alpha_\infty < \frac{1}{1 + \eta}. \tag{6.16}$$

These choices are possible. The first interval is nonempty; its upper endpoint is strictly smaller than $1/(1 + \eta)$. Also $\gamma > 1$ gives $1 - \gamma\eta < 1/(1 + \eta)$. For positive constants $a, c$, to be chosen sufficiently small, define on the whole real line

$$\tilde{g}(w) = (w + 1)_+^{\alpha_0} + c(w + 1)_+^{\alpha_\infty} + a(w + 1)_+, \qquad \tilde{h}(w) = \tilde{g}(w)^{1+\eta}. \tag{6.17}$$

For $w > 0$, put $g_-(w) = \tilde{g}(w)$, and extend $g_-$ oddly with its jump at zero. Denote by $h_-$ the odd extension of $g_-^{1+\eta}$.

The reference functions vanish for $w \le -1$. Their corners at $-1$ stay a positive distance from every evaluation height considered below. They are smooth and strictly positive on a neighborhood of $[0,1]$.

**Proposition 6.8** (A strict reference margin and the second mode)**.** *The constants $a, c > 0$ can be chosen so that, uniformly for $0 \le z \le 1$ and every $\rho > 0$,*

$$Q(\rho, z) := -\mathcal{T}_{\rho,\gamma,s}\tilde{g}(z) \ge c_s \rho^{-\gamma} \min\{\rho, 1\}^{1-s} > 0, \tag{6.18}$$

$$\left(\mathcal{T}_{\rho,\kappa,s}\tilde{h}(z)\right)_+ \le C_s Q(\rho, z). \tag{6.19}$$

*In addition,*

$$g_-' > 0, \quad g_-'' < 0 \quad \text{on } (0,\infty), \qquad h_-'' \le 0 \quad \text{on } (0,1],$$

*and*

$$(h_-''(t))_+ \le C_s t^{p\eta}(-g_-''(t)) \qquad (t > 0). \tag{6.20}$$

*Proof.* Write $v_\alpha(w) = (w+1)_+^\alpha$, including $v_1(w) = (w+1)_+$. Corollary 6.7, used with $\lambda = \gamma$, gives

$$-\mathcal{T}_{\rho,\gamma,s}v_{\alpha_0}(z) \ge c_s A_\gamma(\rho), \qquad A_\gamma(\rho) = \rho^{-\gamma}\min\{\rho,1\}^{1-s}.$$

For any $0 < \alpha \le 1$, the positive part of the paired second difference of $v_\alpha$ at $z \in [0,1]$ is supported on $\tau \ge 1$, and is at most $C_\alpha(1+\tau)^\alpha$ there. For $0 < \alpha < 1$ this follows from the single-sign-change calculation in Lemma 6.6; for $\alpha = 1$ the difference is zero up to $\tau = z+1$ and then equals $\tau - (z+1)$. Hence

$$\begin{aligned}\left(\mathcal{T}_{\rho,\gamma,s}v_\alpha(z)\right)_+ &\le C_s \int_1^\infty \tau^{\alpha-2-s}(\rho+\tau)^{-\gamma}\,d\tau \\ &\le C_s(1+\rho)^{-\gamma}.\end{aligned}$$

The last integral is finite because $\alpha < 1+s$. Since $A_\gamma(\rho) \ge (1+\rho)^{-\gamma}$, choosing $a, c$ sufficiently small proves (6.18).

We can decrease $a, c$ further so that

$$\tilde{g}'' \le -c_0 < 0, \qquad \tilde{h}'' \le -c_0 < 0 \quad \text{on } [-1/4, 5/4].$$

Indeed, as $a, c \downarrow 0$, the functions converge in $C^2$ on this interval to $(w+1)^{\alpha_0}$ and $(w+1)^{\alpha_0(1+\eta)}$, respectively. Both limiting exponents lie strictly between zero and one, so both limiting second derivatives are uniformly negative on this compact interval.

Choose $r_0 \in (0, 1/4)$. At $0 \le z \le 1$, the paired second difference of $\tilde{h}$ is at most $-c_0\tau^2$ for $0 < \tau < r_0$. For larger $\tau$ its positive part is at most $C_s(1+\tau)^{1+\eta}$. When $0 < \rho \le r_0$, the negative local integral is at most $-c_s\rho^{1-s-\kappa}$. The positive part over $\tau \ge r_0$ is bounded independently of $\rho \le 1$; at infinity its majorant is $C_s\tau^{\eta-1-s-\kappa}$, which is integrable since $\eta < 1 < \kappa + s$. It follows that $\mathcal{T}_{\rho,\kappa,s}\tilde{h}(z) \le 0$ for $0 < \rho \le \rho_0$, with one fixed $\rho_0 > 0$.

For $\rho \ge 1$, the positive part of the integral on $[r_0, 1]$ is $O_s(\rho^{-\kappa})$. On $[1, \rho]$ it is at most $C_s\rho^{-\kappa}\int_1^\rho \tau^{\eta-1-s}\,d\tau$, and on $[\rho,\infty)$ it is at most $C_s\rho^{\eta-s-\kappa}$. In particular,

$$\left(\mathcal{T}_{\rho,\kappa,s}\tilde{h}(z)\right)_+ \le C_s\rho^{-\kappa}\begin{cases}1, & \eta < s,\\ 1+\log\rho, & \eta = s,\\ \rho^{\eta-s}, & \eta > s.\end{cases}$$

Each displayed bound is at most $C_s\rho^{-\gamma}$. For $\eta > s$, this uses

$$\kappa - \gamma - \eta + s = \gamma\eta + s > 0.$$

For $\eta = s$, the logarithm is absorbed by the strictly positive power $\kappa - \gamma$. On the remaining compact range $\rho_0 \le \rho \le 1$, the positive integral is bounded and (6.18) gives a fixed lower bound for $Q$. This proves (6.19).

The monotonicity and concavity of $g_-$ on the positive axis follow by differentiating its three summands. The preceding compact-interval choice gives $h_-'' \le 0$ on $(0,1]$. For $t \ge 1$, the small coefficient $a$ is positive and fixed, so

$$g_-(t) \asymp_s 1+t, \qquad |g_-'(t)| \le C_s, \qquad -g_-''(t) \ge c_s(1+t)^{\alpha_\infty - 2}.$$

The last inequality uses $c > 0$ and $\alpha_\infty > \alpha_0$. Since

$$h_-'' = (1+\eta)g_-^{\eta-1}[\eta(g_-')^2 + g_- g_-''],$$

the negative second summand can be discarded in estimating the positive part. Thus

$$(h_-''(t))_+ \le C_s(1+t)^{\eta-1}.$$

Dividing by the lower bound for $-g_-''$, and using $\eta + 1 - \alpha_\infty < p\eta$, proves (6.20) for $t \ge 1$; it is immediate for $0 < t \le 1$. □

**Lemma 6.9** (Height margins for the actual odd lower scalar)**.** *For the scalar $g_-$ just chosen, let $Q_-(\rho, z) = -\mathcal{T}_{\rho,\gamma,s} g_-(z)$. For every $\rho, z > 0$,*

$$Q_-(\rho, z) \ge c_s z^{-1-s}(\rho + z)^{-\gamma}. \tag{6.21}$$

*It also satisfies $Q_-(\rho, z) \ge c_s(1+z)^{\alpha_\infty - 1 - s}(\rho + 1 + z)^{-\gamma}$.*

*Proof.* Put $g_0 = g_-(0+) = 1 + c + a > 0$. Concavity gives a nonpositive paired second difference for $0 < \tau < z$. The value at $\tau = z$ is immaterial. For $\tau > z$, oddness and concavity give

$$\begin{aligned} g_-(z+\tau) + g_-(z-\tau) - 2g_-(z) &= g_-(\tau+z) - g_-(\tau - z) - 2g_-(z) \\ &\le g_-(2z) - g_0 - 2g_-(z) \le -2g_0. \end{aligned}$$

Thus all paired differences are nonpositive. On $z < \tau < 2z$, $K_{\rho,\gamma,s}(\tau) \ge c_s z^{-2-s}(\rho + z)^{-\gamma}$. Integration over this interval proves (6.21).

For $z \ge 1$, on $z/2 < t < 3z/2$ one has $-g_-''(t) \ge c_s z^{\alpha_\infty - 2}$. The integral form of the second difference therefore gives

$$-[g_-(z+\tau) + g_-(z-\tau) - 2g_-(z)] \ge c_s z^{\alpha_\infty - 2}\tau^2 \quad (0 < \tau < z/2).$$

Integrating over $z/4 < \tau < z/2$ proves $Q_- \ge c_s z^{\alpha_\infty - 1 - s}(\rho + z)^{-\gamma}$. For $0 < z \le 1$, use the jump estimate on $1 < \tau < 2$, which instead gives $Q_- \ge c_s(\rho + 1)^{-\gamma}$. These two ranges prove the second assertion. □

6.6. **The upper scalar and domination of the second mode.** Choose

$$0 < \delta < \min\{1, \gamma\eta\}, \qquad q = 12, \qquad k = 11/\delta, \qquad g_+(z) = az(1 + z^{-\delta})^{-k} \quad (z > 0),$$

where $a > 0$ is the coefficient fixed in Proposition 6.8. Extend $g_+$ oddly and continuously, and let $h_+$ be the odd extension of $g_+^{1+\eta}$. The lower functions $g_-, h_-$ retain their odd extensions with positive right-hand limits at zero. Values at zero have no effect on any of the integrals. We have

$$0 < g_+(z) < az < g_-(z) \qquad (z > 0). \tag{6.22}$$

In addition to $Q_-$ from Lemma 6.9, write $Q_+(\rho, z) = \mathcal{T}_{\rho,\gamma,s} g_+(z)$.

For later use, the scalar sizes and derivatives are

$$\begin{gathered} g_+(z) \asymp_s z^q \quad (0 < z \le 1), \qquad g_+(z) \asymp_s z \quad (z \ge 1), \qquad g_-(z) \asymp_s 1 + z, \\ |g_+^{(j)}(z)| \le C_{s,j} g_+(z) z^{-j}, \qquad |g_-^{(j)}(z)| \le C_{s,j}(1+z)^{1-j}, \\ |h_\pm^{(j)}(z)| \le C_{s,j} h_\pm(z) z^{-j} \quad (z > 0). \end{gathered} \tag{6.23}$$

For $g_+$, write $g_+ = az^q(1 + z^\delta)^{-k}$ on $z \le 1$ and use its original expression on $z \ge 1$. In each expression the argument $u = z^{\pm\delta}$ satisfies $0 < u \le 1$ and $|u^{(j)}(z)| \le C_j z^{-j}$. The derivatives of $(1+u)^{-k}$ are bounded on this range, and the factor itself is bounded below. The product and chain rules give the displayed bounds. The same calculation applies to the power $h_+ = g_+^{1+\eta}$, multiplying all its exponents by $1 + \eta$. For $g_-$, each summand is a power of $1 + z$ of exponent at most one, and the linear term yields its lower bound. Differentiating $h_- = g_-^{1+\eta}$ then gives

$|h_-^{(j)}(z)| \le C_{s,j}(1+z)^{1+\eta-j}$, which implies (6.23). The upper odd extension is $C^{11}$ because the derivatives of orders $0 \le j \le 11$ are $O(z^{12-j})$ at zero. The lower scalar derivatives here are taken on the positive axis.

With $u = t^{-\delta}$ for $t > 0$, differentiation gives

$$g_+'(t) = a\frac{1+qu}{(1+u)^{k+1}}, \qquad g_+''(t) = a\frac{k\delta u}{t(1+u)^{k+2}}[(1-\delta)+qu] > 0. \tag{6.24}$$

Thus Proposition 6.4 applies to $g_+$: the paired difference is positive for every $z, \tau > 0$. We need its quantitative form

$$Q_+(\rho, z) \ge c_s z(1+z)^{-1-s-\delta}(\rho+1+z)^{-\gamma}. \tag{6.25}$$

For $0 < z \le 1$ and $2 \le \tau \le 4$, oddness and monotonicity of $g_+'$ give

$$\begin{aligned} g_+(z+\tau) + g_+(z-\tau) - 2g_+(z) &= \int_{\tau-z}^{\tau+z} g_+'(w)\,dw - 2g_+(z) \\ &\ge 2z[g_+'(1) - g_+(1)] > 0. \end{aligned}$$

Here $\tau - z \ge 1$, $g_+(z) \le zg_+(1)$ by convexity, and $g_+(1) = \int_0^1 g_+'(t)\,dt < g_+'(1)$. Integration over $2 \le \tau \le 4$ gives $Q_+ \ge c_s z(\rho+1)^{-\gamma}$. For $z \ge 1$ and $2z \le \tau \le 4z$, the increment comparison in Proposition 6.4 bounds the paired difference below by

$$g_+(2z) - 2g_+(z) = 2az\big[(1+2^{-\delta}z^{-\delta})^{-k} - (1+z^{-\delta})^{-k}\big] \ge c_s z^{1-\delta}.$$

The last bound follows from the mean-value theorem, since $|((1+u)^{-k})'| \ge k2^{-k-1}$ on $[0,1]$. On this increment interval $K_{\rho,\gamma,s}(\tau) \ge c_s z^{-2-s}(\rho+z)^{-\gamma}$. Thus $Q_+ \ge c_s z^{-s-\delta}(\rho+z)^{-\gamma}$, which completes (6.25).

**Lemma 6.10** (Second differences, including an odd jump)**.** *Let $v \in C^2([0,\infty))$, with derivatives at zero taken from the right, and put $v_0 = v(0+)$. Reflect $v$ oddly across zero away from the origin. Its value at zero is immaterial. For $z, \tau > 0$, $z \ne \tau$,*

$$\begin{aligned} D_v(z,\tau) &:= v(z+\tau) + v(z-\tau) - 2v(z) \\ &= -2v_0\mathbf{1}_{\{\tau>z\}} + \int_0^{z+\tau} v''(t)\Psi_{z,\tau}(t)\,dt, \end{aligned} \tag{6.26}$$

*where*

$$\Psi_{z,\tau}(t) = (z+\tau-t)_+ + \operatorname{sign}(z-\tau)(|z-\tau|-t)_+ - 2(z-t)_+ \ge 0.$$

*For the scalars above, with $d = \kappa - \gamma = p\eta$,*

$$D_{h_+}(z,\tau) \le C_s(z+\tau)^d D_{g_+}(z,\tau), \tag{6.27}$$

$$D_{h_-}(z,\tau) \le C_s(z+\tau)^d[-D_{g_-}(z,\tau)]. \tag{6.28}$$

*Moreover,*

$$\sup_{z/2<t<3z/2} (h_\pm''(t))_+ \le C_s h_\pm(z) z^{-2}. \tag{6.29}$$

*Proof.* For $x \ge 0$, write $v(x) = v_0 + v'(0)x + \int_0^x (x-t)v''(t)\,dt$. Substituting this identity at the three arguments gives (6.26); the linear terms cancel. If $\tau \le z$, $\Psi$ is the usual triangular weight. If $\tau > z$, the difference of its first two terms is $2z$ on $t \le \tau - z$, and $z+\tau-t$ on $\tau - z < t < z+\tau$. Subtracting $2(z-t)_+$ leaves a nonnegative function. This proves the representation and its sign.

For the upper scalar, (6.24) with $u = t^{-\delta}$ gives

$$\frac{(g_+')^2}{g_+g_+''} = \frac{(1+qu)^2}{k\delta u\,[(1-\delta)+qu]}.$$

This is $O_s(1)$ when $u \geq 1$ and $O_s(u^{-1})$ when $u \leq 1$. Using (6.23) in the differentiation of $h_+ = g_+^{1+\eta}$ gives

$$\frac{h_+''}{g_+''} = (1+\eta)g_+^{\eta}\left(1+\eta\frac{(g_+')^2}{g_+g_+''}\right) \leq C_s\begin{cases} t^{q\eta}, & t \leq 1, \\ t^{\eta+\delta}, & t \geq 1. \end{cases}$$

Because $q > p$ and $\delta < \gamma\eta$, this is at most $C_s t^d$. Since $h_+(0) = g_+(0) = 0$, integration against $\Psi$ proves (6.27). For the lower scalar use (6.20). In (6.26), the jump contribution to $D_{h_-}$ is negative. Discard it and the negative part of $h_-''$, and compare the remaining integral with $\int(-g_-'')\Psi \leq -D_{g_-}$. This proves (6.28); Lemma 6.9 already gives $Q_- > 0$. For the local bound, (6.23) gives $h_\pm(t) \asymp_s h_\pm(z)$ when $z/2 < t < 3z/2$, by the size bounds on $g_\pm$. Applying the second derivative estimate there gives (6.29). □

**Proposition 6.11** (Domination of the adverse height term)**.** *For each $\vartheta > 0$, there is a finite $C_{s,\vartheta}$ such that, for every $\rho, z > 0$,*

$$\mathcal{T}_{\rho,\kappa,s}h_\pm(z) \leq C_{s,\vartheta}Q_\pm(\rho,z) + \vartheta c_s^{\mathrm{v}}\Delta_\kappa\rho^{-\kappa-1-s}h_\pm(z). \tag{6.30}$$

*Proof.* First fix $K \geq 2$. If $z \leq K\rho$, then $z+\tau \leq (K+1)(\rho+\tau)$. By (6.27), (6.28), and (6.9),

$$D_{h_\pm}(z,\tau)K_{\rho,\kappa,s}(\tau) \leq C_{s,K}|D_{g_\pm}(z,\tau)|K_{\rho,\gamma,s}(\tau).$$

For a positive left-hand side, the factor $(z+\tau)^{\kappa-\gamma}$ is bounded by $C_K(\rho+\tau)^{\kappa-\gamma}$, and $K_{\rho,\kappa,s}/K_{\rho,\gamma,s} \leq C_s(\rho+\tau)^{-(\kappa-\gamma)}$. The comparison also holds for a nonpositive left-hand side. Integrating it bounds the height term by $C_{s,K}Q_\pm$, since $D_{g_+} \geq 0$ and $D_{g_-} \leq 0$ almost everywhere.

If $z > K\rho$, the same comparison applies to $\tau \geq z/2$, since $z+\tau \leq 3\tau$. On $0 < \tau < z/2$ the interval of integration in

$$D_{h_\pm}(z,\tau) = \int_{-\tau}^{\tau}(\tau - |v|)h_\pm''(z+v)\,dv$$

lies in $(z/2, 3z/2)$. Using (6.29) and $\int_{-\tau}^{\tau}(\tau-|v|)\,dv = \tau^2$ gives

$$\begin{aligned}\int_0^{z/2} D_{h_\pm}(z,\tau)K_{\rho,\kappa,s}(\tau)\,d\tau &\leq \frac{C_s h_\pm(z)}{z^2}\int_0^\infty \tau^{-s}(\rho+\tau)^{-\kappa}\,d\tau \\ &\leq C_s K^{-2}h_\pm(z)\rho^{-\kappa-1-s}.\end{aligned}$$

The substitution $\tau = \rho u$ evaluates the integral as $\rho^{1-s-\kappa}\mathrm{B}(1-s, \kappa+s-1)$, and the bound uses $z^{-2} \leq K^{-2}\rho^{-2}$. Choose $K$ so large that $C_sK^{-2} \leq \vartheta c_s^{\mathrm{v}}\Delta_\kappa$, which is possible for our fixed positive gap. The two regions prove the proposition. □

**Corollary 6.12** (Both strict linear margins)**.** *Fix positive $A_\pm, B_\pm$, and set*

$$\begin{aligned} f^+(\rho,z) &= -A_+\varepsilon\rho^{-\gamma}g_+(z) + B_+\varepsilon^{1+\eta}\rho^{-\kappa}h_+(z), \\ f^-(\rho,z) &= -A_-\varepsilon\rho^{-\gamma}g_-(z) - B_-\varepsilon^{1+\eta}\rho^{-\kappa}h_-(z). \end{aligned} \tag{6.31}$$

*For sufficiently small $\varepsilon > 0$, uniformly in $\rho, z > 0$,*

$$-2\mathcal{J}_s f^+ \geq A_+L_1^+ + B_+L_2^+ > 0, \qquad -2\mathcal{J}_s f^- \leq -A_-L_1^- - B_-L_2^- < 0, \tag{6.32}$$

*where*

$$L_1^\pm = \varepsilon Q_\pm(\rho,z), \qquad L_2^\pm = \varepsilon^{1+\eta}c_s^{\mathrm{v}}\Delta_\kappa\rho^{-\kappa-1-s}h_\pm(z).$$

*Proof.* Separation gives

$$-2\mathcal{J}_s f^+ = 2A_+\varepsilon Q_+ + 2B_+L_2^+ - 2B_+\varepsilon^{1+\eta}\mathcal{T}_{\rho,\kappa,s}h_+,$$

and

$$-2\mathcal{J}_s f^- = -2A_-\varepsilon Q_- - 2B_-L_2^- + 2B_-\varepsilon^{1+\eta}\mathcal{T}_{\rho,\kappa,s}h_-.$$

Apply (6.30) with $\vartheta = 1/2$, and choose $\varepsilon$ so that $2B_\pm C_{s,1/2}\varepsilon^\eta \leq A_\pm$. The separation theorem applies at each positive evaluation height. Both modes are $C^2$ near every evaluation height

$z > 0$. The first grows at most linearly and the second at most as $1+|w|^{1+\eta}$, with $1 < 1+s+\gamma$ and $1+\eta < 1+s+\kappa$. □

6.7. **Normal coordinates and the nonlinear curvature formula.** The cylinder $\mathcal{C} = \Sigma \times \mathbb{R}$ has dimension eight and the ambient kernel is $|Z-X|^{-9-s}$. At a fixed height we use the base normal coordinates of (2.1). Thus $Y = Q + t\nu(Q)$, where $r = |Q|$ and $-r < t < r$, and $D = \{t < 0\}$ in these coordinates. The volume formula (2.3) gives

$$dY = J(r,t)\,dt\,d\sigma(Q), \qquad J(r,t) = (1-t^2/r^2)^3. \tag{6.33}$$

The factor axes have measure zero. The radius $r$ is related to the source position by $|Y|^2 = r^2 + t^2$.

Let $F(Q,w)$ be radial in $Q$, with $|F| \le r/100$, and let $E_F$ denote the set $t < F(Q,w)$. Write $F_0 = F(P,z)$ and $X_\theta = (P + \theta F_0 \nu(P), z)$. Fix a smooth radial cutoff $\chi_1$ equal to one on $B_1(0)$ and zero outside $B_2(0)$, and set $\chi_\ell(U) = \chi_1(U/\ell)$. The translated far kernel is

$$W_\theta(Z) = \frac{1-\chi_\ell(Z-X_\theta)}{|Z-X_\theta|^{9+s}}.$$

The far curvature of $E_{\theta F}$ is exactly

$$\begin{aligned}\mathcal{H}^{\mathrm{far}}_s[E_{\theta F}](X_\theta) = &\int_{\mathbb{R}^9} (\mathbf{1}_{\mathcal{C}_D^c} - \mathbf{1}_{\mathcal{C}_D})(Z) W_\theta(Z)\,dZ \\ &- 2\int_{\Sigma\times\mathbb{R}} \int_0^{\theta F(Q,w)} W_\theta(Q+t\nu(Q),w) J(r,t)\,dt\,d\sigma(Q)\,dw,\end{aligned} \tag{6.34}$$

Here $\mathcal{C}_D = D \times \mathbb{R}$. A positive displacement changes the sign integrand by $-2$ on the added strip. A negative displacement removes a strip and is represented by reversing the limits in the inner integral. Absolute convergence follows from the cutoff and the integrable far kernel.

**Lemma 6.13** (Local Taylor estimate and far-band cancellation)**.** *Suppose $P \neq 0$, $0 < \ell \le c|P|$ with $c$ a small absolute constant, and the $C^5$ norm of $F/\ell$ in cylinder coordinates rescaled by $\ell$ is at most $b \ll 1$ on the ball of radius $4\ell$. The field is radial in $Q$. After subtracting the constant and linear terms in $\theta$, the local curvature and the first integral of (6.34) have combined remainder bounded by*

$$C_s b^2 \ell^{-s}. \tag{6.35}$$

*The derivative of the complete curvature at $\theta = 0$ is $-2\mathcal{J}_s F$. The strip remainders below are written at $\theta = 1$. Write*

$$D_\ell(Q,w) = \max\{\ell, (|P-Q|^2 + (w-z)^2)^{1/2}\}.$$

*Suppose the angular region is invariant under the factor-interchange isometry $S$ fixing $P$. If the kernel distances, with a lower cutoff at $\ell$, remain comparable to $D_\ell$ along the intermediate normal displacements, then the remaining far-band terms have the following bounds:*

$$C_s \iint (|F_0|^2|F| + |F_0||F|^2) D_\ell^{-11-s}\,d\sigma\,dw, \tag{6.36}$$

$$C_s |F|^3 \operatorname{Av}_{Q/r}(D_\ell^{-11-s} + r^{-2} D_\ell^{-9-s}), \tag{6.37}$$

$$C_s |F|^3 \operatorname{Av}_{Q/r}(D_\ell^{-12-s} + r^{-2} D_\ell^{-10-s}). \tag{6.38}$$

*The last two displays bound the pure-source remainder at fixed radius and height and its derivative in the height separation. Here $\operatorname{Av}_{Q/r}$ denotes integration against normalized measure on the link, with an indicator inserted if the angular region is restricted. In taking the height derivative, the angular region is held fixed. The pure-source remainder is odd in the source amplitude and even in the height separation. If $F(r,-w) = -F(r,w)$, pairing opposite heights over the same angular region gives a difference bounded by $2|z|$ times the supremum of the*

*height derivative on the intervening interval. For the entire strip remainder, the bound without angular symmetry is*

$$C_s \iint \left[ (|F_0 F| + F^2) D_\ell^{-10-s} + \frac{F^2}{r} D_\ell^{-9-s} \right] d\sigma \, dw. \tag{6.39}$$

*Proof.* Work first in units scaled by $\ell$, with $F_0$ denoting the scaled displacement in the local calculation. The moving graph construction in the proof of Lemma 5.2 applies in a tangent frame of $\mathcal{C}$ with horizontal variable $\eta \in \mathbb{R}^8$. The undeformed chart is a product of the base chart with a height interval. Its bounds are uniform since $\ell/|P| \le c$. The fixed-frame height through the moving center is $h_\theta = U_\theta \circ S_\theta - \theta F_0$, using the horizontal inverse and vertical coordinate of that construction, with $F$ in place of the scaled base field. At the center $h_\theta(0) = 0$, since the horizontal component of the normal there vanishes. The implicit differentiation leading to (5.4) gives

$$\|h_\theta\|_{C^2} \le C, \qquad \|\dot{h}_\theta\|_{C^2} \le Cb, \qquad \|\ddot{h}_\theta\|_{C^2} \le Cb^2.$$

Indeed, $T_\theta = \mathrm{Id} + \theta \widehat{F} \nu_h$ has $C^1$ distance $O(b)$ from the identity and its inverse has first and second $\theta$-derivatives of orders $b$ and $b^2$ in $C^2$. These estimates use only the local $C^5$ bound. The chart accounts for the whole set on the support of $\chi_1$. In fact, projection along base normals together with retention of the height is smooth near $(P, z)$, and its differential there is the orthogonal projection onto the tangent plane. On the unscaled ball of radius $3\ell$ its Lipschitz constant is at most $3/2$ by taking $c$ small. It sends the ball of radius $2\ell$ about $X_\theta$ into the stipulated $4\ell$ patch about $(P, z)$. The set is below the graph, as follows from the positive normal component in the fixed vertical direction.

Subtract the tangent halfspace of the deformed graph at the moving point, as in (5.5). Let $L_\theta(\eta) = Dh_\theta(0)\eta$ and $B_\theta = h_\theta - L_\theta$. The tangent-halfspace integral is zero under centered radial truncation. The local curvature is therefore

$$-2 \int_{\mathbb{R}^8} B_\theta(\eta) \int_0^1 k(\eta, L_\theta(\eta) + u B_\theta(\eta)) \, du \, d\eta, \qquad k(\eta, t) = \frac{\chi_1(\eta, t)}{(|\eta|^2 + t^2)^{(9+s)/2}}.$$

For $r = |\eta|$, spatial Taylor estimates give $\partial_\theta^j B_\theta = O(b^j r^2)$ for $j = 0, 1, 2$. Also $t_\theta = L_\theta + u B_\theta$ satisfies $\partial_\theta^j t_\theta = O(b^j r)$ for $j = 1, 2$. With $|\partial_t^j k| \le C_s r^{-9-s-j}$, the same product differentiation as in (5.5) bounds the first and second parameter derivatives by $C_s b^j r^{-7-s}$ for $j = 1, 2$. For example, the term $B_\theta k_{tt} (\partial_\theta t_\theta)^2$ is bounded by $C r^2 r^{-11-s} b^2 r^2$. The factor $r^7 \, dr$ in $d\eta$ gives the integrable radial majorant $C_s b^j r^{-s} \, dr$. The same bounds hold uniformly with a smooth radial deletion at radius $a$ inserted in $k$, since its derivatives of order $j$ on its transition annulus are $O(r^{-j})$. These bounds justify differentiating and passing to the principal value. Taylor's theorem and rescaling by $\ell^{-s}$ give the local part of (6.35).

The second derivative of the first integral in (6.34) with respect to $\theta$ is bounded by $C F_0^2 \ell^{-2-s} \le C b^2 \ell^{-s}$: the $L^1$ norm of the second spatial derivatives of the far cutoff kernel is $O_s(\ell^{-2-s})$. This proves the remaining part of (6.35).

To identify the full linear term, use the smooth kernel $K_a(Z) = (1 - \chi_a(Z))|Z|^{-9-s}$ for the curvature before differentiating at $\theta = 0$. The derivative of the boundary strip is

$$-2 \int_{\Sigma \times \mathbb{R}} F(Q, w) K_a(Q - P, w - z) \, d\sigma(Q) \, dw.$$

For the displacement term, the derivative with respect to the center is the negative source derivative of $K_a$. Integration by parts against $\mathbf{1}_{\mathcal{C}_D^c} - \mathbf{1}_{\mathcal{C}_D}$, whose distributional gradient is $2(\nu, 0)$ times surface measure, gives

$$2F_0 \int_{\Sigma \times \mathbb{R}} \nu(P) \cdot \nu(Q) K_a(Q - P, w - z) \, d\sigma(Q) \, dw.$$

The distributional formula follows from the divergence theorem away from the vertex line. On bounded height intervals the boundary of the removed base tube $|Y| \le d$ has measure $O(d^7)$, so its contribution tends to zero. The integration by parts for $K_a$ extends to the full cylinder by its decay at infinity. In differentiating the source strip, a tail majorant is $Cr(1 + r + |w - z|)^{-9-s}$

against $d\sigma(Q)\,dw$, by $|F| \le r/100$. After height integration, its radial tail is bounded by $C\int_1^\infty r^{-1-s}\,dr$. Combining both derivatives before letting $a \downarrow 0$ produces the numerator

$$-2\big[F(Q,w) - F_0 + F_0(1-\nu(P)\cdot\nu(Q))\big].$$

In the surface integral the linear part of $F(Q,w)-F_0$ cancels in a centered tangent chart, by the same argument as for (2.8). The other terms vanish quadratically since $1-\nu(P)\cdot\nu(Q) = \frac{1}{2}|\nu(P)-\nu(Q)|^2$. Thus the derivative converges to $-2\mathscr{J}_s F$. The local differentiated strip bounds justify passage to the limit in the derivative of curvature. Its constant term is zero since the chamber-exchanging isometry of (2.4) extends to the cylinder by fixing the height.

For the remaining strip integral put $u = F_0$, and let $\overline{W}(u,t,\zeta)$ be the angular average of the far kernel with source displacement $t$ and height separation $\zeta = w - z$. We may restrict to a fixed angular region preserved by the factor interchange followed by rotations fixing $P$. For the resulting isometry $S$, $S\nu(P) = -\nu(P)$ and $S\nu(Q) = -\nu(SQ)$, so $\overline{W}(u,t,\zeta) = \overline{W}(-u,-t,\zeta)$. The radial cutoff preserves this identity. In particular, the first derivatives in $u, t$ vanish at $(0,0)$.

The cutoff kernel and its derivatives of order $j$ are bounded by $C_s \max\{\ell, |Z - X_\theta|\}^{-9-s-j}$. Under the distance comparison in the lemma, this gives

$$|\partial_{u,t}^\beta \partial_\zeta^k \overline{W}(u,t,\zeta)| \le C_s \operatorname{Av}_{Q/r} D_\ell(Q,w)^{-9-s-|\beta|-k} \qquad (|\beta| + k \le 3).$$

The angular integral is taken after the pointwise bound, since $D_\ell$ can vary with the source angle. Subtract the linear strip term $F\overline{W}(0,0,\zeta)$. The remaining terms, up to the common factor $-2$, are

$$\int_0^F [\overline{W}(u,t,\zeta) - \overline{W}(0,t,\zeta)]J(r,t)\,dt \quad \text{and} \quad \mathcal{R}(F,r;\zeta) = \int_0^F [\overline{W}(0,t,\zeta)J(r,t) - \overline{W}(0,0,\zeta)]\,dt.$$

For the first term, $\partial_u \overline{W}(0,0,\zeta) = 0$ and integration of the second derivatives show that

$$|\overline{W}(u,t,\zeta) - \overline{W}(0,t,\zeta)| \le C_s(|u|^2 + |u||t|)\operatorname{Av}_{Q/r} D_\ell^{-11-s}.$$

Integrate in $t$ and then in $r, w$ to obtain (6.36). For the second term the kernel difference at $u = 0$ is $O(t^2)$ with weight $\operatorname{Av}_{Q/r} D_\ell^{-11-s}$, and $|J(r,t) - 1| \le Ct^2/r^2$. Integrating the two bounds gives (6.37), while the same calculation with a height derivative gives (6.38). Both $J$ and $\overline{W}(0,t,\zeta)$ are even in $t$, so $\mathcal{R}$ is odd in $F$. It is also even in $\zeta$ because the cutoff and kernel are radial. For an odd field in $w$ the paired expression at $w > 0$ is

$$\mathcal{R}(F(r,w),r;w-z) - \mathcal{R}(F(r,w),r;w+z).$$

The mean value theorem gives the asserted height gain. Finally, before angular averaging, the first-order kernel difference is bounded by $C_s(|u| + |t|)D_\ell^{-10-s}$. Using $|J-1| \le C|t|/r$ and integrating this bound along the strip gives (6.39). □

6.8. **The full bounded-height exterior estimate.** In this and the following estimates, $\chi : [0,\infty) \to [0,1]$ is a fixed smooth nondecreasing function, equal to zero on $[0,1]$ and to one on $[2,\infty)$.

Fix $A, B > 0$. Set

$$m = \frac{8}{p}, \quad \lambda(w) = (\varepsilon\tilde{g}(w))^{1/p}, \quad F(Q,w) = -A\varepsilon|Q|^{-\gamma}\tilde{g}(w) - B\varepsilon^{1+\eta}|Q|^{-\kappa}\tilde{h}(w).$$

Let $E_\varepsilon^{\mathrm{ref}}$ be a measurable set whose section at height $w$ agrees with the normal sublevel set $t < F(Q,w)$ for $r(Y) > C_0\lambda(w)$. The radius $r(Y) = (|Y_x| + |Y_y|)/\sqrt{2}$ is the radius $|Q|$ in (2.1). Inside the core the section may be arbitrary. When $\lambda(w) = 0$, the hypothesis means that the section is the undeformed chamber $D$, up to null sets.

**Proposition 6.14** (Lower sign outside the bounded-height core)**.** *For every fixed $0 < s < 1$, there exist $T_s < \infty$ and $\varepsilon_s > 0$, depending on the fixed coefficients and $C_0$, such that*

$$\mathcal{H}_s[E_\varepsilon^{\mathrm{ref}}](P + F(P,z)\nu(P), z) < 0$$

*whenever*

$$0 \le z \le 1, \qquad \rho = |P| \ge T_s\lambda(z), \qquad 0 < \varepsilon < \varepsilon_s.$$

*The constants are independent of the measurable choices inside the permitted core.*

*Proof.* The reference scalars are smooth near $z \in [0,1]$ and grow at most like $1 + |w|$ and $(1+|w|)^{1+\eta}$, respectively. The separated formula applies, since $1 < 1+s+\gamma$ and $1+\eta < 1+s+\kappa$. With the notation of Proposition 6.8 it gives

$$\begin{aligned} -2\mathcal{J}_s F(P,z) = &-2A\varepsilon Q(\rho,z) + 2B\varepsilon^{1+\eta}\mathcal{T}_{\rho,\kappa,s}\tilde{h}(z) \\ &- 2B\varepsilon^{1+\eta}c_s^{\mathrm{v}}\Delta_\kappa\rho^{-\kappa-1-s}\tilde{h}(z). \end{aligned}$$

Choose $\varepsilon$ so small that $2BC_s\varepsilon^\eta \le A$. Since $\tilde{h}$ is bounded below on $[0,1]$, we retain both margins:

$$-2\mathcal{J}_s F(P,z) \le -c_s\varepsilon\rho^{-\gamma}\min\{\rho,1\}^{1-s} - c_s\varepsilon^{1+\eta}\rho^{-\kappa-1-s}. \tag{6.40}$$

Choose a fixed $R_c \ge C_0$ and use the cutoff $\chi$ to define

$$f(Q,w) = \begin{cases} \chi(|Q|/(R_c\lambda(w)))F(Q,w), & \lambda(w) > 0, \\ 0, & \lambda(w) = 0. \end{cases}$$

Here $\varepsilon^{1+\eta}\tilde{h}(w) = \lambda(w)^{\kappa+1}$ by $p(1+\eta) = \kappa+1$. Thus, for $r = |Q|$,

$$\frac{|F(Q,w)|}{r} = A\left(\frac{\lambda(w)}{r}\right)^p + B\left(\frac{\lambda(w)}{r}\right)^{\kappa+1}.$$

On the support of the cutoff $r \ge R_c\lambda(w)$. Choose $R_c$ large enough that $|f| \le r/100$. Since $\kappa+1 > p$, the first power also bounds the second there up to a constant. With the scalar value bound this gives

$$|f(Q,w)| \le C_s\varepsilon r^{-\gamma}\tilde{g}(w) \le C_s\varepsilon r^{-\gamma}(1+w_+).$$

Use the local radius $\ell = c_0\min\{\rho,1\}$, where $c_0 < 1/32$ is also small enough for Lemma 6.13. In the $4\ell$ patch, $r \asymp \rho$ and $w \in [-1/4, 5/4]$. On this height interval the scalars are positive and smooth with bounded derivatives, and $\lambda(w) \asymp_s \lambda(z)$. Taking $T_s$ large makes $r > 2R_c\lambda(w)$ on the patch, so $f = F$ there and the actual and cutoff sets agree near the evaluation point. For the rescaled $C^5$ norm, each base derivative of order $j$ on $r^{-\gamma}$ or $r^{-\kappa}$ costs at most $C_j\rho^{-j}$, which is compensated by $\ell^j$. The height derivatives are bounded on the patch, and $\ell \le 1$. The second mode is bounded relative to the first by $C_s(\lambda(z)/\rho)^{\kappa-\gamma}$, also after these scaled differentiations. The resulting norm of $f/\ell$ is at most $C_s\varepsilon\rho^{-\gamma}/\ell$. For $\rho \le 1$ this is bounded by $C_sT_s^{-p}$ because $\rho^p \ge T_s^p\varepsilon\tilde{g}(z)$ and $\inf_{[0,1]}\tilde{g} > 0$. For $\rho \ge 1$ it is bounded by $C_s\varepsilon$. The local and moving-center estimate (6.35) then gives

$$C_s\varepsilon^2\rho^{-2\gamma}\ell^{-2-s}.$$

Write $d = |(P,z) - (Q,w)|$ and $\tau = |w - z|$. On the distant strip we can use $\ell + d$ in place of $D_\ell$ in (6.39), since they differ by at most a factor two. If $r \notin [\rho/2, 2\rho]$, the base distance is bounded below by $|r - \rho| \ge (\rho + r)/3$, and the shifts of sizes at most $r/100$ and $\rho/100$ preserve this bound up to a constant. If $r \in [\rho/2, 2\rho]$, then

$$|f(P,z)| + |f(Q,w)| \le C_s\varepsilon\rho^{-\gamma}(1+\tau).$$

Since $\ell(1+\tau) \le \ell + d$, the local smallness bound makes these shifts a small fraction of $\ell + d$. This proves the distance comparison for all interpolated kernel arguments, with the cutoff controlling distances smaller than $\ell$.

For $d < c\rho$, with $c > 0$ fixed and sufficiently small, the source lies on a tangent chart of $\mathcal{C}$ with $r \asymp \rho$. In tangent polar coordinates $\xi = u\omega$ the graph height $h_0$ has size $O(u^2/\rho)$ and gradient $O(u/\rho)$. Hence $d = (u^2 + h_0(u\omega)^2)^{1/2}$ is comparable to $u$ and has $u$-derivative bounded

above and below by positive constants. Together with the bounded area density this gives, for every nonnegative radial weight $\Phi$,

$$\int_{\{d<c\rho\}} \Phi(d)\, d\sigma(Q)\, dw \le C \int_0^{c\rho} \Phi(u) u^7\, du. \tag{6.41}$$

Here the patch size and all constants are uniform after scaling by $\rho$.

In this chart $|f(Q,w)|+|f(P,z)| \le C_s\varepsilon\rho^{-\gamma}(1+d)$. For $\rho \le 1$, $1+d$ is bounded and $\ell \asymp \rho$. In (6.39), the kernel factors are then bounded by $C_s\rho^{-10-s}$, including the $r^{-1}$ from the Jacobian term. The chart volume is $O(\rho^8)$ by (6.41), giving $C_s\varepsilon^2\rho^{-2\gamma-2-s}$. For $\rho \ge 1$, we have $\ell = c_0$. The part $d \le 1$ is bounded by $C_s\varepsilon^2\rho^{-2\gamma}$. For $1 < d < c\rho$, the numerator has an extra factor at most $C_s d^2$, so the two shell contributions are bounded by

$$C_s\varepsilon^2\rho^{-2\gamma}\int_1^{c\rho} u^{-1-s}\, du, \qquad C_s\varepsilon^2\rho^{-2\gamma-1}\int_1^{c\rho} u^{-s}\, du.$$

If $c\rho > 1$, both are bounded by $C_s\varepsilon^2\rho^{-2\gamma}$, using $0 < s < 1$. If $c\rho \le 1$, this shell is empty.

On $d \ge c\rho$, one has $\rho + r + \tau \le 2\rho + 2d$, so the denominator is comparable to $\rho + r + \tau$. Use the Jacobian term to bound the source-square kernel term as well, since $r \le \rho + r + \tau$. Writing $R = \rho + \tau$, the radial integrals used for the source-square and mixed terms respectively are

$$\int_0^\infty \frac{r^{5-2\gamma}}{(R+r)^{9+s}}\, dr = C_s R^{-3-2\gamma-s}, \qquad \int_0^\infty \frac{r^{6-\gamma}}{(R+r)^{10+s}}\, dr = C_s R^{-3-\gamma-s}.$$

These identities follow by $r = Ru$, with convergence at zero since $\gamma < 3$ and at infinity by the denominator powers. The remaining bound is

$$C_s\varepsilon^2\left[\int_0^\infty (1+\tau)^2(\rho+\tau)^{-3-2\gamma-s}\, d\tau + \rho^{-\gamma}\int_0^\infty (1+\tau)(\rho+\tau)^{-3-\gamma-s}\, d\tau\right].$$

For $\rho \le 1$, split at $\tau = 1$. The tails are uniformly bounded inside each integral, and on $[0,1]$ the bounds are $C_s\rho^{-2-2\gamma-s}$ and $C_s\rho^{-2-\gamma-s}$, respectively. With the mixed prefactor the total is $C_s\varepsilon^2\rho^{-2\gamma-2-s}$. For $\rho \ge 1$, use $1 + \tau \le \rho + \tau$ and integrate to bound the total by $C_s\varepsilon^2\rho^{-2\gamma-s}$.

The difference between the cutoff set and the actual set is supported at height $w$ in $r(Y) \le 2R_c\lambda(w)$, so its base measure is $O_s(\lambda(w)^8)$ by $|Y| \le \sqrt{2}r(Y)$. The Jacobi defect $F - f$ vanishes at the evaluation point, so only its source integral remains. Its source mass at height $w$ has the same bound

$$\int_0^{2R_c\lambda(w)} r^6|F(r,w)|\, dr \le C_s(\lambda(w)^{p+7-\gamma} + \lambda(w)^{\kappa+1+7-\kappa}) = C_s\lambda(w)^8. \tag{6.42}$$

The identity uses $p = 1 + \gamma$ and the convergence of both radial integrals at zero. It also holds for $\lambda(w) = 0$, when both masses vanish.

For either the geometric core or the Jacobi source defect, the distance to the respective evaluation point is bounded below by $c_s(\rho + \tau)$. If $\tau \ge \rho/2$, use the height distance. If $\tau < \rho/2$ and $\rho \le 1$, the value bound and the lower bound at $z$ give $\lambda(w) \le C_s\lambda(z) \le C_s\rho/T_s$. If $\tau < \rho/2$ and $\rho \ge 1$, the growth bound gives $\lambda(w) \le C_s\varepsilon^{1/p}(1+\rho)^{1/p}$. Taking $T_s$ large and then $\varepsilon$ small ensures the base source radius is at most $\rho/2$ in both cases. The displaced base center has norm at least $\rho$, by orthogonality of $P$ and $\nu(P)$. This proves the distance bound for both integrals.

Using $m = 8/p$, the two errors are bounded by

$$C_s\varepsilon^m\int_{\mathbb{R}} \frac{\tilde{g}(w)^m}{(\rho + |w-z|)^{9+s}}\, dw \le C_s\varepsilon^m\int_0^\infty \frac{(1+\tau)^m}{(\rho+\tau)^{9+s}}\, d\tau.$$

For $\rho \le 1$, the part $\tau \le 1$ is $O_s(\rho^{-8-s})$ and the rest is bounded since $m < 8 + s$. For $\rho \ge 1$, $1 + \tau \le \rho + \tau$ gives $O_s(\rho^{m-8-s})$. Thus the values of the reference scalars near $w = -1$ enter only through absolutely convergent integrals.

Combining the estimates gives

$$|\mathcal{H}_s[E_\varepsilon^{\mathrm{ref}}](P+F\nu,z)+2\mathcal{J}_sF(P,z)|\le\begin{cases}C_s(\varepsilon^2\rho^{-2\gamma-2-s}+\varepsilon^m\rho^{-8-s}), & \rho\le 1,\\ C_s(\varepsilon^2\rho^{-2\gamma}+\varepsilon^m\rho^{m-8-s}), & \rho\ge 1.\end{cases}$$

For $\rho\le 1$, divide by the second negative margin in (6.40). The ratios are fixed multiples of

$$\left(\frac{\varepsilon}{\rho^p}\right)^{1-\eta},\qquad \left(\frac{\varepsilon}{\rho^p}\right)^{m-1-\eta}.$$

Indeed $p(1-\eta)=2\gamma+1-\kappa$ and $p(m-1-\eta)=7-\kappa$, both positive. Since $\varepsilon/\rho^p\le C_sT_s^{-p}$, taking $T_s$ large makes both ratios smaller than any prescribed positive constant. For $\rho\ge 1$, divide by the first margin instead. The ratios are bounded by $C_s\varepsilon\rho^{-\gamma}$ and $C_s\varepsilon^{m-1}\rho^{m+\gamma-8-s}$. The latter power of $\rho$ is negative since $m+\gamma-5=(\gamma-1)(\gamma-3)/(\gamma+1)<0$. Thus decreasing $\varepsilon_s$ absorbs both errors into that margin. In each region the corresponding margin strictly dominates the error, so the curvature is negative. □

6.9. **A radial bound for the cubic source term.**

**Lemma 6.15.** *Let $R,\lambda>0$. If $F$ is radial on $\Sigma$ and $|F(r)|\le C\min\{r,\lambda^pr^{-\gamma}\}$, then*

$$\int_\Sigma |F(r)|^3\left[(R+r)^{-12-s}+r^{-2}(R+r)^{-10-s}\right]d\sigma\le C_s\lambda^{2p}R^{-4-2\gamma-s}.$$

*Proof.* Use the bound by $Cr$ for one factor of $|F|$ and the bound by $C\lambda^pr^{-\gamma}$ for the other two. Since $(R+r)^{-2}\le r^{-2}$, the normal-coordinate surface measure (2.3) reduces the estimate to

$$C\lambda^{2p}\int_0^\infty r^{5-2\gamma}(R+r)^{-10-s}\,dr=C\lambda^{2p}R^{-4-2\gamma-s}\int_0^\infty t^{5-2\gamma}(1+t)^{-10-s}\,dt.$$

The last integral converges at zero because $\gamma<3$ and at infinity because its integrand is $O(t^{-5-2\gamma-s})$. □

6.10. **Full nonlinear estimates on the exterior profile regions.** Fix the positive coefficients in the fields (6.31) and put $\lambda_\pm(w)=(\varepsilon|g_\pm(w)|)^{1/p}$. Let $E_\varepsilon^\pm$ have the exact normal sections $t<f^\pm(r,w)$ when $r>R_1\lambda_\pm(w)$, with arbitrary measurable inner sections. Assume double rotational invariance and complement symmetry under $(x,y,w)\mapsto(y,x,-w)$. Values on the height slice $w=0$ are immaterial.

**Theorem 6.16** (Fixed-exponent nonlinear graph bounds)**.** *Fix $K\ge 4$. For the upper sign allow every $z>0$; for the lower sign in this theorem restrict to $z\ge 1$. Write $\rho=|P|$,*

$$X_\varepsilon^\pm=(P+f^\pm(\rho,z)\nu(P),z),\qquad L_1^\pm=\varepsilon Q_\pm(\rho,z).$$

*For sufficiently small $\varepsilon>0$, throughout $z\le K\rho$ one has*

$$|\mathcal{H}_s[E_\varepsilon^\pm](X_\varepsilon^\pm)+2\mathcal{J}_sf^\pm(P,z)|\le C_{s,K}\varepsilon^2Q_\pm(\rho,z).\tag{6.43}$$

*The evaluation point automatically lies in the prescribed normal tail in this region.*

*There is a fixed $T_0<\infty$ such that, whenever $z\ge K\rho$ and $\rho\ge T_0\lambda_\pm(z)$,*

$$|\mathcal{H}_s[E_\varepsilon^\pm](X_\varepsilon^\pm)+2\mathcal{J}_sf^\pm(P,z)|\le C_s\left[\rho^{-s}(\lambda_\pm(z)/\rho)^{2p}+\varepsilon^{2/p}L_1^\pm\right].\tag{6.44}$$

*The constants do not increase if $T_0$ is subsequently increased. They may depend on the fixed coefficients and $R_1$.*

*Proof.* Omit the sign subscript except where the two scalars require different bounds. Put $\mu=8/p$ and $G=g(z)>0$. Fix the small chart radii $c,c_1\in(0,1)$ in the conditions $|P-Q|<c\rho$ and $|w|<c_1\rho$ used below before selecting the radial cutoff. Set

$$F(r,w)=\chi(r/(L\lambda(w)))f(r,w)$$

for a fixed $L\ge R_1$, setting $F=0$ where $\lambda=0$. The power identities $p=1+\gamma$ and $p(1+\eta)=1+\kappa$ give

$$|F|\le C_s\lambda(w)^pr^{-\gamma}\mathbf{1}_{\{r\ge L\lambda(w)\}},\qquad |F|/r\le C_s(L^{-p}+L^{-\kappa-1}).$$

Take $L$ large enough that $|F(r,w)| \le (c/100)r$. In particular, for $r \in [\rho/4, 4\rho]$ and $|P-Q| \ge c\rho$, the total normal displacement of the source and center is at most $c(r+\rho)/100 \le |P-Q|/20$. At each height, the difference between the prescribed set and $E_F$ is supported in $|Y| \le 2\sqrt{2}L\lambda(w)$, by the normal coordinates. Its volume is at most $C_s\lambda(w)^8$. The Jacobi cutoff has the same mass bound $\int_\Sigma |f-F|\,d\sigma \le C_s\lambda(w)^8$, by (6.42) applied with $\lambda = (\varepsilon|g(w)|)^{1/p}$ and with absolute values of the two model coefficients.

Local scales and kernel distances. Use $\ell = c_0 \min\{\rho, z\}$ in (6.43) and $\ell = c_0\rho$ in (6.44), for a fixed sufficiently small $c_0 > 0$. In the first region $\ell \asymp_K z$. For $g_+$, if $\rho \le 1$ then $g_+(z) \le C_{s,K} z^q$ and $\rho^{-\gamma} g_+(z)/\ell \le C_{s,K}\rho^{q-1-\gamma}$. If $\rho \ge 1$, use $g_+(z) \le C_s z$. For $g_-$, use $g_-(z) \le C_s z$ for $z \ge 1$ and $\rho \ge 1/K$. Thus

$$\rho^{-\gamma} G/\ell \le C_{s,K} \quad \text{and} \quad \lambda(z)/\rho \le C_{s,K}\varepsilon^{1/p} \qquad (z \le K\rho).$$

The second inequality follows from $(\lambda(z)/\rho)^p = \varepsilon G \rho^{-\gamma-1}$ and $\ell \le \rho$. In either region, the radii and heights on the cylinder patch of radius $4\ell$ are comparable to $\rho, z$. There $F = f$ and the prescribed set agrees locally with $E_F$, by decreasing $\varepsilon$ in the first region and increasing $T_0$ in the second. The ratio of the second normal mode to the first is bounded there by $C_s(\lambda(z)/\rho)^{\kappa-\gamma}$. Using the scalar derivative bounds, each derivative of order $j \le 5$ costs at most $C_{s,j}\ell^{-j}$ on that patch, in cylinder coordinates. Hence the smallness parameter $b$ of Lemma 6.13 is bounded by

$$C_{s,K}\varepsilon\rho^{-\gamma} G/\ell \quad \text{in the first region,} \qquad C_s(\lambda(z)/\rho)^p \quad \text{in the second.}$$

Hereafter $F_0 = F(P,z) = f(P,z)$.

For the far integrals use $D_\ell$ from Lemma 6.13. If $r \le \rho/4$ or $r \ge 4\rho$, the base distance is comparable to $\rho + r$ throughout normal interpolation, by the smallness of $|F|/r$ and $|F_0|/\rho$. At comparable radii and $z \le K\rho$, when $|w - z| \le z/2$, both displacements are $O_{s,K}(\varepsilon\ell)$. When $\tau = |w-z| > z/2$, the height alone controls the distance if $\tau \ge c\rho$ for a fixed small $c > 0$. Otherwise $|w|, z \le 3\tau$. For the upper scalar with $\rho \le 1$, $g_+(|w|) + g_+(z) \le C_s\tau^q$ and $\rho^{-\gamma}\tau^{q-1} \le C_s$. For $\rho \ge 1$ use linear growth. For the lower scalar, $\tau > z/2 \ge 1/2$ and $\rho \ge 1/K$, so $1 + |w| + z \le C\tau$ gives the same bound $O_{s,K}(\varepsilon\tau)$ for the displacements. In the second region, $z \ge K\rho$, the height distance controls the distance at comparable radii when $|w| > 2z$. On $|w| \le 2z$, $\lambda(w) \le C_s\lambda(z)$, so the comparable-radius displacements are at most $C_s\rho(\lambda(z)/\rho)^p \ll \ell$. These comparisons give the required bounds on the derivatives of the cutoff kernel, including on its transition annulus. The kernel and its derivatives vanish inside the cutoff ball. Every angular region used below is either the full link or one of the two regions defined by $|P-Q| < c\rho$. They are invariant under the chamber-exchanging isometry fixing $P$.

We use the following forms of the floors (6.25) and (6.21). If $z \le K\rho$, then $\rho + 1 + z \le (K+1)(1+\rho)$ and $\rho + z \le (K+1)\rho$, so

$$Q_+ \ge c_{s,K}(1+\rho)^{-\gamma} z(1+z)^{-1-s-\delta}, \qquad Q_- \ge c_{s,K}\rho^{-\gamma} z^{-1-s}.$$

In particular $Q_+ \ge c_{s,K} z$ whenever $\rho \le 1$ in this region, including $1 < z \le K\rho$. If $z \ge K\rho$, then $\rho + 1 + z \le 2(1+z)$ and $\rho + z \le 2z$. Thus $Q_+ \ge c_s z$ for $z \le 1$, while for $z \ge 1$

$$Q_+ \ge c_s z^{-s-\delta-\gamma}, \qquad Q_- \ge c_s z^{-1-s-\gamma}.$$

We will repeatedly use the following elementary estimates for integrating in the height. For $m > 0$, $n > m + 1$, set

$$I_{m,n}(\rho) = \int_0^\infty g(w)^m (\rho + w)^{-n}\,dw.$$

For $g_+$ and $\rho \le 1$, this is bounded whenever $qm > n - 1$, by integrating $w^{qm-n}$ on $(0,1)$ and $w^{m-n}$ on $(1,\infty)$. For $\rho \ge 1$ it is at most $C_s\rho^{m+1-n}$ for either scalar: bound $g(w) \le C_s(1+w)$ and set $w = \rho u$, using $\int_0^\infty (1+u)^{m-n}\,du < \infty$. For $g_-$ the same proof gives $C_{s,K}\rho^{m+1-n}$ on $\rho \ge 1/K$, since $1 + \rho u \le C_K\rho(1+u)$. The integral of $g(w)^m w^{-n}$ on $w > 2z$ is bounded in the small upper-height case $z \le 1$ if $qm > n - 1$, and by $C_s z^{m+1-n}$ for $z \ge 1$, by the same power integrations.

Core errors. Let $b_{\mathrm{c}} = \chi_{E_\varepsilon^+} - \chi_{E_F}$. For $P = (P_x, P_y)$ choose $O \in O(4)$ with $OP_y = P_x$ and put $S(x,y) = (Oy, O^{-1}x)$. This involution fixes $P$ and reverses $\nu(P)$. By rotational invariance and complement symmetry, $b_{\mathrm{c}}(SY, -w) = -b_{\mathrm{c}}(Y,w)$ almost everywhere. Set

$$k_{u,v}(Y,w) = (|P + u\nu(P) - Y|^2 + (v-w)^2)^{-(9+s)/2}.$$

Changing variables by $(Y,w) \mapsto (SY,-w)$ on $w<0$ writes the geometric core contribution as

$$-2\int_0^\infty \int_{\mathbb{R}^8} b_{\mathrm{c}}(Y,w)[k_{F_0,z}(Y,w) - k_{-F_0,-z}(Y,w)]\,dY\,dw. \tag{6.45}$$

The Jacobi cutoff defect has no evaluation-point term, since $f(P,z) = F_0$. Its source is odd in $w$, so the same pairing applies with $F_0 = 0$ and source measure $(f-F)d\sigma\,dw$. A lower distance bound $cR$ from every interpolated center gives a kernel difference at most $C_s(|F_0| + z)R^{-10-s}$. In the first region the local bounds give $|F_0| \le C_{s,K}\varepsilon z$. In the second they give $|F_0| \le C_s\rho(\lambda(z)/\rho)^p \le C_s z$.

In the first region, for $w \le (2K+2)\rho$ the core radius is at most $C_{s,K}\varepsilon^{1/p}\rho$. For the upper scalar, use $\lambda_+(w) \le C_s\varepsilon^{1/p}w$, which follows from $q > p > 1$. For the lower scalar use $\lambda_-(w) \le C_s\varepsilon^{1/p}(1+w)^{1/p}$ and $\rho \ge 1/K$. The base distance from the interpolated center to the core is therefore at least $c\rho$. If $w > (2K+2)\rho$, every interpolated height has distance at least $w/2$ from $w$. Thus $R = \rho + w$ is valid in both cases. Either core error is bounded by $C_{s,K}\varepsilon^\mu z I_{\mu,10+s}(\rho)$. For the upper scalar the height integral is bounded at $\rho \le 1$, since $q\mu > 9 + s$. The quotient by $Q_+$ there is $O_{s,K}(\varepsilon^\mu)$ by (6.25). At $\rho \ge 1,\ z \le 1$, the quotient has the additional factor $\rho^{\mu+\gamma-9-s}$. At $z \ge 1,\ \rho \ge 1$, use $z \le K\rho$ to bound the additional factor by $\rho^{\mu+\gamma+\delta-8}$. For the lower scalar $\rho \ge 1/K,\ z \ge 1$, (6.21) bounds the factor by $C_{s,K}\rho^{\mu+\gamma-9-s}z^{2+s} \le C_{s,K}\rho^{\mu+\gamma-7}$. All these large-radius powers are negative, since

$$\mu + \gamma - 5 = \frac{(\gamma-1)(\gamma-3)}{\gamma+1} < 0. \tag{6.46}$$

This proves the required near-height core bound $C_{s,K}\varepsilon^\mu Q$.

In the second region, for $|w| \le 2z$ take $T_0$ large enough to separate the core of radius $C_s\lambda(z)$ from the evaluation base by $c\rho$. Bounding the kernel by $C_s(\rho + |w-z|)^{-9-s}$ and integrating in height gives $C_s\lambda(z)^8\rho^{-8-s}$ for either core error, without using pairing on these heights. For $w > 2z$, use (6.45) with $R = w$, giving

$$C_s\varepsilon^\mu z\int_{2z}^\infty g(w)^\mu w^{-10-s}\,dw.$$

This is $O_s(\varepsilon^\mu z)$ for the upper scalar at $z \le 1$, and $O_s(\varepsilon^\mu z^{\mu-8-s})$ for $z \ge 1$. In the first case use $Q_+ \ge c_s z$. In the second, the floors above give the extra factors $z^{\mu+\gamma+\delta-8}$ and $z^{\mu+\gamma-7}$. Both powers are negative by (6.46). In either case the quotient by $L_1 = \varepsilon Q$ is bounded by $C_s\varepsilon^{\mu-1}$. Local and mixed terms at $z \le K\rho$. The local curvature and the second-order error from moving the center of the cone integral are bounded by

$$C_{s,K}\varepsilon^2\rho^{-2\gamma}G^2\ell^{-2-s}.$$

Using $\ell \asymp_K z$, divide by the upper floor. After removing $C_{s,K}\varepsilon^2$, the respective bounds in the ranges $\rho \le 1,\ \rho \ge 1,\ z \le 1$, and $\rho \ge 1,\ z \ge 1$ are

$$\rho^{-2\gamma}z^{2q-3-s}, \qquad \rho^{-\gamma}z^{2q-3-s}, \qquad \rho^{-\gamma}z^{\delta}.$$

The first uses $g_+(z) \le C_{s,K}z^q$ on $z \le K$. These quantities are bounded since $z \le K\rho$, $2q-3-s > 2\gamma$, and $\delta < 1 < \gamma$. The analogous quotient for the lower scalar is bounded by $C_{s,K}\rho^{-\gamma}z$, using $G \le C_s z$ and (6.21). It is bounded because $\rho \ge 1/K$.

For the mixed terms use (6.36). On $|P-Q| < c\rho$ we have $r \asymp \rho$. Integrating $D_\ell^{-11-s}$ over the seven base tangent variables gives a bound of $C_s\max\{\ell, |w-z|\}^{-4-s}$. Indeed, scaling the tangent variables by $\max\{\ell, |w-z|\}$ gives the exponent $7-(11+s) = -4-s$. For $j = 1, 2$,

$$\int_{\mathbb{R}} |g_+(w)|^j \max\{\ell, |w-z|\}^{-4-s}\,dw \le C_s[G^j\ell^{-3-s} + \mathbf{1}_{\{z\le 1\}}]. \tag{6.47}$$

On $|w-z|<z/2$ this follows from comparability with $G$ and direct integration of the cutoff kernel. Outside this interval $|w|\le 3|w-z|$. For $z\le 1$, use the integrable bounds $C_s|w|^{qj-4-s}$ on $|w|<1$ and $C_s|w|^{j-4-s}$ on $|w|\ge 1$. For $z\ge 1$, linear growth instead gives $C_s z^{j-3-s}$, bounded by $C_s G^j \ell^{-3-s}$. The estimate for $g_-$, where $z\ge 1$, has no indicator term: on $|w-z|\ge z/2$ one has $1+|w|\le C|w-z|$, and the same calculation applies. The resulting mixed error is bounded by

$$C_{s,K}\varepsilon^3\rho^{-3\gamma}[G^3\ell^{-3-s}+(G+G^2)\mathbf{1}_{\{z\le 1\}}]$$

for the upper scalar, with the indicator term absent for the lower. The first term is the local error bound multiplied by $C_s\varepsilon\rho^{-\gamma}G/\ell\le C_{s,K}\varepsilon$. For the indicator term the quotient by $Q_+$ is at most $C_{s,K}\varepsilon^3\rho^{-3\gamma}(1+\rho)^\gamma z^{q-1}$. It is bounded by $C_{s,K}\varepsilon^3$: if $\rho\le 1$ use $q-1>3\gamma$ and $z\le K\rho$, and otherwise use $z\le 1$.

On $|P-Q|\ge c\rho$ the kernel distance with cutoff is bounded below by $c_K(\rho+r+|w|)$. At comparable radii the displacement bound by a small multiple of $r$ preserves the base separation. For $|w|>2K\rho$ the height distance is at least $|w|/2$. For $j=1,2$, integration of $r^{6-j\gamma}(\rho+r+|w|)^{-11-s}$ in $r$ gives $C_s(\rho+|w|)^{-4-j\gamma-s}$, since $j\gamma<7$. Thus the mixed error here is at most

$$C_{s,K}\varepsilon^3[\rho^{-2\gamma}G^2 I_{1,4+\gamma+s}(\rho)+\rho^{-\gamma}GI_{2,4+2\gamma+s}(\rho)].$$

In the upper region $\rho\le 1$ the height integrals are bounded, by $qj>3+j\gamma+s$. The quotient by $Q_+$, with the factor $C_{s,K}\varepsilon^3$ removed, is bounded by $\rho^{-2\gamma}z^{2q-1}+\rho^{-\gamma}z^{q-1}$, which is bounded by $z\le K\rho$. At larger radii the height integral with index $j$ is bounded by $C_s\rho^{j-3-j\gamma-s}$. For $g_+$ and $z\le 1$ this gives the quotient bound $C_s\rho^{-1-2\gamma-s}$. For $g_+$ and $z\ge 1$ the two factors are at most $\rho^{-2-2\gamma-s}z^{2+s+\delta}$ and $\rho^{-1-2\gamma-s}z^{1+s+\delta}$. Both are bounded by $C_{s,K}\rho^{-2\gamma+\delta}$. For $g_-$ the same calculation uses the lower floor and replaces $\delta$ by 1, giving $C_{s,K}\rho^{1-2\gamma}$, bounded for $\rho\ge 1/K$.

Local and mixed terms at $z\ge K\rho$. The local estimate is $C_s\rho^{-s}(\lambda(z)/\rho)^{2p}$. In the far band with $|w|\le 2z$ we use the quadratic bound (6.39), which controls both mixed and pure terms on this part. The kernel distance with cutoff is comparable to $\rho+r+|w-z|$: for comparable radii use $\ell=c_0\rho$, and for the remaining radii use base separation. Also $|F(r,w)|\le C_s\lambda(z)^p r^{-\gamma}$. The mixed quadratic term and the source-square term with the Jacobian factor are bounded respectively by

$$C_s|F_0|\lambda(z)^p\int_0^\infty r^{6-\gamma}(\rho+r)^{-9-s}\,dr,$$
$$C_s\lambda(z)^{2p}\int_0^\infty r^{5-2\gamma}(\rho+r)^{-8-s}\,dr.$$

Here the height integration lowers each inverse power of $\rho+r+|w-z|$ by one. The remaining quadratic source-square term is bounded by the second expression since $r\le\rho+r+|w-z|$. Scaling $r=\rho u$ and using $|F_0|\le C_s\lambda(z)^p\rho^{-\gamma}$ bounds both by $C_s\lambda(z)^{2p}\rho^{-2\gamma-2-s}$. The most singular radial power is $r^{5-2\gamma}$, integrable at zero because $\gamma<3$.

At $|w|>2z$, the distance is comparable to $r+|w|$ and the mixed terms from (6.36) are bounded by

$$C_sF_0^2\varepsilon\int_{2z}^\infty g(w)w^{-4-\gamma-s}\,dw,\qquad C_s|F_0|\varepsilon^2\int_{2z}^\infty g(w)^2w^{-4-2\gamma-s}\,dw.$$

These follow from the same radial integration as the far-base estimate above. Use $|F_0|\le C_s\lambda(z)$, since $\rho\ge T_0\lambda(z)$. For the upper scalar at $z\le 1$, the height integrals are bounded and $Q_+\ge c_sz$, so the quotients by $L_1^+$ are bounded by

$$C_s\varepsilon^{2/p}z^{2q/p-1},\qquad C_s\varepsilon^{1+1/p}z^{q/p-1}.$$

For $z\ge 1$, the integrals are bounded by $C_sz^{-2-\gamma-s}$ and $C_sz^{-1-2\gamma-s}$. Using $\rho\le z/K$ in the floors gives the quotients

$$C_s\varepsilon^{2/p}z^{\delta-2\gamma/p},\qquad C_s\varepsilon^{1+1/p}z^{\delta-\gamma-\gamma/p}$$

for the upper scalar, and the same bounds with 1 in place of $\delta$ for the lower. These powers of $z$ are negative since $\delta<1$ and $2\gamma/p>1$.

Pure-source term near the base at $z \le K\rho$. The remainder and its height derivative are those in (6.37) and (6.38). In this region split off the band $|P-Q| < c\rho,\ |w| < c_1\rho$ with $c, c_1$ as fixed above. For $|w| \le 2z$ in the band, use $|g(w)| \le C_s G$ and $r \asymp \rho$. The two cutoff integrals in (6.37) are bounded by $C_s\ell^{-3-s}$ and $C_s\rho^{-2}\ell^{-1-s}$, by integration on the eight-dimensional cylinder chart. Since $\ell \le \rho$, the resulting error is bounded by $C_s\varepsilon^3\rho^{-3\gamma}G^3\ell^{-3-s}$. This is the local quadratic error bound multiplied by $C_s\varepsilon\rho^{-\gamma}G/\ell$.

For $w > 2z$ in the band, pair $w$ and $-w$ using the height derivative of the odd pure-source remainder. The interpolated height distance is at least $w/2$. Integration over the seven base tangent directions bounds the derivative kernel in (6.38) by $C_s(w^{-5-s} + \rho^{-2}w^{-3-s}) \le C_s w^{-5-s}$, since $w < c_1\rho$. The band error is therefore at most

$$C_{s,K}\varepsilon^3\rho^{-3\gamma} z \int_{2z}^{c_1\rho} g(w)^3 w^{-5-s}\, dw,$$

with zero contribution when this interval is empty. For $g_+$ and $\rho \le 1$ the integral is bounded by $C_s\rho^{3q-4-s}$, so division by the upper floor gives a bounded factor $\rho^{3q-4-s-3\gamma}$. For $\rho \ge 1,\ z \le 1$ the integral is bounded by $C_s$, and the factor is at most $C_s\rho^{-2\gamma}$. For $z \ge 1$ the integral is bounded by $C_s z^{-1-s}$. The resulting factors are bounded by $C_{s,K}\rho^{-2\gamma}z^{\delta}$ and $C_{s,K}\rho^{-2\gamma}z$, for the upper and lower scalars, respectively. Thus this band contributes at most $C_{s,K}\varepsilon^3 Q$.

Pure-source term on the remaining regions. In the complement of the preceding band, for $z \le K\rho$, the kernel distance with cutoff is bounded below by $c_K(\rho + r + w)$ along the pairing of $w > 0$ with $-w$. On $|P-Q| \ge c\rho$ this follows from base separation and from the height separation when $w > 2K\rho$. On $|P-Q| < c\rho,\ w \ge c_1\rho$, the height separates the source if $w > 2z$. When $w \le 2z$, we have $z \ge c_1\rho/2$ and $\ell \asymp \rho$, so the cutoff gives the bound. Apply Lemma 6.15 to the two terms of (6.38), with $R = \rho + w$, and multiply by the height gain $C_K z$. This bounds the near-height contribution by

$$C_{s,K}\varepsilon^2 z I_{2,4+2\gamma+s}(\rho).$$

For the upper scalar at $\rho \le 1$ the height integral is bounded, by $2q > 3 + 2\gamma + s$. For $\rho \ge 1$ it is bounded by $C_s\rho^{-1-2\gamma-s}$, and for the lower scalar this estimate holds on $\rho \ge 1/K$ with $C_{s,K}$. Dividing by the upper floor gives a bounded factor if $\rho \le 1$, and at most $C_s\rho^{-1-\gamma-s}$ if $\rho \ge 1,\ z \le 1$. If $z \ge 1$, the factors for the upper and lower scalars are bounded by $C_{s,K}\rho^{\delta-\gamma}$ and $C_{s,K}\rho^{1-\gamma}$, using $z \le K\rho$. These are bounded because $\delta < 1 < \gamma$.

In the far-height region the quadratic bound already handles $|w| \le 2z$. For $w > 2z$, the height separation along the pairing is at least $w/2$, and the base separates the kernel when $r$ is large relative to $w$. Thus the distance with cutoff is bounded below by $c(r + w)$. The same radial lemma, now with $R = w$, bounds the paired contribution by

$$C_s\varepsilon^2 z \int_{2z}^{\infty} g(w)^2 w^{-4-2\gamma-s}\, dw.$$

For the upper scalar at $z \le 1$ this is $O_s(\varepsilon^2 z)$. For $z \ge 1$ it is $O_s(\varepsilon^2 z^{-2\gamma-s})$. Division by the respective floors and by $\varepsilon$ bounds the quotient by $C_s\varepsilon$, since $z^{\delta-\gamma}$ and $z^{1-\gamma}$ are bounded for $z \ge 1$.

Conclusion. The linear term of the cutoff calculation is $-2\mathcal{J}_s F$, by Lemma 6.13. Restoring the geometric core and the Jacobi cutoff gives the stated estimates. In the near-height region, the bounds sum to $C_{s,K}(\varepsilon^2 + \varepsilon^{\mu})Q$, where $\mu = 8/p > 2$ because $p < 7/2$. In the far-height region the two core contributions are absorbed because $2p < 8$, $\lambda/\rho \le 1/T_0 \le 1$, and $\mu - 1 - 2/p = 6/p - 1 > 0$. For the remaining terms use $2/p < 1 < 1 + 1/p$, so that $\varepsilon$ and $\varepsilon^{1+1/p}$ are bounded by $\varepsilon^{2/p}$ when $0 < \varepsilon \le 1$. Increasing $T_0$ after these choices only strengthens the distance and smallness bounds used in the proof. □

6.11. **Ordered geometric labels and explicit graph barriers.** Let $\mathfrak{h}_{\pm}$ be the ordered base profiles of Theorem 5.10, with $E^0_{\pm} = \{(x,y) : |y| > \mathfrak{h}_{\pm}(|x|)\}$. Use their dilation labels $V_{\pm}$ from (5.40) with the odd extensions specified there. In particular $0 < V_- < V_+$ in $D$. Lemma 5.11 gives their $C^{1,\eta}_{\mathrm{loc}}$ regularity, homogeneity of degree $p$, and the estimates (5.41).

Put $g_0 = g_-(0+) = 1 + c + a$. For $X \in D$, set

$$U_\varepsilon(X) = g_+^{-1}(V_+(X)/\varepsilon),$$
$$L_\varepsilon(X) = \begin{cases} 0, & V_-(X) \le \varepsilon g_0, \\ g_-^{-1}(V_-(X)/\varepsilon), & V_-(X) > \varepsilon g_0. \end{cases} \tag{6.48}$$

The inverses are on the positive axis. Extend both graphs oddly under $T$, and set them equal to zero on $\Sigma$. Denote their subgraphs by $E_\varepsilon^+$ and $E_\varepsilon^-$, respectively. At $w > 0$ their sections are given by $V_\pm(Y) > \varepsilon g_\pm(w)$. For the lower graph, $g_-(w) > g_0$ at positive height, so this formula also accounts for the plateau. At $w < 0$, reflection and oddness give the reflected complement of the positive section at $-w$, up to the height level set. For nonzero heights these level sets are smooth leaves and have measure zero by Lemma 5.11. It follows that

$$E_\varepsilon^\pm = \{(Y, w) : V_\pm(Y) > \varepsilon g_\pm(w)\} \quad \text{up to null sets.}$$

For $w > 0$ this is the dilation of $E_\pm^0$ by $\lambda_\pm(w) = (\varepsilon g_\pm(w))^{1/p}$. Formula (5.44) with $v = \varepsilon g_\pm(w)$ gives the normal field (6.31) outside $\rho < R_* \lambda_\pm(w)$. Factor interchange reverses the normal coordinate and preserves $\rho$. Taking the reflected complement at negative heights therefore gives the same formula with the odd scalars there, and the core radius is $R_*(\varepsilon|g_\pm(w)|)^{1/p}$.

**Proposition 6.17** (Strict curvature signs away from the plateau edge)**.** *For sufficiently small $\varepsilon > 0$, the explicit graphs satisfy*

$$\mathcal{H}_s[E_{U_\varepsilon}](X, U_\varepsilon(X)) > 0 \quad (X \in D),$$

*and*

$$\mathcal{H}_s[E_{L_\varepsilon}](X, L_\varepsilon(X)) < 0 \quad (X \in D,\ L_\varepsilon(X) > 0).$$

*On the interior of the zero plateau the lower curvature is nonpositive.*

*Proof.* Fix $K \ge 4$. At an evaluation point of positive height $z$, write $\rho = (|x| + |y|)/\sqrt{2}$ and $\lambda = \lambda_\pm(z)$. The radial part of the linear margin in (6.32) has weight

$$L_2^\pm = c_s^{\mathrm{v}} \Delta_\kappa \rho^{-\kappa-1-s} \lambda^{\kappa+1},$$

since $h_\pm(z) = g_\pm(z)^{1+\eta}$ and $p(1+\eta) = \kappa + 1$. In particular,

$$\rho^{-s}(\lambda/\rho)^{2p} = \frac{(\lambda/\rho)^{2p-\kappa-1}}{c_s^{\mathrm{v}} \Delta_\kappa} L_2^\pm, \qquad 2p - \kappa - 1 = 2\gamma - \tfrac{3}{2} > 0. \tag{6.49}$$

Choose a single $T_* \ge \max\{T_0, R_*\}$ so large that the first term of the bound (6.44), including its constant, is at most $B_\pm L_2^\pm/2$ on $\rho \ge T_* \lambda$ for both signs. This follows from (6.49). Also take $T_*$ at least the threshold in Proposition 6.14 with the coefficients $A_-, B_-$ and core radius factor $R_*$. The remaining errors in (6.43) and (6.44) are bounded respectively by $C_{s,K} \varepsilon L_1^\pm$ and $C_s \varepsilon^{2/p} L_1^\pm$. Choose $\varepsilon$ small enough that each is at most $A_\pm L_1^\pm/2$, and so that the linear and nonlinear estimates apply. Thus on $\rho \ge T_* \lambda$ the upper sign holds for every $z > 0$, and the lower sign holds for $z \ge 1$.

For the lower sign at smaller positive heights, introduce

$$E_\varepsilon^{\mathrm{ref}} = \{(Y, w) : V_-(Y) > \varepsilon \tilde{g}(w)\}. \tag{6.50}$$

For $w > 0$, the reference agrees with $E_{L_\varepsilon}$, because $\tilde{g}(w) = g_-(w)$. For $w < 0$, $\tilde{g}(w) \ge 0 > g_-(w)$, so $E_\varepsilon^{\mathrm{ref}} \subset E_{L_\varepsilon}$ up to null sets. At a point $Z_0 = (X, z)$ of height $z > 0$, the extra region is at distance at least $z$ from $Z_0$. Thus

$$\mathcal{H}_s[E_{L_\varepsilon}](Z_0) - \mathcal{H}_s[E_\varepsilon^{\mathrm{ref}}](Z_0) = -2 \int_{E_{L_\varepsilon} \setminus E_\varepsilon^{\mathrm{ref}}} |Z_0 - Z|^{-9-s}\, dZ \le 0.$$

At every height with $\tilde{g}(w) > 0$, the reference section is the leaf superlevel with $v = \varepsilon \tilde{g}(w)$. Formula (5.44) gives the reference field of Proposition 6.14. If $\tilde{g}(w) = 0$, the section is $D$. That proposition now gives the lower sign for $0 < z \le 1$ and $\rho \ge T_* \lambda$.

For the remaining region $\rho \le T_*\lambda$, translate the height $z$ to zero and dilate by $1/\lambda$ in all nine variables. By homogeneity of $V_\pm$, the scaled upper set and scaled lower reference set have defining inequalities

$$V_+(Y) > \frac{g_+(z+\lambda w)}{g_+(z)}, \qquad V_-(Y) > \frac{\tilde g(z+\lambda w)}{g_-(z)},$$

respectively. In each case the scaled evaluation point is $(X/\lambda, 0)$ with $V_\pm(X/\lambda) = 1$. Its base point belongs to the compact part of the leaf $V_\pm = 1$ where $(|Y_x|+|Y_y|)/\sqrt{2} \le T_*$. This compact set is contained in $D$ and can include the smooth factor axis.

For the upper scalar, $g_+(z) \le C_s z^q$ when $z \le 1$ and $g_+(z) \le C_s z$ when $z \ge 1$. As $q > p > 1$, these give $\lambda/z \le C_s\varepsilon^{1/p}$ for all $z > 0$. Fix a bounded scaled height interval $|w| \le M$. For small $\varepsilon$, the argument $z + \lambda w$ is between $z/2$ and $3z/2$, where $g_+$ is comparable to $g_+(z)$. The derivative bounds for the scalar give, for $j = 1, 2$,

$$\left|\partial_w^j \frac{g_+(z+\lambda w)}{g_+(z)}\right| \le C_{s,j}(\lambda/z)^j.$$

For the lower reference the estimate $g_-(z) \le C_s(1+z)$ gives $\lambda/(1+z) \le C_s\varepsilon^{1/p}$. On the same scaled interval, $1 + z + \lambda w$ stays between $(1+z)/2$ and $3(1+z)/2$. Applying the derivative bound to each power of $1 + z + \lambda w$ in $\tilde g$ gives

$$\left|\partial_w^j \frac{\tilde g(z+\lambda w)}{g_-(z)}\right| \le C_{s,j}(\lambda/(1+z))^j, \qquad j = 1, 2.$$

Both ratios equal one at $w = 0$. By the first-derivative bounds their values converge uniformly to one on $[-M, M]$, and the second-derivative bounds give $C^2$ convergence there. All these estimates are uniform in $z > 0$.

The scaled sets converge locally in measure to the vertical cylinders over $E_\pm^0$. Indeed, on a bounded set a change of indicator requires $V_\pm(Y)$ to lie between one and one of the preceding ratios. The ratios tend to one uniformly, and the leaf $V_\pm = 1$ has measure zero. Near the compact part of this leaf the nonzero gradient from (5.41) permits $V_\pm$ to be used as a local base coordinate. In these charts the boundary graph in that coordinate is the height ratio itself. This gives uniform local $C^2$ convergence, with a common bound on a finite covering of the compact family of evaluation points. Lemma 5.6, applied in dimension nine to these charts and the local convergence of the indicators, gives uniform convergence of the rescaled curvatures to the cylindrical curvatures.

Subtracting the tangent halfspace, extended vertically, allows absolute integration of the curvature of the cylinder near its evaluation point. Integrating the kernel in the vertical variable then gives $c_s^{\mathrm{v}}|P-Q|^{-8-s}$. The cylindrical curvature is therefore $c_s^{\mathrm{v}}\mathcal{H}_s[E_\pm^0]$. The uniform strict margins on the compact leaf portions follow from (5.36). The scaled upper and lower reference curvatures have the corresponding signs for small $\varepsilon$. Scaling back multiplies the curvature by the positive factor $\lambda^{-s}$. The reference inclusion gives the same lower sign for $E_{L_\varepsilon}$.

Finally, at an interior point $X \in D$ of the plateau the lower height vanishes near $X$. For $Y \in D$, the two heights at $Y, TY$ are $L_\varepsilon(Y)$ and $-L_\varepsilon(Y)$. The identity

$$d^{-8-s}G_s(b/d) = \int_0^b (d^2+h^2)^{-p_s}\,dh$$

in (6.2), applied to these two interactions, gives after angular averaging

$$\mathcal{H}_s L_\varepsilon(X) = 2\int_{\mathcal{D}}\int_0^{L_\varepsilon(b)} [K_{\mathrm{c}}(a,b;h) - K_{\mathrm{d}}(a,b;h)]\,dh\,d\mu(b) \le 0,$$

where $a$ is the radial pair of $X$ and $b$ that of $Y$. This integral converges absolutely because the height is zero near the evaluation point and the integrated kernels are bounded by $C_s d^{-8-s}$ away from it. The sign follows from Lemma 6.1. The thresholds $K, T_*$ were fixed before decreasing $\varepsilon$, so one choice of $\varepsilon$ gives all the signs. □

6.12. **Sobolev regularity and the weak signs.**

**Lemma 6.18** (Regularity of the explicit barriers)**.** *The functions in* (6.48) *are continuous, locally bounded, invariant under $O(4)\times O(4)$, and odd under factor interchange. Moreover*

$$L_\varepsilon \in W^{1,\infty}_{\mathrm{loc}}(\mathbb{R}^8), \qquad U_\varepsilon \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^8).$$

*They obey $0 \le L_\varepsilon \le U_\varepsilon$ in $D$, and $L_\varepsilon$ is positive somewhere there.*

*Proof.* By Lemma 5.11, the odd labels are locally Lipschitz. The scalar map defining $L_\varepsilon$ is Lipschitz with constant $(a\varepsilon)^{-1}$. Off the plateau this follows from $g_-' \ge a$. The inverse joins the plateau continuously at $\varepsilon g_0$ because $g_-(0+) = g_0$. The odd extension has the same Lipschitz bound by adding the bounds on the intervals separated by the plateau. Its composition with $V_-$ gives the lower regularity.

For the upper regularity, set $F(v) = g_+^{-1}(v/\varepsilon)$ for $v > 0$, with the continuous odd extension. Near zero $g_+(z) \asymp_s z^q$ and $g_+'(z) \asymp_s z^{q-1}$, the latter by differentiation of $az^q(1+z^\delta)^{-k}$. Thus on any bounded positive interval

$$|F(v)| \le Cv^{1/q}, \qquad 0 < F'(v) \le Cv^{1/q-1}.$$

The constants here may depend on $\varepsilon$ and the interval, and away from zero the bounds follow from smooth invertibility. The value bound gives continuity of $U_\varepsilon = F(V_+)$ at the cone. In the chambers, (5.41) gives

$$|DU_\varepsilon(X)| \le C|X|^{p-1}|V_+(X)|^{1/q-1}.$$

Write $X = R(\cos\theta\,\xi, \sin\theta\,\zeta)$ with $\xi,\zeta \in S^3$, $0 \le \theta \le \pi/2$. Since $|V_+(X)| \asymp_s R^p|\sin\theta - \cos\theta|$ by (5.41) and oddness, we obtain

$$|DU_\varepsilon(X)| \le CR^{p/q-1}|\sin\theta - \cos\theta|^{1/q-1}. \tag{6.51}$$

The polar measure has factor $R^7(\cos\theta\sin\theta)^3\,dR\,d\theta$. The angular bound is integrable since $|\sin\theta - \cos\theta|$ has a simple zero at $\pi/4$ and $1/q - 1 > -1$. The radial bound is integrable on bounded intervals since $6 + p/q > -1$.

To obtain the weak gradient, replace $F$ on $[-\delta_0, \delta_0]$ by its odd linear interpolant. The interpolant has slope at most $C\delta_0^{1/q-1}$. The compositions are locally Lipschitz and converge uniformly on compact sets because the change in value is at most $C\delta_0^{1/q}$. Their gradients agree eventually with $DU_\varepsilon$ at each point off the cone. On $0 < |V_+| < \delta_0$ the interpolant slope is bounded by $C|V_+|^{1/q-1}$. Thus (6.51) dominates the gradients, and dominated convergence proves $W^{1,1}_{\mathrm{loc}}$ convergence. The rotation and interchange symmetries follow from the corresponding properties of the labels.

Where $L_\varepsilon > 0$, (6.22) and $V_- < V_+$ give

$$L_\varepsilon < \frac{V_-}{a\varepsilon} < \frac{V_+}{a\varepsilon} < U_\varepsilon.$$

Where $L_\varepsilon = 0$, positivity of $U_\varepsilon$ gives the order. For $X \in D$, $V_-(RX) = R^pV_-(X) \to \infty$. The lower graph is consequently positive for all sufficiently large $R$ along this ray. □

**Lemma 6.19** (Removal of the plateau edge)**.** *For the choices of Proposition 6.17,*

$$\mathcal{H}_s L_\varepsilon \le 0 \le \mathcal{H}_s U_\varepsilon \quad \textit{weakly in } D.$$

*Both barriers belong to $W^{s,1}_{\mathrm{loc}}(\mathbb{R}^8)$.*

*Proof.* For $f \in W^{1,1}(\mathbb{R}^8)$ the translation formula gives

$$[f]_{W^{s,1}(\mathbb{R}^8)} = \int_{\mathbb{R}^8} \frac{\|f(\cdot+h) - f\|_{L^1}}{|h|^{8+s}}\,dh, \qquad \|f(\cdot+h) - f\|_{L^1} \le \min\{2\|f\|_{L^1},\, |h|\|Df\|_{L^1}\}.$$

The two bounds give integrability for large and small $|h|$, respectively. Applying this to local cutoffs of the barriers proves their local $W^{s,1}$ regularity.

The upper barrier is smooth in $D$, and the lower one is smooth on either side of

$$S_\varepsilon = \{X \in D : V_-(X) = \varepsilon g_0\}.$$

This is a regular level surface by (5.41). The pointwise signs give the weak signs on these smooth regions, by the identity between the pointwise and weak operators following (6.4).

For a nonnegative test $\psi \in C_c^\infty(D)$ that can meet $S_\varepsilon$, take a smooth $0 \le \chi \le 1$ equal to one on $[-1,1]$ and zero outside $[-2,2]$. Set $f_{\delta_0} = \psi\,\chi((V_- - \varepsilon g_0)/\delta_0)$. On a fixed compact neighborhood of the support of $\psi$, regular level coordinates with $V_- - \varepsilon g_0$ as one coordinate show that the strip $|V_- - \varepsilon g_0| \le 2\delta_0$ has volume $O(\delta_0)$. The cutoff derivative costs $O(\delta_0^{-1})$ on that strip, so $\|f_{\delta_0}\|_{L^1} \le C\delta_0$ and $\|Df_{\delta_0}\|_{L^1} \le C$. The translation formula above, split at $|h| = \delta_0$, yields

$$[f_{\delta_0}]_{W^{s,1}} \le C\left(\int_0^{\delta_0} r^{-s}\,dr + \delta_0\int_{\delta_0}^{\infty} r^{-1-s}\,dr\right) \le C_s\delta_0^{1-s}.$$

The test $\psi - f_{\delta_0}$ is nonnegative and has support away from the edge. Hence its weak lower curvature is nonpositive. Letting $\delta_0 \downarrow 0$ in (6.4) gives the sign for $\psi$. □

**Theorem 6.20** (Global ordered barriers)**.** *For every fixed $s \in (0,1)$, there are continuous, locally bounded functions $L, U : \mathbb{R}^8 \to \mathbb{R}$, given explicitly by (6.48), such that*

$$L, U \in W^{s,1}_{\mathrm{loc}}(\mathbb{R}^8), \qquad 0 \le L \le U \text{ on } D, \qquad \mathcal{H}_s L \le 0 \le \mathcal{H}_s U \text{ weakly on } D.$$

*They are $O(4)\times O(4)$-invariant and odd under factor interchange, and $L$ is positive somewhere in $D$. For every fixed $X \in D$,*

$$\frac{L(RX)}{R} \sim \frac{R^{p-1}V_-(X)}{a\varepsilon} \longrightarrow +\infty \qquad (R \to \infty). \tag{6.52}$$

*Proof.* Take the ordered profiles of Theorem 5.10, the scalars chosen above, and $\varepsilon$ as in Proposition 6.17. Lemmas 6.18 and 6.19 give the regularity, order, and weak signs. For fixed $X \in D$, the value $R^pV_-(X)$ tends to infinity. Hence for large $R$ the lower graph is past its plateau and $\varepsilon g_-(L(RX)) = R^pV_-(X)$, with $L(RX) \to \infty$. Since $g_-(z)/z \to a$, dividing this equality by $R$ and using $p > 1$ gives (6.52). □

## 7. Dirichlet exhaustion and the entire minimal graph

The barriers are ordered on the chamber $D$, while oddness reverses their order on $TD$. We therefore use the angular kernel ordering of Lemma 6.1 to prove a comparison principle on one chamber. A Dirichlet exhaustion then produces the entire graph.

### 7.1. The published existence and regularity inputs.

For an open set $\Omega \subset \mathbb{R}^8$ and $\ell > 0$, write $\Omega_\ell = \{X : \mathrm{dist}(X,\Omega) < \ell\}$.

**Proposition 7.1** (Dirichlet graphs and interior bounds)**.** *The following results hold for each fixed $0 < s < 1$.*

1. *If $\Omega$ is a bounded Lipschitz domain and $g : \Omega^c \to \mathbb{R}$ is measurable and essentially bounded on every bounded subset of $\Omega^c$, the graph Dirichlet problem has a unique variational graph minimizer $v = g$ almost everywhere on $\Omega^c$. Minimization is in the energy-difference sense of* [7, Definition 1.2]*. The minimizer belongs to $W^{s,1}(\Omega) \cap L^\infty(\Omega)$, solves $\mathcal{H}_s v = 0$ weakly in $\Omega$, and has a smooth representative there. Its subgraph is locally $s$-minimal in $\Omega \times \mathbb{R}$.*
2. *If the subgraph of a measurable function $v$ is locally $s$-minimal in $B_{2r}(X_0)\times\mathbb{R}$ and $v$ is essentially bounded on $B_r(X_0)$, then $v$ has a smooth representative on $B_r(X_0)$ satisfying*

$$\|Dv\|_{L^\infty(B_r(X_0))} \le C_{8,s}\left(1 + \frac{\mathrm{osc}_{B_r(X_0)}\, v}{r}\right)^{9+s}. \tag{7.1}$$

   *Here the oscillation is the essential oscillation.*
3. *On any open set $\Omega$, a measurable finite function $v \in W^{s,1}_{\mathrm{loc}}(\Omega)$ which weakly solves $\mathcal{H}_s v = 0$ has a smooth representative there which solves the equation pointwise. Its subgraph is locally $s$-minimal in $\Omega\times\mathbb{R}$, with finite fractional perimeter and comparison against arbitrary measurable set competitors in every open set compactly contained in the cylinder.*

*Verification of the quoted hypotheses.* We use [7, Theorems 1.3 and 1.4] with the graph kernel $g_s(q) = (1+q^2)^{-p_s}$ from Section 6. It is even, continuous, and takes values in $(0,1]$, with

$$\int_{\mathbb{R}} |q| g_s(q)\, dq = \int_1^\infty t^{-p_s}\, dt = \frac{1}{p_s - 1}.$$

The exterior-tail condition [7, (1.13)] requires that

$$X \longmapsto \int_{\Omega_{\Theta \operatorname{diam}\Omega}\setminus\Omega} \frac{|g(Y)|}{|X-Y|^{8+s}}\, dY \quad \text{belong to } L^1(\Omega)$$

for a fixed $\Theta > 0$ depending on the kernel, the dimension, and $s$. If $M$ bounds $|g|$ on this collar, the $L^1$ norm is at most $MI_s(\Omega)$. With $d_\Omega(X) = \operatorname{dist}(X, \Omega^c)$, we have

$$I_s(\Omega) := \int_\Omega \int_{\Omega^c} |X-Y|^{-8-s}\, dY\, dX \le \frac{|S^7|}{s} \int_\Omega d_\Omega(X)^{-s}\, dX < \infty. \tag{7.2}$$

The $Y$-integral is bounded by integrating over $|X-Y| \ge d_\Omega(X)$. A finite cover by Lipschitz boundary charts gives $|\{X \in \Omega : d_\Omega(X) < t\}| \le C_\Omega t$ for $0 < t < t_0$, since the points in this layer lie in strips of vertical thickness $O(t)$ over the boundary graphs. Thus

$$\int_\Omega d_\Omega(X)^{-s}\, dX \le t_0^{-s}|\Omega| + s\int_0^{t_0} t^{-s-1} |\{X \in \Omega : d_\Omega(X) < t\}|\, dt < \infty$$

because $s < 1$. Theorem 1.3 now yields the unique minimizer in $W^{s,1}(\Omega)$. Taking the larger of the collar constants in the two theorems, Theorem 1.4 gives

$$\|v\|_{L^\infty(\Omega)} \le \operatorname{diam}\Omega + \|g\|_{L^\infty(\Omega_{\Theta\operatorname{diam}\Omega}\setminus\Omega)}.$$

Our pointwise and weak operators (6.2) and (6.3) agree with [7, (1.5)–(1.6)] for $n = 8$. With the boundedness just proved, the bounded-domain part of [7, Theorem 1.10] applies to the Dirichlet minimizer in (i). For (iii), use the equivalence of weak solutions, smooth pointwise solutions, and locally $s$-minimal subgraphs in the first part of that theorem, which applies to any open $\Omega$. By [7, Definition 1.1], the set minimality compares with every measurable $F$ agreeing with the subgraph outside the relatively compact open set of comparison.

For (ii), apply [2, Theorem 1.1] with base dimension 8 and its perimeter exponent $\alpha = s$. Its definition of minimality in the cylinder uses the same relatively compact open sets. After translating $X_0$ to the origin, the hypothesis is minimality over $B_{2r}$, and the oscillation in the estimate is over $B_r$. □

7.2. **Positivity of symmetric Dirichlet solutions.** We establish nonnegativity on $D$ before using the comparison principle, which requires nonnegative chamber values. For $d > 0$ write $F_d(h) = d^{-7-s}\mathcal{G}_s(h/d)$. This function is even and satisfies $F_d(0) = F_d'(0) = 0$. Differentiation gives

$$F_d'(h) = d^{-8-s} G_s(h/d), \qquad F_d''(h) = d^{-9-s}(1 + h^2/d^2)^{-p_s} = (d^2+h^2)^{-p_s}. \tag{7.3}$$

For a bounded Lipschitz base domain $\Omega$, set $Q_\Omega = (\mathbb{R}^8 \times \mathbb{R}^8) \setminus (\Omega^c \times \Omega^c)$. For functions $u, v$ with the same exterior datum and restrictions in $W^{s,1}(\Omega)$, put

$$\Delta_\Omega(u,v) = \iint_{Q_\Omega} \left[F_{|z-\zeta|}(u(z) - u(\zeta)) - F_{|z-\zeta|}(v(z) - v(\zeta))\right] dz\, d\zeta. \tag{7.4}$$

This integral of the difference is absolutely convergent by [7, Lemma 2.8]. The minimization condition in [7, Definition 1.2] is $\Delta_\Omega(u,v) \le 0$ for each such competitor $v$.

**Lemma 7.2** (Folding into the positive chamber)**.** *Let $\Omega$ be a bounded Lipschitz domain invariant under $O(4) \times O(4)$ and $T$. Suppose its exterior datum $g$ is locally bounded, invariant, odd, and nonnegative on $D \setminus \Omega$. The unique Dirichlet graph minimizer $u$ with exterior datum $g$ is invariant and odd, and satisfies $u \ge 0$ on $D$.*

*Proof.* By Proposition 7.1(i), $u$ belongs to $W^{s,1}(\Omega)\cap L^\infty(\Omega)$. If $Q\in O(4)\times O(4)$, then $u\circ Q$ is another minimizer with the same datum because the change of variables $(z,\zeta)\mapsto(Qz,Q\zeta)$ preserves distances and $Q_\Omega$. The same reasoning, using the evenness of $F_d$, applies to $-u\circ T$. Uniqueness gives the two symmetries almost everywhere. For each symmetry the identity holds everywhere in $\Omega$ for the smooth representative by continuity. Use the prescribed representative $g$ outside $\Omega$.

Set $v=|u|$ on $D$, $v(z)=-|u(Tz)|$ on $TD$, and $v=u$ on the cone. Since $g$ is odd and nonnegative on $D\setminus\Omega$, $v=u$ on $\Omega^c$. To check admissibility, put $M=\|u\|_{L^\infty(\Omega)}$. For almost all $z,\zeta\in\Omega$,

$$|v(z)-v(\zeta)|\le|u(z)-u(\zeta)|+2M|\chi_D(z)-\chi_D(\zeta)|.$$

We verify the finiteness of the seminorm of $\chi_D$ on any ball $B_R$. For $z=(x,y)\in D$ with $r=|x|$ and $t=|y|$,

$$\operatorname{dist}(z,D^c)=\frac{t-r}{\sqrt{2}}. \tag{7.5}$$

Indeed, the radii $(r',t')$ of a point of $D^c$ satisfy $r'\ge t'$, so

$$t-r\le(t-t')+(r'-r)\le\sqrt{2}\big((r-r')^2+(t-t')^2\big)^{1/2}.$$

The distance between the points is at least the distance between their radii. Equality is attained by taking both radii equal to $(r+t)/2$ along the directions of $x$ and $y$, with an arbitrary direction when $x=0$. Integrating first in $\zeta$ gives

$$\int_{B_R\setminus D}|z-\zeta|^{-8-s}\,d\zeta\le\frac{|S^7|}{s}\operatorname{dist}(z,D^c)^{-s}.$$

The integral of the right-hand side over $B_R\cap D$ is finite, since

$$\int_0^R\int_0^t r^3t^3(t-r)^{-s}\,dr\,dt\le R^6\int_0^R\frac{t^{1-s}}{1-s}\,dt=\frac{R^{8-s}}{(1-s)(2-s)}.$$

This proves $\chi_D\in W^{s,1}(B_R)$ and hence $v\in W^{s,1}(\Omega)$.

For $a,b\in\mathcal{D}$, let $J_{\mathrm{d}}(a,b;h)$ and $J_{\mathrm{c}}(a,b;h)$ be the normalized angular averages of $F_d(h)$ with the direct and crossed distances, respectively. They are even, and both their values and first derivatives vanish at zero. Off the radial diagonal both distances are bounded away from zero for the fixed radial pair, so (7.3) yields

$$\partial_h^2J_{\mathrm{d}}(a,b;h)=K_{\mathrm{d}}(a,b;h),\qquad\partial_h^2J_{\mathrm{c}}(a,b;h)=K_{\mathrm{c}}(a,b;h). \tag{7.6}$$

Let $\mathcal{D}_\Omega\subset\mathcal{D}$ consist of radii of points in $D\cap\Omega$. By invariance of $\Omega$, the indicator of $Q_\Omega$ becomes $\mathbf{1}_{\{a\in\mathcal{D}_\Omega\text{ or }b\in\mathcal{D}_\Omega\}}$. For an odd invariant function $q$, its same-chamber terms on $D\times D$ and $TD\times TD$ both average to $J_{\mathrm{d}}(a,b;q(a)-q(b))$. The mixed terms on $D\times TD$ and $TD\times D$ both average to $J_{\mathrm{c}}(a,b;q(a)+q(b))$, by oddness of $q$ and evenness of $F_d$. Thus $\Delta_\Omega(u,v)$ is twice the integral over the indicated radial pairs, with measure $d\mu(a)d\mu(b)$, of the change from $q=v$ to $q=u$ in

$$J_{\mathrm{d}}(a,b;q(a)-q(b))+J_{\mathrm{c}}(a,b;q(a)+q(b)).$$

The absolute convergence of (7.4) justifies averaging the differences in the angular variables first.

If $u(a)u(b)\ge0$, the displayed sum is unchanged by folding. If $u(a)u(b)<0$, put $A=|u(a)|+|u(b)|$ and $B=\big||u(a)|-|u(b)|\big|$. Before folding, the absolute height arguments for the direct and crossed terms are $A$ and $B$, respectively. After folding they are $B$ and $A$. The sum before folding minus the sum after folding is therefore

$$\begin{aligned}&[J_{\mathrm{d}}(a,b;A)-J_{\mathrm{d}}(a,b;B)]-[J_{\mathrm{c}}(a,b;A)-J_{\mathrm{c}}(a,b;B)]\\&\quad=\int_B^A\int_0^\tau[K_{\mathrm{d}}(a,b;h)-K_{\mathrm{c}}(a,b;h)]\,dh\,d\tau\ge0\end{aligned}$$

by (7.6) and Lemma 6.1. It follows that $\Delta_\Omega(u,v) \geq 0$. Minimality gives the reverse inequality, so $\Delta_\Omega(u,v) = 0$. The additivity of the integrable differences gives

$$\Delta_\Omega(v,w) = \Delta_\Omega(v,u) + \Delta_\Omega(u,w) \leq 0 \qquad \text{for every competitor } w.$$

Thus $v$ is also a minimizer, and uniqueness gives $u = v$ almost everywhere. This proves nonnegativity on $D$ for the chosen representative by continuity inside $\Omega$ and the hypothesis on $g$ outside. □

### 7.3. **A comparison principle on one chamber.**

**Lemma 7.3** (Chamber comparison)**.** *Let $\Omega = B_R^8$. Suppose $u, v$ are finite measurable, invariant, and odd functions, locally bounded on $\mathbb{R}^8$, with $u, v \in W^{s,1}(\Omega)$ and $u, v \geq 0$ on $D$. Assume*

$$u \leq v \quad \text{on } D \setminus \Omega, \qquad \mathcal{H}_s u - \mathcal{H}_s v \leq 0 \quad \text{weakly on } D \cap \Omega.$$

*Then $u \leq v$ almost everywhere on $D$.*

*Proof.* Put $w = u - v$, $A = D \cap B_R$, and $\eta = w_+ \mathbf{1}_D$. The exterior order gives $\eta = 0$ outside $A$. On $\Omega$, the positive-part map gives $[w_+]_{W^{s,1}(\Omega)} \leq [w]_{W^{s,1}(\Omega)}$. Multiplication by $\mathbf{1}_D$ adds at most $\|w_+\|_{L^\infty(\Omega)}[\mathbf{1}_D]_{W^{s,1}(\Omega)}$ to this bound. The latter seminorm is finite by the proof of Lemma 7.2. Since $\eta$ vanishes outside $\Omega$, its whole-space seminorm has only the additional term

$$2\int_\Omega \int_{\Omega^c} \frac{|\eta(X)|}{|X-Y|^{8+s}}\, dY\, dX,$$

which is finite by boundedness of $\eta$ and (7.2). Thus $\eta \in W^{s,1}(\mathbb{R}^8)$.

To justify testing on $A$ with $\eta$, we first estimate the volume of the $2\delta$-neighborhood of $\partial A$. For $0 < \delta < 1$ it is contained in the union of the spherical annulus $\{||X| - R| < 2\delta\}$ and the cone strip $\{X \in B_{R+2\delta} : \operatorname{dist}(X, \Sigma) < 2\delta\}$. The annulus has volume at most $C_R\delta$. By (7.5) and its reflected version, the strip is contained in $\{0 \leq r, t \leq R+2,\ |t-r| < 2\sqrt{2}\,\delta\}$ in the radial variables. Its volume is at most

$$|S^3|^2 \int_0^{R+2}\int_0^{R+2} \mathbf{1}_{\{|t-r|<2\sqrt{2}\,\delta\}} r^3 t^3\, dt\, dr \leq C_R\delta.$$

Choose a Lipschitz cutoff $0 \leq h_\delta \leq 1$ that equals one at distance at most $\delta$ from $\partial A$, vanishes at distance at least $2\delta$, and satisfies $|Dh_\delta| \leq C/\delta$ almost everywhere. For instance, take a piecewise linear cutoff of the distance to $\partial A$. The volume bound yields $\|h_\delta\|_{L^1} \leq C_R\delta$ and $\|Dh_\delta\|_{L^1} \leq C_R$. The translation estimate in the proof of Lemma 6.19, split at translation length $\delta$, gives $[h_\delta]_{W^{s,1}} \leq C_{R,s}\delta^{1-s}$. Moreover,

$$[\eta h_\delta]_{W^{s,1}} \leq \|\eta\|_\infty [h_\delta]_{W^{s,1}} + \iint_{\mathbb{R}^8\times\mathbb{R}^8} \frac{h_\delta(X)|\eta(X)-\eta(Y)|}{|X-Y|^{8+s}}\, dX\, dY.$$

The integral tends to zero by dominated convergence because $[\eta]_{W^{s,1}} < \infty$ and $\partial A$ has measure zero. Also $\|\eta h_\delta\|_{L^1} \leq \|\eta\|_\infty C_R\delta$. Thus $\eta(1-h_\delta) \to \eta$ in $W^{s,1}(\mathbb{R}^8)$. Each $\eta(1-h_\delta)$ is nonnegative and supported a positive distance inside $A$. Convolution with a nonnegative smooth mollifier of sufficiently small support keeps these properties and approximates the truncated function in $W^{s,1}$. For the seminorm convergence, apply continuity of $L^1$ translations to $(f(X)-f(Y))/|X-Y|^{8+s}$ under simultaneous translation of $X, Y$, where $f = \eta(1-h_\delta)$, and then average the translated differences against the mollifier. By (6.4) the weak inequality therefore holds with test $\eta$.

Use the energy primitives $J_{\mathrm{d}}, J_{\mathrm{c}}$ from the folding proof, with the radial variables $a, b \in \mathcal{D}$. For $q = u, v$, the weak integral on $D \times D$ equals the integral of $J'_{\mathrm{d}}(a, b; q(a) - q(b))(\eta(a) - \eta(b))$ against $d\mu(a)d\mu(b)$, where primes differentiate the height argument. The term on $TD \times TD$ vanishes. On $D \times TD$, write the second point as $TY$, $Y \in D$. Then the height difference is $q(a) + q(b)$ and the test difference is $\eta(a)$. The term on $TD \times D$ has the same integral by swapping the base points and using the oddness of $G_s$. The crossed kernel is symmetric in $a, b$ because $|X - TY| = |Y - TX|$. Hence the sum of the crossed terms equals the integral of

$J_{\mathrm{c}}'(a,b;q(a)+q(b))(\eta(a)+\eta(b))$. All these operations use absolute integrability from (6.4) on the corresponding pieces of the weak integral.

For $0\le\theta\le 1$ put $u_\theta=v+\theta w$ and define

$$A_{\mathrm{d}}(a,b)=2\int_0^1 K_{\mathrm{d}}(a,b;u_\theta(a)-u_\theta(b))\,d\theta,$$
$$A_{\mathrm{c}}(a,b)=2\int_0^1 K_{\mathrm{c}}(a,b;u_\theta(a)+u_\theta(b))\,d\theta.$$

The kernels are positive and finite away from the radial diagonal. Since $u_\theta\ge 0$ on $D$, we have $|u_\theta(a)-u_\theta(b)|\le u_\theta(a)+u_\theta(b)$. The monotonicity of $K_{\mathrm{d}}$ in the absolute height argument, followed by Lemma 6.1 at height $u_\theta(a)+u_\theta(b)$, gives

$$A_{\mathrm{d}}(a,b)\ge A_{\mathrm{c}}(a,b)>0. \tag{7.7}$$

Taking the difference of the two weak integrals and applying the fundamental theorem of calculus to $J_{\mathrm{d}}'$ and $J_{\mathrm{c}}'$ with (7.6) yields

$$\langle \mathcal{H}_s u-\mathcal{H}_s v,\eta\rangle=\frac12\iint_{\mathcal{D}\times\mathcal{D}}\big[A_{\mathrm{d}}(w(a)-w(b))(\eta(a)-\eta(b))+A_{\mathrm{c}}(w(a)+w(b))(\eta(a)+\eta(b))\big]\,d\mu(a)d\mu(b). \tag{7.8}$$

Each linearized term is absolutely integrable since it equals the difference of the corresponding absolutely integrable terms before linearization. Pointwise the expression in brackets equals

$$(A_{\mathrm{d}}-A_{\mathrm{c}})(w(a)-w(b))(\eta(a)-\eta(b))+2A_{\mathrm{c}}((w_+(a))^2+(w_+(b))^2).$$

The first line is nonnegative by (7.7) and monotonicity of the positive-part map. The second is nonnegative and is strictly positive wherever $w_+(a)>0$ and $A_{\mathrm{c}}>0$. If $\{a : w_+(a)>0\}$ had positive radial measure, its product with any bounded radial set of positive measure would give a positive contribution to (7.8), contrary to the weak inequality. This proves $w_+=0$ almost everywhere on $D$. □

### 7.4. **The entire graph obtained by exhaustion.**

**Theorem 7.4** (Existence between chamber barriers)**.** *Let $0<s<1$. Suppose $L,U:\mathbb{R}^8\to\mathbb{R}$ are continuous, belong to $W^{s,1}_{\mathrm{loc}}(\mathbb{R}^8)$, are $O(4)\times O(4)$-invariant, and are odd under $T$. Assume*

$$0\le L\le U \quad \text{on } D,\qquad \mathcal{H}_s L\le 0\le \mathcal{H}_s U \quad \text{weakly on } D,\qquad L(X_*)>0 \quad \text{for some } X_*\in D. \tag{7.9}$$

*Both curvature inequalities use the full functions on $\mathbb{R}^8$. Then there is an entire smooth nonaffine function $u$ such that*

$$\mathcal{H}_s u=0,\qquad L\le u\le U \text{ on } D,\qquad u>0 \text{ on } D,\qquad u(TX)=-u(X),\qquad u|_\Sigma=0. \tag{7.10}$$

*It is $O(4)\times O(4)$-invariant and its subgraph locally minimizes the fractional $s$-perimeter against arbitrary set perturbations. If in addition*

$$\lim_{R\to\infty}\frac{L(RX)}{R}=+\infty\qquad (X\in D), \tag{7.11}$$

*then the entire family of dilated subgraphs converges:*

$$\chi_{R^{-1}E_u}\longrightarrow\chi_{D\times\mathbb{R}}\quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^9)\quad \text{as } R\to\infty. \tag{7.12}$$

*Proof.* For each positive integer $j$, let $u_j$ be the Dirichlet graph minimizer on $B_j^8$ with exterior value $L$. The datum is continuous and bounded on bounded sets, so Proposition 7.1(i) gives

$$u_j=L\quad \text{on } \mathbb{R}^8\setminus B_j,\qquad u_j\in W^{s,1}(B_j)\cap L^\infty(B_j)\cap C^\infty(B_j).$$

It also gives the weak equation on $B_j$ and local $s$-minimality of the subgraph in $B_j\times\mathbb{R}$. By Lemma 7.2, $u_j$ is invariant, odd, and nonnegative on $D$.

Both barriers belong to $W^{s,1}(B_j)$ by their local Sobolev regularity. The interior bound for $u_j$ and the boundedness of $L$ on bounded subsets of $B_j^c$ make $u_j$ locally bounded across $\partial B_j$, as needed for comparison. On $D \cap B_j$ the weak signs give

$$\mathcal{H}_s L - \mathcal{H}_s u_j \le 0, \qquad \mathcal{H}_s u_j - \mathcal{H}_s U \le 0.$$

Since $u_j = L \le U$ on $D \setminus B_j$, the two applications of Lemma 7.3 yield

$$L \le u_j \le U \qquad \text{on } D. \tag{7.13}$$

The almost-everywhere inequalities extend to every point of $D \cap B_j$ by continuity, and the exterior inequalities hold for the chosen representative $u_j = L$ on $B_j^c$. For $X \in TD$, applying (7.13) to $TX$ and using oddness gives $U(X) \le u_j(X) \le L(X) \le 0$. If $X \in \Sigma$, then $TX$ is in the rotational orbit of $X$ because the two factors have the same length. Hence $u_j(TX) = u_j(X) = -u_j(X)$, so $u_j(X) = 0$. This proves the uniform height bound

$$|u_j(X)| \le |U(X)| \qquad (X \in \mathbb{R}^8). \tag{7.14}$$

Fix an integer $K \ge 1$. For $j > 2K$ the subgraph of $u_j$ is locally $s$-minimal in $B_{2K} \times \mathbb{R}$, and its height oscillation on $B_K$ is at most $2\|U\|_{L^\infty(B_K)}$. Proposition 7.1(ii) therefore gives

$$\|Du_j\|_{L^\infty(B_K)} \le C_{8,s} \left(1 + \frac{2\|U\|_{L^\infty(B_K)}}{K}\right)^{9+s}. \tag{7.15}$$

Integrating along line segments shows that the right-hand side is a common Lipschitz constant on $\overline{B_K}$. Arzelà–Ascoli gives uniform convergence on $\overline{B_K}$ along a subsequence, and a diagonal subsequence over the integers $K$ converges locally uniformly to a locally Lipschitz function $u$. For each fixed $X$, the symmetries, the bounds (7.13), and the identity on $\Sigma$ pass to the limit.

To pass to the weak equation, take $\psi \in C_c^\infty(\mathbb{R}^8)$. For $j$ large enough that $\operatorname{supp}\psi \subset B_j$, the weak integral in (6.3) is zero. At each pair $X \ne Y$, its integrand converges to the corresponding integrand for $u$. The absolute values are bounded by

$$\|G_s\|_\infty \frac{|\psi(X) - \psi(Y)|}{|X - Y|^{8+s}}.$$

This bound is integrable over all pairs because the translation estimate in the proof of Lemma 6.19 gives $W^{1,1}(\mathbb{R}^8) \subset W^{s,1}(\mathbb{R}^8)$. Thus dominated convergence gives $\langle \mathcal{H}_s u, \psi\rangle = 0$, even for unbounded exterior heights. If $M_K$ is the Lipschitz constant of $u$ on $B_K$, then

$$[u]_{W^{s,1}(B_K)} \le M_K |B_K| |S^7| \int_0^{2K} r^{-s}\, dr = \frac{M_K |B_K| |S^7| (2K)^{1-s}}{1-s}.$$

Thus $u \in W^{s,1}_{\mathrm{loc}}(\mathbb{R}^8)$. Proposition 7.1(iii) gives smoothness, the pointwise equation, and local $s$-minimality of $E_u$ against arbitrary measurable competitors on every bounded open set in $\mathbb{R}^9$. The smooth representative agrees with the continuous limit everywhere.

It remains to prove strict positivity. We have $u(X_*) \ge L(X_*) > 0$. If $u(X_0) = 0$ for some $X_0 \in D$, then $X_0$ is a local minimum, since $u \ge 0$ in $D$. Taylor's formula gives $Du(X_0) = 0$ and $|u(Y)| \le C|Y - X_0|^2$ near $X_0$. Write $d = |Y - X_0|$. Since $|G_s(q)| \le |q|$, the local integral in (6.2) has absolute integrand at most $Cd^{-7-s}$, whose integral over $B_\delta(X_0)$ is at most $C|S^7|\delta^{1-s}/(1-s)$. For $d \ge \delta$, boundedness of $G_s$ gives an integrable bound. In the part over $TD$, change variables $Y \mapsto TY$. The two height differences are $-u(b)$ on the direct pairs and $+u(b)$ on the crossed pairs, where $b$ denotes the radial pair in the integration variable. By (7.3) the angular averages of the two fluxes are $\partial_h J_{\mathrm{d}}(a, b; -u(b))$ and $\partial_h J_{\mathrm{c}}(a, b; u(b))$, where $a$ is the radial pair of $X_0$. Integrating (7.6) from zero, where both derivatives vanish, gives

$$\mathcal{H}_s u(X_0) = -2 \int_{\mathcal{D}} \int_0^{u(b)} \left[K_{\mathrm{d}}(a, b; h) - K_{\mathrm{c}}(a, b; h)\right] dh\, d\mu(b). \tag{7.16}$$

The inner integral in (7.16) is nonnegative by Lemma 6.1. It is strictly positive on a set of positive $\mu$-measure. In fact, the radial representative $u(\rho e_1, \sigma e_1)$ is continuous and positive near the radii of $X_*$. Thus there is a rectangle of radial pairs $0 < \rho < \sigma$, away from $a$, on which

$u(b) > 0$. The weight $\rho^3\sigma^3$ is positive there, and the kernel comparison is strict there for every $h$. Consequently $\mathcal{H}_s u(X_0) < 0$, contradicting the pointwise equation. This proves $u > 0$ on $D$.

If $u$ were affine, invariance under reflection in each factor would force both linear parts to vanish. Oddness would then force the constant part to vanish, contrary to $u(X_*) > 0$. Hence $u$ is nonaffine.

Finally, under (7.11), the lower barrier gives $u(RX)/R \to +\infty$ for every $X \in D$. For $X \in TD$, oddness gives $u(RX)/R = -u(RTX)/R \to -\infty$. Therefore, for $X \notin \Sigma$,

$$\chi_{R^{-1}E_u}(X, z) = \mathbf{1}_{\{z<u(RX)/R\}} \longrightarrow \mathbf{1}_D(X).$$

The exceptional set $\Sigma \times \mathbb{R}$ has nine-dimensional Lebesgue measure zero. For any sequence $R_k \to \infty$, the difference of the indicators is bounded by one and tends to zero almost everywhere. Dominated convergence on each bounded set proves (7.12) for every such sequence, hence for the full limit. □

### 7.5. **Proof of the main theorem.**

*Proof of Theorem 1.1.* Fix $s \in (0, 1)$. Theorem 6.20 supplies $L, U$ with all the hypotheses of Theorem 7.4. In particular, their weak signs hold throughout $D$, and (6.52) gives (7.11). Theorem 7.4 gives a smooth nonaffine graph with the asserted minimality, symmetries, and blow-down limit. It also gives $u_s > 0$ on $D$ and $u_s = 0$ on $\Sigma$. For $X \in TD$, we have $TX \in D$, so $u_s(X) = -u_s(TX) < 0$. Thus the zero set is exactly $\Sigma$. Since $D$ is open, the boundary of the limiting set is

$$\partial(D \times \mathbb{R}) = (\overline{D} \times \mathbb{R}) \setminus (D \times \mathbb{R}) = \Sigma \times \mathbb{R}.$$ □

## References


[1] E. Bombieri, E. De Giorgi, and E. Giusti, *Minimal cones and the Bernstein problem*, Invent. Math. **7** (1969), no. 3, 243–268. https://doi.org/10.1007/BF01404309.

[2] X. Cabré and M. Cozzi, *A gradient estimate for nonlocal minimal graphs*, Duke Math. J. **168** (2019), no. 5, 775–848. https://doi.org/10.1215/00127094-2018-0052. arXiv:1711.08232.

[3] L. Caffarelli, J.-M. Roquejoffre, and O. Savin, *Nonlocal minimal surfaces*, Comm. Pure Appl. Math. **63** (2010), no. 9, 1111–1144. https://doi.org/10.1002/cpa.20331. arXiv:0905.1183.

[4] L. Caffarelli and E. Valdinoci, *Regularity properties of nonlocal minimal surfaces via limiting arguments*, Adv. Math. **248** (2013), 843–871. https://doi.org/10.1016/j.aim.2013.08.007. arXiv:1105.1158.

[5] M. Caselli, E. Florit-Simon, and J. Serra, *Yau's conjecture for nonlocal minimal surfaces*, J. Eur. Math. Soc. (2026), published online first. https://doi.org/10.4171/JEMS/1785. arXiv:2306.07100v3.

[6] M. Cozzi, A. Farina, and L. Lombardini, *Bernstein–Moser-type results for nonlocal minimal graphs*, Comm. Anal. Geom. **29** (2021), no. 4, 761–777. https://doi.org/10.4310/CAG.2021.v29.n4.a1. arXiv:1807.05774.

[7] M. Cozzi and L. Lombardini, *On nonlocal minimal graphs*, Calc. Var. Partial Differential Equations **60** (2021), no. 4, article 136. https://doi.org/10.1007/s00526-021-02002-9. arXiv:2010.16206.

[8] J. Dávila, M. del Pino, and J. Wei, *Nonlocal s-minimal surfaces and Lawson cones*, J. Differential Geom. **109** (2018), no. 1, 111–175. https://doi.org/10.4310/jdg/1525399218. Accepted manuscript.

[9] B. Dyda, A. Kuznetsov, and M. Kwaśnicki, *Fractional Laplace operator and Meijer G-function*, Constr. Approx. **45** (2017), no. 3, 427–448. https://doi.org/10.1007/s00365-016-9336-4. arXiv:1509.08529v1.

[10] J.-C. Felipe-Navarro and T. Sanz-Perela, *Semilinear integro-differential equations, I: odd solutions with respect to the Simons cone*, J. Funct. Anal. **278** (2020), no. 2, article 108309. https://doi.org/10.1016/j.jfa.2019.108309. arXiv:1903.05158v2.

[11] A. Figalli and E. Valdinoci, *Regularity and Bernstein-type results for nonlocal minimal surfaces*, J. Reine Angew. Math. **729** (2017), 263–273. https://doi.org/10.1515/crelle-2015-0006. arXiv:1307.0234.

[12] O. Savin and E. Valdinoci, *Regularity of nonlocal minimal cones in dimension 2*, Calc. Var. Partial Differential Equations **48** (2013), nos. 1–2, 33–39. https://doi.org/10.1007/s00526-012-0539-7. arXiv:1202.0973.

[13] R. Tsiamis, *Stable s-minimal cones in $\mathbb{R}^3$ are flat for s close to zero*, arXiv:2608.21340v1 (2026).